

Introduction to Psychology

PSYC 110 — Yale University

Professor Paul Bloom

Complete Lecture Notes — All 20 Lectures

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PSYC 110 — Lecture 1 — Introduction

Overview

In this opening lecture, Professor Paul Bloom welcomes students to Introduction to Psychology and lays out the course as "a comprehensive introduction to the study of the human mind." He previews a wide range of topics — brains, children, language, sex, memory, madness, disgust, racism, love, and more — and stresses that scientific psychology has real-world relevance to everyday problems such as studying, persuading others, and choosing therapies. Bloom outlines course logistics (lectures, readings, exams, weekly questions, a book review, and an experimental participation requirement) and describes the five traditional sub-areas of psychology the course will cover: neuroscience, developmental, cognitive, social, and clinical. He argues that the mind cannot be studied through psychology alone and that the course will draw on evolution, economics, game theory, philosophy, computer science, anthropology, literature, and theology. He closes by previewing five example topics — the brain, children, attractiveness, morality (good and evil), and mental illness — and announces four guest lecturers.

Key Concepts & Definitions

- **Psychology (as framed in this course):** "A comprehensive introduction to the study of the human mind" — how the human mind works, how we think, and what makes us what we are.
- **Neuroscience:** "The study of the mind by looking at the brain."
- **Developmental psychology:** The area "trying to learn about how people develop and grow and learn." (Bloom notes this is the area he focuses on.)
- **Cognitive psychology:** "A sort of computational approach to studying the mind, often viewing the mind on analogy with a computer and looking at how people do things like understand language, recognize objects, play games, and so on."
- **Social psychology:** "The study of how people act in groups, how people act with other people."
- **Clinical psychology:** "The study of mental health and mental illness" — the aspect many people think of immediately when they hear "psychology."
- **Capgras syndrome:** Typically the result of some sort of stroke; you "develop a particular delusion" that "the people you love the most have been replaced" — by aliens, robots, Martians, CIA agents, or trained actors/actresses. Associated with a very high level of violence. Hundreds of cases exist.

- **Cotard's syndrome:** A related disorder involving the very same parts of the brain as Capgras.
"You believe that you're dead; you are persuaded that you're dead." You're walking around, you know you're walking around, you know there are people around, but you think that you're dead.
- **Psychopathy:** "People who, either due to brain damage or because they are born that way, have no moral understanding."
- **Multiple personality disorder:** Cases "where people have more than one personality" — with debate over whether such cases are a real phenomenon or made up.

Detailed Notes

Chapter 1. Introduction to and Requirements for the Course [00:03:40]

- Professor: **Dr. Paul Bloom**, professor of the course.
- The course is "a comprehensive introduction to the study of the human mind."
- Topics will include: brains, children, language, sex, memory, madness, disgust, racism, love, and many others.
- Sample questions the course will address:
 - The proper explanation for differences between men and women.
 - Whether animals can learn language.
 - The puzzle of what grosses us out.
 - Why some of us eat too much and what we could do to stop.
 - Why people go crazy in groups.
 - Whether you can trust your childhood memories.
 - Why some of us get depressed and others don't.

Format & expectations:

- Two lectures a week plus course readings.
- "To do well in the course, you have to attend both the lectures and do the readings."
- Some overlap between lectures and readings; some readings will not appear in lectures, and some entire lectures will not connect to the readings.
- If you miss class, get notes from a friend or the person next to you.
- Slides will be posted online in a black-and-white, easy-to-print format. Slides are **not** a substitute for attending class.

Texts (both required):

- **Peter Gray, *Psychology*, 5th edition** (textbook).
- **The Norton Reader, edited by Gary Marcus** (collection of short readings).
- Available at **Labyrinth bookstore on York Street**, or online.
- Bloom previously used the Marcus Reader; Professor **Marvin Chun** used Peter Gray's 5th edition last semester — so used copies may be available.

Evaluation:

- **Midterm and Final.**
- The Final will **not** be held in the exam period — "because I like to take long vacations" — and will be held on the **last day of class**.
- Exams: multiple choice and short answer, fill in the blank.
- Previous exams will be posted online prior to the exams; review sessions will be held.

Weekly questions:

- Starting the **third week of class** (not next week, the week after), each Monday a brief question or set of questions will be posted.
- Answers must be sent to your **teaching fellow** (assigned by Friday).
- Estimated 5–10 minutes, then revised to **15–20 minutes** of work.
- Purpose: motivate students to keep up with material and readings.
- Graded **pass/fail**; most everybody should be able to pass.

Other requirements:

- A **short book review**, due near the end of the class; details later.
- An **experimental participation requirement**; handout next week. Purpose: give students experience seeing what psychological research is about, and give researchers hundreds of subjects.

How to do well:

1. Attend all classes.
2. Keep up with the readings.
3. Ideally, do the readings **before** class.
4. Form study groups (formally or informally).
5. Bloom will set aside a few minutes at the start of a future class for students to introduce themselves to the person next to them.

Contact with the instructor:

- The class is large; staying anonymous is an option.
- Bloom comes at least 10 minutes early and stays late.
- Office hours listed in the syllabus.
- Students may send email and set up an appointment.
- He is willing to talk about intellectual ideas, course problems, etc., and welcomes introductions on campus.

Chapter 2. General Goals for the Course [00:10:03]

- Common (sometimes joking) motivations students bring:
 - "Maybe you're crazy and hope to become less crazy."
 - "Maybe you want to learn how to study better, improve your sex life, interpret your dreams, and win friends and influence people."
- Bloom: with the exception of the sex part, the course might actually help with some of these. Scientific psychology has real-world relevance — for studying, interacting with people, persuading others, choosing therapy.
- Larger goal: "a state of the art introduction to the most important topic that there is: us."

Five traditional sub-areas of psychology (all to be covered):

Sub-area	Definition (as given)
Neuroscience	Study of the mind by looking at the brain
Developmental	How people develop, grow, and learn (Bloom's focus)
Cognitive	Computational approach to the mind; mind on analogy with a computer; language, object recognition, games
Social	How people act in groups and with other people
Clinical	Study of mental health and mental illness

Related disciplines the course will draw on:

- "You cannot study the mind solely by looking at the discipline of psychology."
- Evolution of the mind.
- Economics and game theory ("now essential tools for understanding human thought and human behavior").

- Philosophy, computer science, anthropology, literature, theology, "and many, many other domains."

Chapter 3. Examples of Materials Covered in the Course [00:13:08]

Bloom previews **five example topics**.

Example 1: The Brain (covered next week, Monday)

- Shows an image of a specific brain — **Terri Schiavo's brain**, with a "little white mark."
- Schiavo was in a coma without consciousness as a result of brain damage.
- The case is "a stark illustration of the physical nature of mental life" — the physical basis for "everything that we normally hold dear, like free will, consciousness, morality and emotions."
- The course will begin by asking how a physical thing can give rise to mental life.

Example 2: Children

- Shows a photo of his younger son **Zachary**, dressed as Spider-Man.
- Why study child development:
 - **The question of development:** Everyone in the room can speak English, has some understanding of how the physical world works, and how other people behave. How do we come to have this knowledge? How much is "hard-wired, built-in, innate," and how much is "the product of culture, of language, of schooling"?
 - Developmental psychologists use ingenious methods to figure out "the basic components of human nature."
 - **The question of continuity:** To what extent is your fate sealed? If we met you at age five, could we describe the way you are now?
 - Quotes poet **William Wordsworth**: "The child is father to the man" — meaning you can see in every child the adult he or she will become.
 - Is this true for personality? Interests? Intelligence?
 - **The question of individual differences:** People differ in taste in food, IQ, aggressiveness/shyness, sexual attraction (males, females, both, or neither), musical ability, political liberal/conservative orientation.
 - Translated as a **genes vs. environment** question, including their interaction.
 - One common theory: we are shaped by our parents.

- Quotes British poet **Philip Larkin**:

> They mess you up, your mum and dad.

> They may not mean to but they do.

> They fill you with the faults they had

> And add some extra just for you.

- Whether parents matter has been a huge controversy in popular culture; it will occupy much of the course.

Example 3: Attractiveness

- Question: "What makes for a pretty face?"
- Shows highly-rated attractive **computer-generated** Caucasian male and female faces — not real people.
- By generating faces and asking people to rate them, scientists have learned what makes a face attractive, both **within cultures and across cultures**.
- To be discussed when covering **social behavior** and **sex**.
- Not all attractiveness is linked to sex — example: **pandas** are "notoriously cute."

Example 4: Morality — Good and Evil

- "Morality is extremely central to our lives."
- The deep question: good and evil, evil and good.
- Three pictures exemplify different sorts of evil:
 1. **Institutional evil** — somebody behaving cruelly toward another, "perhaps not due to malice but because of the situation that she's in."
 2. **Osama bin Laden** — "a mass murderer or driven by political cause?"
 3. **Ted Bundy** — an example before getting into "technical stuff like crazy-evil."
 - Same question: is being good or bad part of your nature, or largely due to the situation you fall into?
 - "There's a lot of some quite spectacular experiments that try to tease that apart."
 - **Good** — two notoriously good men:
 - **Oskar Schindler** (Holocaust).
 - **Paul Rusesabagina** (Rwanda).
- Both saved many lives at great risk; both had good movies made about them.

- "You couldn't have predicted ahead of time that they would be heroes."
- Personal question: what would we do in such situations?

Example 5: Mental Illness

- A full week near the end of the course will be devoted to **major disorders** like **depression** and **anxiety**, because of their profound social importance.
- "Such disorders are reasonably common in college students."
- Many people in the room are currently suffering from a mood disorder, an anxiety disorder, or both; many are on medication.
- The course will discuss current research on causes and the best ways to treat them.

Less common disorders (also covered):

- **Memory disorders** — including disorders that prevent forming new memories, and **amnesia** (forgetting the past).
- **Phineas Gage:**
 - A construction worker about 100 years ago.
 - An explosion drove a metal pipe through his head; pipe ended up "100 feet away, covered with brains and blood."
 - He was not killed. He sat down and said "uh, oh," then stopped by a pub for cider on the way to the hospital.
 - He was not blind, not deaf, not retarded.
 - But: "He lost his sense of right and wrong. He lost his control."
 - Was previously a hard-working family man; afterward he could not hold a job, could not stay faithful to his wife, could not speak for five minutes without cursing, got into fights and brawls, got drunk.
 - Ended up on a circus sideshow traveling the country with the steel pipe that went through his head.
 - "An extraordinary example of how the brain can give rise to the mind."
- **Multiple personality disorder** — people with more than one personality; debated whether real or made up.
- **Capgras syndrome:**
 - Typically results from a stroke; hundreds of cases.
 - Delusion: "the people you love the most have been replaced" — by aliens, robots, Martians, CIA agents, trained actors and actresses.

- Associated with very high levels of violence.
- Example: a man in Australia, under the delusion that his father had been replaced with a robot, cut off his head.
 - **Cotard's syndrome:**
- Involves the same parts of the brain as Capgras.
- You believe you are dead, even while walking around aware of others.
 - These syndromes are "located ... at a pinpoint level to certain parts of your brain." The course will discuss "the best modern theories as to why these syndromes occur."

Why study extreme/abnormal cases:

- Not because they are frequent, and not from "gruesome, morbid curiosity."
- "By looking at extreme cases, they can help us best understand normal life."
- Extremes throw into sharp contrast things we take for granted.
- **Psychopathy** helps with questions of free will and responsibility, and the relationship/difference between **mental illness and evil**.
- **Multiple personality** cases force the question: **What is a self?** Are we composed of multiple people, or unified across time?
- **Capgras** tells us how we see the world — that "there is a difference between recognizing something in the sense that you could name it, and knowing what it is."

Guest lecturers (four):

| Guest | Role / Topic |

|---|---|

| **Dr. Marvin Chun** | Teaches Intro Psychology in the fall; "my competition." Lecture on **cognitive neuroscience**, especially the cognitive neuroscience of faces. |

| **Dr. Susan Nolen-Hoeksema** | "The world's authority on depression," especially **sex differences and depression**; lectures toward the end of the course. |

| **Kelly Brownell** | Head of the **Rudd Center**; focuses on obesity, eating disorders, dieting; will talk about the **psychology of food**. |

| **Dr. Peter Salovey** | Dean of Yale College; coming on **Valentine's Day** to talk about "the mysteries of love." |

- All details are in the **syllabus**. Bloom will stick around to answer questions.

Notable Quotes

"What this is going to be is a comprehensive introduction to the study of the human mind."

"We are going to cover a very, very wide range of topics including brains, children, language, sex, memory, madness, disgust, racism and love, and many others."

"Cognitive ... refers to a sort of computational approach to studying the mind, often viewing the mind on analogy with a computer."

"I am convinced that you cannot study the mind solely by looking at the discipline of psychology."

"What I want to do is provide a state of the art introduction to the most important topic that there is: us."

"The child is father to the man" — William Wordsworth (quoted by Bloom).

"They mess you up, your mum and dad. / They may not mean to but they do. / They fill you with the faults they had / And add some extra just for you." — Philip Larkin (quoted by Bloom).

"By looking at extreme cases, they can help us best understand normal life. Often by looking at extremes it throws into sharp contrast things we naturally take for granted."

"There is a difference between recognizing something in the sense that you could name it, and knowing what it is."

Assigned Readings / References

- **Peter Gray**, *Psychology*, 5th edition (textbook).
- **The Norton Reader**, edited by **Gary Marcus** (collection of short readings).

- Both available at **Labyrinth bookstore on York Street** or online.
- Syllabus referenced for office hours, full topic list, and guest-lecturer details.

PSYC 110 - Lecture 2 - Foundations: This Is Your Brain

Overview

This lecture introduces the brain as the foundation of mental life and contrasts two opposing positions: dualism (the common-sense, religiously and philosophically widespread view that the mind/soul is separate from the body) and the modern scientific consensus that "the mind is what the brain does." Professor Paul Bloom opens with Francis Crick's "Astonishing Hypothesis" and Rene Descartes' arguments for dualism, then explains why scientists reject dualism in favor of a materialist account. The bulk of the lecture builds the brain from the bottom up: starting with the neuron as the basic unit, moving to neurotransmitters and how drugs alter them, then to massively parallel wiring, then to subcortical structures, the cortex, its lobes, topographic maps, and disorders. The lecture closes with hemispheric lateralization (via Michael Gazzaniga's split-brain work) and two notes of humility: the unsolved "hard problem" of consciousness, and the tension between a mechanistic conception of mind and humanist values such as free will and intrinsic value.

Key Concepts & Definitions

- **The Astonishing Hypothesis (Francis Crick):** "You, your joys and your sorrows, your memories and your ambitions, your sense of personal identity and free will are in fact no more than the behavior of a vast assembly of nerve cells and their associated molecules... you're nothing but a pack of neurons."
- **Dualism:** The doctrine that humans consist of two separate things — a material body and an immaterial mind/soul. Found "in every religion and in most philosophical systems throughout history"; explicit in Plato; most articulately defended by Rene Descartes.
- **Beast machines (Descartes):** Descartes' term for non-human animals, which he claimed were merely physical robots.
- **Method of doubt (Descartes):** Asking "What can I be sure of?" — concluding one cannot doubt that one is thinking, supporting the separability of mind from body.
- **Materialist consensus:** "The mind is what the brain does. The mind reflects the workings of the brain just like computation reflects the working of a computer."
- **Existence proof:** Modern evidence that physical objects (machines) can do complicated things like play chess, manipulate symbols, do mathematical/logical reasoning — making it possible that humans are physical machines.

- **Neuron:** The basic unit of the brain; a specialized cell with three major parts (dendrites, cell body, axon).
- **Dendrites:** Tentacle-like extensions that receive signals from other neurons; signals can be excitatory or inhibitory.
- **Excitatory signal:** Raises the likelihood the neuron will fire.
- **Inhibitory signal:** Lowers the likelihood the neuron will fire.
- **Cell body:** Sums incoming signals; if a threshold (e.g., "+60 or something") is reached the neuron fires.
- **Axon:** The output fiber along which the neuron fires; can be many feet long (e.g., spinal cord to big toe).
- **Myelin sheath:** Insulation surrounding the axon; helps firing work quicker.
- **Sensory neurons:** Take information from the world (e.g., retina to brain).
- **Motor neurons:** Tell muscles what to do (e.g., raising a hand).
- **Interneurons:** Connect sensory and motor neurons; "do the thinking."
- **All-or-nothing principle:** A neuron either fires or it doesn't, like a gun — squeezing harder/faster doesn't increase the firing.
- **Coding of intensity:** Achieved by (1) the number of neurons firing and (2) the frequency of firing.
- **Synapse:** The tiny gap (~1/10,000 of a millimeter wide) between the axon of one neuron and the dendrite of another.
- **Neurotransmitters:** Chemicals released into the synapse when a neuron fires; can excite or inhibit the next neuron.
- **Agonists:** Drugs that increase the effect of neurotransmitters — by making more, stopping cleanup, or mimicking them.
- **Antagonists:** Drugs that slow neurotransmitter activity — by destroying them, blocking creation, or pasting over dendrites so neurotransmitters can't connect.
- **Parallel distributed processing:** How the brain is wired, in contrast to a serial PC/Mac.
- **Neural networks:** Computational systems modeled after how neurons are wired together, studied in engineering and CS departments to build smart machines.
- **Subcortical structures:** Structures below/underneath the cortex (e.g., medulla, cerebellum, hypothalamus).
- **Medulla:** Responsible for heart rate and respiration; damage likely fatal.
- **Cerebellum:** Responsible for body balance and muscular coordination; contains approximately 30 billion neurons.
- **Hypothalamus:** Responsible for feeding, hunger, thirst, and to some extent sleep.

- **Cortex:** The outer, wrinkled layer; if flattened, about two feet square; ~80% of the volume of the human brain.
- **Lobes of the cortex:** Frontal, parietal, occipital, temporal.
- **Topographic maps:** Maps of the body in the brain; nearby body parts map to nearby cortical regions.
- **Motor cortex / somatosensory (sensory) cortex:** Cortical areas where body maps reside; map size reflects degree of motor command or sensory control, not real-world body size.
- **Projection areas:** The map-containing parts of cortex; less than one quarter of the cortex.
- **Lateralization:** The unequal distribution of functions across hemispheres (e.g., for right-handers: language on the left; math and music on the right).
- **Crossover:** Left visual field → right hemisphere; right visual field → left hemisphere; right hemisphere controls left body, left hemisphere controls right body.
- **Corpus callosum:** The huge web connecting the two hemispheres.
- **Hard problem of consciousness:** The puzzle of subjective experience — how a physical brain gives rise to feelings/what-it's-like-ness.
- **Mechanistic conception of mental life:** Treating the mind as an information processor / elaborate computer to be explained, rather than as something beautiful or mysterious.

Detailed Notes

Chapter 1. The Brain, the Mind and Dualism [00:00:00]

- The class begins with Francis Crick's "Astonishing Hypothesis": that all of mental life is the activity of nerve cells.
 - Bloom predicts most students will not believe it at first; "Most people, in fact, hold a different view. Most people are dualists."
 - **Dualism:** present in every religion, most philosophical systems, explicit in Plato; most articulately defended by Rene Descartes.
 - Descartes asked: "Are humans merely physical machines, merely physical things?" Answer: No.
- Animals = "beast machines" / robots.
- Humans = immaterial souls possessing physical bodies — two separate things.
- **Descartes' two arguments for dualism:**
1. **From human action:** Descartes lived in a sophisticated time with hydraulic robots in the French Royal Gardens (a "seventeenth-century Disneyland" — stepping on a panel made a swordsman jump out, or a bathing beauty cover herself). Bodies can be reflexive (knee-tap reflex). But humans do

coordinated, creative, spontaneous things — e.g., free use of language. "Somebody says, 'How are you?' And I say, 'I am fine. How are you?' But sometimes I could say what I choose to be... 'Pretty damn good.'" Machines, Descartes argued, cannot make such choices.

2. Method of doubt: He could doubt God, country, friends, family, even his body (madmen think they have extra limbs; he might be dreaming; an evil demon — the modern analog is *The Matrix* — might be deluding him). But he could not doubt that he was thinking. Thus body and mind are separable. Descartes concluded: "I knew that I was a substance, the whole essence or nature of which is to think, and that for its existence, there is no need of any place nor does it depend on any material thing. That is to say, the soul by which I am, when I am, is entirely distinct from body."

- **Common-sense dualism — five illustrations:**

1. Language: We talk of "my arm, my heart, my child, my car," and also "my body" and "my brain" — as if we own and are separate from them.

2. Personal identity in fiction: We easily follow stories where a teen wakes up as an older Jennifer Garner, or where a man dies and is reborn into a child's body.

3. Transformations: Kafka's *The Metamorphosis* — "As Gregor Samsa woke one morning from uneasy dreams, he found himself transformed in his bed into a gigantic insect." Homer's *Odyssey* — Odysseus' companions stuck (by the witch) into the bodies of pigs: "They had the head and voice and bristles and body of swine but their minds remained unchanged as before, so they were penned there weeping."

4. Many minds, one body: Slapstick comedy (*All of Me*, with Steve Martin and Lily Tomlin); one analysis of multiple personality disorder; exorcism beliefs across cultures.

5. Surviving bodily destruction: Most cultures and religions hold this. In a Chicago survey: ~96% of Christians said they would go to Heaven; most Jewish respondents said the same (despite Judaism's less-clear afterlife story); most respondents with no religion *also* said they would go to Heaven.

Chapter 2. Scientific Consensus against Dualism [00:12:06]

- Scientific consensus: dualism is wrong; "there is no 'you' separable or separate from your body. In particular, there is no 'you' separable from your brain."

- **The mind is what the brain does.**

- **Problems with dualism:**

- It is "profoundly unscientific" — fails to explain how children learn language, what we find attractive, the basis of mental illness.

- The conduit problem: Descartes couldn't explain how an immaterial soul connects to a physical body, though he knew the connection exists (banging your toe → pain; alcohol → reasoning effects).

- **Existence proof against dualism:** Descartes thought physical objects couldn't, e.g., play chess. Now we know machines can play chess, manipulate symbols, do limited math/logic, recognize things, compute. So it's at least possible humans are physical machines / brains.
- **Brain–mind correspondence:** Long known that head blows, syphilis, caffeine, and alcohol affect mental life. Newer: **CAT scans, PET, fMRI** show different parts of brain active during different mental tasks (seeing vs. hearing vs. reading vs. generating words). With fMRI we can to some extent tell whether someone is thinking about music vs. sex, or solving a moral dilemma vs. something else.
- The brain itself looks "disgusting" — "like an old meat loaf"; gray outside the head ("gray matter") but bright red inside because it's pulsing with blood. Aside: "It's good with cream sauce but everything's good with cream sauce."
- Bloom expresses sympathy for Descartes: it does seem ridiculous that such a fleshy lump produces free will, love, and consciousness. The goal of neuroscience is to make it less ridiculous.
- Plan: build up from neurons → larger structures → whole brain.

Chapter 3. The Neuron: The Basic Building Blocks of Thought [00:19:28]

- **Basic unit:** the neuron — a specialized cell. Neurons look diverse, but a typical one has:
 - **Dendrites** — receive excitatory (+) or inhibitory (–) signals.
 - **Cell body** — sums signals; at threshold (+60 or so), the neuron fires.
 - **Axon** — output fiber; can be many feet long (spinal cord → big toe).
 - **Myelin sheath** — insulation around the axon; speeds firing.
- **Numbers:** ~1,000 billion neurons; each connected to thousands, perhaps tens of thousands of others.
- **Three flavors of neurons:**
 - Sensory neurons (e.g., retina → brain).
 - Motor neurons (brain → muscles).
 - Interneurons (do the thinking; link sensation and action).
- **Regeneration:** It was thought neurons never grow back. This is "actually not true" — some parts of the brain can regrow neurons.
- **All-or-nothing:** "A neuron is like a gun. It either fires or it doesn't."
- **Coding intensity** despite all-or-nothing firing:

1. Number of neurons firing (more neurons → more intense).

2. Frequency of firing (faster bangs → more intense).

- **Synaptic communication:**

- Tiny gap (~1/10,000 mm) between axon and next dendrite = **synapse**.

- When a neuron fires, the axon releases **neurotransmitters** across the gap; they excite or inhibit the next neuron.

- **Psychopharmacology (medical and recreational) consists of fiddling with neurotransmitters.**

- **Agonists:** increase neurotransmitter effect (make more, block cleanup, or mimic).

- **Antagonists:** decrease neurotransmitter effect (destroy them, prevent creation, or block dendrite receptors).

| Drug / disorder | Mechanism / effect |

|---|---|

| **Curare** | Antagonist; blocks motor neurons from affecting muscle fibers → paralysis → death (breathing depends on motor neurons). |

| **Alcohol** | Inhibitory. Paradox: drinkers feel disinhibited because alcohol *inhibits the inhibitory parts of the brain* (largely in the frontal lobes). Enough alcohol then inhibits the excitatory parts → fall down, pass out. |

| **Amphetamines** ("speed," "coke") | Increase arousal by increasing **norepinephrine**. |

| **Prozac** | Works on **serotonin**; makes serotonin more prevalent — depression involves too little serotonin. |

| **Parkinson's disease** | Involves loss of motor control; too little **dopamine**. |

| **L-DOPA** | Increases dopamine; temporarily alleviates Parkinson's symptoms. |

- **Wiring vs. computers:** The brain is *not* wired like a PC. Two reasons:

1. **Damage-resistant:** snip a laptop with pliers and it dies, but the brain can sustain a lot of damage with preserved function; other parts can take over.

2. **Speed paradox:** brain tissue is slow, computer wires are fast, yet the brain is extremely fast (faces recognized quickly). Solution: **massively parallel distributed processing**.

- **Neural network modeling:** Research (often outside psychology — engineering, CS) tries to build machines using neuron-like components. Caveat: this is a young field. "There is no machine yet that can recognize faces or understand sentences at the level of a two-year-old human."

- The visual cortex schematic shown is "extraordinarily simplified" — the brain is hugely more complicated than any current neural network.

Chapter 4. The Different Parts of the Brain [00:32:58]

- **Things you don't need a brain for** (often studied in France via post-decapitation reflex testing — "kind of gruesome"):

- Newborn sucking
- Limb flexation / withdrawal from pain
- Erection of the penis
- Vomiting

- **Live demo:** Bloom drops the textbook (5th edition) onto a student's outstretched hand; the hand automatically rebounds. "This is something automatic, instinctive, and does not require your brain."

- **Subcortical structures** (below the cortex):

| Structure | Function |

|---|---|

| **Medulla** | Heart rate, respiration; damage likely lethal. |

| **Cerebellum** | Body balance and muscular coordination; ~30 billion neurons. |

| **Hypothalamus** | Feeding, hunger, thirst, somewhat sleep. |

- **Cortex:**

- Outer, wrinkled layer; if flattened, ~2 feet square.
- "Where all the neat stuff takes place." Fish — little; reptiles/birds — a bit; primates — a lot; humans — "a real lot."
- ~80% of human brain volume.
- **Lobes:** frontal, parietal, occipital, temporal.
- Two hemispheres — mostly duplicated with subtle differences.

- **Topographic maps** in the cortex:

- Brain surgery can be done on conscious patients because the brain has no sense organs.
- **Dr. Penfield (McGill University):** extending earlier dog-brain experiments, stimulated patients' cortex with a small electrical probe. Zapping various points made body parts move, or caused perceived colors, sounds, or touch.
- Built up **maps:** motor cortex (left of diagram) and somatosensory/sensory cortex (right).
- **Two key properties:**

1. **Topographical:** nearby body parts map to nearby cortical regions (tongue closer to jaw than to hip — same on cortex).

2. **Distorted by use:** map size reflects motor command / sensory innervation, not actual body part size. Tongue and face occupy huge cortical area; shoulder barely shows. The "sensory body" homunculus has huge tongue/face, tiny shoulders.

- **Less than one quarter of the cortex** contains these projection maps; the rest handles language, reasoning, moral thought, etc.

- As we move from rat → cat → monkey → human, less cortex is projection, more is "other things."

- **Methods to find what other parts do:**

- **Imaging** (CAT, PET, fMRI) — see active regions during tasks like language.

- **Brain damage studies** — lesions, tumors, strokes, injury. "Neuropsychologists love when motorcyclists drive without helmets because through their horrible accidents we gain great insights." If damage to region X disables face recognition, X may be involved in face recognition.

- **Disorders that illuminate function:**

| Disorder | What it is |

|---|---|

| **Apraxia** | Not paralysis; simple movements fine, but cannot coordinate movements (e.g., wave goodbye, light a cigarette). |

| **Agnosia** ("psychic blindness") | Not blindness; eyes are intact, but cannot recognize things. |

| **Visual agnosia** | Cannot recognize objects. |

| **Prosopagnosia** | Cannot recognize faces. |

| **Sensory neglect** | Not paralysis or blindness; lose the idea of, e.g., the left side of the body or world. |

| **Aphasia** (classic) | Discovered by **Paul Broca, 1861**: a patient with brain damage could only say "tan, tan, tan, tan." |

| **Receptive aphasia** | Speaks fluently but words don't make sense; cannot understand others. |

| **Acquired psychopathy** | Frontal-lobe damage robs the ability to tell right from wrong. |

Chapter 5. Mechanist Conception and the Hard Problem of Consciousness [00:44:47]

- **How many minds do you have?** The brain looks symmetrical but isn't — real differences between right and left hemispheres.

- **Handedness:** ~9/10 are right-handed; right-handers have language in the **left hemisphere**. Left-handers are complicated — some have language right, some left, some elsewhere. Generalizations below assume right-handers.

- **Lateralized functions** (right-hander):

- Left hemisphere: language.
- Right hemisphere: math, music.

- **Crossover:**

- Left visual field → right hemisphere; right visual field → left hemisphere.
- Right hemisphere controls left body; left hemisphere controls right body.

- The two halves are connected by the **corpus callosum**.

- **Michael Gazzaniga** — one of the world's top neuroscientists, leading expert on the two hemispheres; lecture plays a *Scientific American* clip on split-brain patients (man whose hemispheres no longer communicate). Caveat: "people are just extremely pleased with themselves, so you have to ignore that while watching it."

- Methodological lesson: studying extreme/unusual cases (like split-brain patients) reveals how normal brains work.

- **Closing humility — two points:**

1. **The hard problem of consciousness:** Psychology / cognitive science / neuroscience treats the mind as an information processor — find a program that solves a problem (face recognition, language, motor control, logic), then ask how it is instantiated in the brain. But subjective experience remains mysterious: a computer plays chess and recognizes numbers, "but my computer doesn't have feelings in the same sense." Illustrations shown: an old Star Trek angst image; Bloom's son Max being happy. Huxley: "How it is that anything so remarkable as a state of consciousness comes about as a result of irritating nervous tissue, is just as unaccountable as the appearance of the Djinn... when Aladdin rubs his lamp."

2. **The mechanistic conception vs. humanist values:** Bloom is presenting a mechanistic view of mental life — explaining decisions, parental love, falling in love, etc. This may feel "repellant" because it sits uneasily with:

- **Free will and moral/legal responsibility** (cutting someone off in traffic reflects badly; saving a life deserves praise).
- **The intrinsic value of people.**
- **The spiritual value of people.**
- **Three possible responses to the tension:**

1. Reject the scientific conception of mind; embrace dualism.
2. Embrace the scientific worldview and reject humanist values — some philosophers/psychologists hold free will, responsibility, spiritual/intrinsic value are "pre-scientific notions that get washed away."
3. Try to reconcile the scientific view with humanist values — a recurring theme of the course.

Notable Quotes

"You, your joys and your sorrows, your memories and your ambitions, your sense of personal identity and free will are in fact no more than the behavior of a vast assembly of nerve cells and their associated molecules. As Lewis Carroll's Alice might have phrased it, 'you're nothing but a pack of neurons.'" — Francis Crick, The Astonishing Hypothesis

"The mind is what the brain does."

"I knew that I was a substance, the whole essence or nature of which is to think, and that for its existence, there is no need of any place nor does it depend on any material thing. That is to say, the soul by which I am, when I am, is entirely distinct from body." — Descartes

"As Gregor Samsa woke one morning from uneasy dreams, he found himself transformed in his bed into a gigantic insect." — Franz Kafka

"They had the head and voice and bristles and body of swine but their minds remained unchanged as before, so they were penned there weeping." — Homer, on Odysseus' companions

"A neuron is like a gun. It either fires or it doesn't."

"Neuropsychologists love when motorcyclists drive without helmets because through their horrible accidents we gain great insights into how the brain works."

"How it is that anything so remarkable as a state of consciousness comes about as a result of irritating nervous tissue, is just as unaccountable as the appearance of the Djinn... when Aladdin rubs his lamp." — Huxley

Assigned Readings / References

- Francis Crick — *The Astonishing Hypothesis*
- Rene Descartes — arguments for dualism (method of doubt; "beast machines")
- Plato — early dualism
- Franz Kafka — *The Metamorphosis* (Gregor Samsa)
- Homer — Odysseus' companions transformed by the witch
- *The Matrix* — modern analog of Descartes' evil-demon scenario
- *All of Me* (Steve Martin and Lily Tomlin)
- Paul Broca (1861) — patient who could only say "tan"
- Dr. Penfield (McGill University) — cortical stimulation mapping
- Michael Gazzaniga — split-brain research (*Scientific American* clip)
- The Gray book — referenced as covering lateralization in detail
- Huxley — quoted on consciousness

PSYC 110 - Lecture 3 - Foundations: Freud

Overview

This lecture introduces psychoanalysis as one of two "grand theories" in psychology (the other, behaviorism, is covered in the next lecture). Professor Paul Bloom situates Sigmund Freud as the most famous psychologist ever and emphasizes that his theory is unusual in scope: rather than being narrow, it attempts to explain everyday life, child development, mental illness, religion, war, and love. The lecture walks through the core Freudian ideas — unconscious motivation, the conflict between id, ego, and superego, the stages of psychosexual development, defense mechanisms, dreams, and the aims of psychoanalysis. Bloom then turns to the scientific assessment of Freud, drawing on Karl Popper's criterion of falsifiability and Frederick Crews's critique to argue that much of Freudian theory is "not even wrong." The lecture closes by showing that despite the rejection of Freud's particulars, his core idea of the dynamic unconscious has been vindicated by modern empirical work in language comprehension, social psychology, and terror management.

Key Concepts & Definitions

- **Psychoanalysis:** Freud's grand theory of mind, developed over many decades, encompassing everyday life, child development, mental illness, religion, war, and love.
- **Unconscious motivation:** The idea that desires and motivations governing behavior may not be available to the person's awareness; you may not know why you do what you do.
- **Id:** Present at birth; the "animal part of the self" that wants to eat, drink, pee, poop, get warm, and have sexual satisfaction. "Outrageously stupid." Works on the **Pleasure Principle** — it wants pleasure and it wants it now.
- **Polymorphous perversity:** Freud's phrase for the infant's pure desire for pleasure.
- **Ego (the self):** Develops as a reaction to the fact that pleasure isn't always available; plans how to satisfy desires or how to suppress them. Works on the **Reality Principle**. Its emergence symbolizes the origin of consciousness.
- **Superego:** The internalized rules of parents and society — a conscience. Also "stupid" — it tells you not just to avoid bad actions but to avoid bad thoughts ("You should be ashamed of yourself.").
- **Pleasure Principle:** Governs the id.
- **Reality Principle:** Governs the ego.

- **Psychosexual development:** Five stages of personality development, each associated with a particular erogenous zone. Getting "stuck" at a stage produces personality consequences.
 - **Oral stage:** First stage; mouth is associated with pleasure (sucking, chewing). Premature weaning can produce an "oral person" (eats too much, chews gum, smokes; or is dependent and needy).
 - **Anal stage:** Follows the oral stage. Problems with toilet training produce an "anal personality" — literally, unwillingness to part with one's own feces; manifests as compulsive, clean, stingy.
 - **Phallic stage:** Focus of pleasure shifts to the genitals. Fixation can lead to excessive masculinity (in females or males) or, if female, a need for attention or domination.
 - **Oedipus Complex:** Based on the myth of a king who killed his father and married his mother. The boy desires his mother, sees his father as a rival, infers his father will castrate him, and gives up.
 - **Electra complex:** Freud's add-on theory for women; tied to the penis envy story.
 - **Penis envy:** Freud's claim that girls discover they lack a penis, infer they have been castrated, turn to the father as a "penis substitute," and reject the mother.
 - **Latency stage:** After the Oedipal resolution; sex is repressed for several years.
 - **Genital stage:** The healthy adult stage.
 - **Defense mechanisms:** Mechanisms that keep "horrible stuff from the id" from reaching consciousness; part of normal life, not pathological.
- **Sublimation:** Redirecting sexual or aggressive energy into another channel (e.g., Picasso channeling sexual energy into artwork). "Gives up the details and keeps the energy."
- **Displacement:** Refocusing shameful desires from a forbidden target onto a more acceptable one (e.g., a bullied boy kicks the dog instead of his father).
- **Projection:** Attributing one's own uncomfortable impulses to someone else (e.g., projecting homosexual desires onto others).
- **Rationalization:** Giving a socially acceptable explanation for a bad action (e.g., a parent who enjoys smacking his child says "It's for the child's own good").
- **Regression:** Returning to an earlier stage of development under stress (children cry, suck thumb, seek a blanket).
- **Hysteria:** A category popular in Freud's time including hysterical blindness, hysterical deafness, paralysis, trembling, panic attacks, gaps of memory including amnesia. Viewed as symptoms of mechanisms keeping things unconscious.
 - **Free association:** Freud's later technique (after abandoning hypnosis) for accessing repressed material; patients offer **resistance**, and the analyst helps them gain **insight**.
 - **Manifest content:** What you experience in a dream.

- **Latent content:** The hidden implication of a dream.
- **Wish fulfillment:** Freud's view that every dream is a wish (sometimes forbidden) the dreamer has.
- **Falsifiability:** Karl Popper's criterion: what distinguishes science from non-science is that scientific predictions make strong claims that could be proven wrong.
- **Subliminal:** "A fancy term meaning below the level of consciousness, you don't know you're seeing them."
- **Terror management / subliminal death primes:** Studies in social psychology in which words like "corpse," "dead," "dying" are flashed too fast to be consciously seen, affecting attitudes.

Detailed Notes

Chapter 1. Sigmund Freud in a Historical Context

- The course will cover two grand theories — psychoanalysis (Freud) and behaviorism (Skinner) — across this lecture and the next.
- Unlike most psychological theories, which are narrow (e.g., a theory of racial prejudice is not a theory of language acquisition; a theory of schizophrenia is not an explanation of sexual attractiveness), these two are "theories of everything," encompassing day-to-day life, child development, mental illness, religion, war, and love.
- Bloom states this is "not a history course"; the point is that these ideas have critical influence on how we think about the present.
- Biographical facts:
 - Born in the 1850s.
 - Spent most of his life in Vienna, Austria.
 - Died in London after escaping there at the beginning of World War II as the Nazis began to occupy his city.
 - Not known for any single discovery, but for an encompassing theory of mind developed over many decades.
 - A celebrity in his time, recognized on the street.
 - Extraordinarily productive, in part because he was a serious cocaine addict.

- Nominated for Nobel Prizes in medicine and in literature; got neither. Albert Einstein wrote a letter opposing the medicine prize, saying, "Don't give the prize to Freud. He doesn't deserve a Nobel Prize. He's just a psychologist."

- Universally acclaimed as profoundly important, but also disliked:

- Personally: deeply ambitious about promoting psychoanalysis, often dishonest, extremely brutal to friends, terrible to enemies.

- Anecdote: leaving Europe during the rise of the Nazis, the Gestapo demanded he sign a letter saying he had not been harassed; he signed and added, "The Gestapo has not harmed me in any way. In fact, I highly recommend the Gestapo to everybody."

- As a Jewish "sexual renegade" he was identified by the Nazis as out to destroy sacred Christian notions.

- Some of Freud's stupider claims are what critics latch onto:

- **Phallic symbols**: certain architectural monuments as subconscious penile representations.

- **Penis envy**: the developmental claim that every female discovers she lacks a penis, infers she has been castrated, turns to the father as a penis substitute, and rejects the mother.

Chapter 2. Unconscious Motivation: The Id, Ego and Superego

- At the core of Freud's importance are two big ideas:

1. The existence of an **unconscious / unconscious motivation**.

2. The notion of **unconscious dynamics / unconscious conflict** that lead to mental illnesses, dreams, slips of the tongue, etc.

- Unconscious motivation rejects the claim that you know what you're doing. Example: You may give honest reasons for marrying someone ("I love them, they're smart, I want kids"), but a Freudian might say you want to marry John because he reminds you of your father or because you want to get back at someone.

- If you protest, the Freudian replies these processes are unconscious, so of course you don't know what's happening.

- We already accept this unconscious operation in visual perception (we don't know how we see cars, trees, people) — Freud applies it to relationships and major life decisions.

- Everyday examples of unconscious motivation:

- Liking or disliking someone without knowing why.

- Arguing for something without being able to articulate the reasons.

- Forgetting someone's name at exactly the wrong time.

- Calling out the wrong name "in the throes of passion."

- The unconscious is not a rational computer looking out for your interests. Three distinct processes — id, ego, superego — are in violent internal conflict.
- **Id**: present at birth; animal part; wants pleasure (food, drink, elimination, warmth, sex); outrageously stupid; Pleasure Principle; "polymorphous perversity."
- **Ego** (self): emerges because pleasure isn't always available; copes by planning how to satisfy desires or to suppress them; Reality Principle; origin of consciousness.
- **Superego**: internalized rules of parents and society; conscience; "like a little angel above your head." Also stupid — it doesn't reason; it shames ("You should be ashamed of yourself. That's disgusting. Stop doing that.").
- The ego sits between two "screaming creatures" — id pushing for desire, superego shaming.
- Most of this is unconscious — like the workings of kidneys or stomachs; you cannot introspect and find them.

Chapter 3. Personality Development and Psychosexual Development

- Five stages of personality development, each tied to an erogenous zone. Getting stuck at a stage produces lasting personality effects.

| Stage | Focus / Issue | Fixation outcome |

|-----|-----|-----|

| Oral | Mouth (sucking, chewing); premature weaning is the danger | "Oral person" — overeats, chews gum, smokes; or dependent and needy |

| Anal | Toilet training | "Anal personality" — literally unwilling to part with feces; compulsive, clean, stingy |

| Phallic | Genitals; site of Oedipus Complex | Excessive masculinity (in either sex); in females, need for attention or domination |

| Latency | Post-Oedipal; sex is repressed | — |

| Genital | Healthy adult stage | — |

- **Oedipus Complex** narrative (for boys, per Freud):

1. Around age 3–4, in the phallic stage, the boy seeks an external object of pleasure.
2. He turns to his mother ("Mom is nice, I love Mom").
3. Father is in the way; the boy plans to kill the father. ("Every three- and four-year-old boy thinks this.")

4. Because children don't have a good sense of the boundary between mind and world, the boy assumes the father can tell, and that the father is angry.
5. The worst thing the father could do is castration; the boy fears this.
6. The boy concedes — "Dad wins" — and gives up sexual interest, entering **latency**.
 - Bloom's aside: as the father of two sons, both went through a phase where they explicitly said they wanted to marry Mommy.
 - **Genital stage**: healthy adult stage. But you are "not out of the woods" — the id/ego/superego dynamic continues. The superego represses not only bad actions but bad thoughts; the id sends up "weird, sick stuff" (sexual and violent desires), and the superego pushes back. Material is repressed and does not even reach consciousness.

Chapter 4. Defense Mechanisms, the Aims of Psychoanalysis, Dreams

- Freud held a "**hydraulic**" theory: repressed material slips out in dreams, in slips of the tongue, and in clinical symptoms.
- **Defense mechanisms** (normal, non-pathological):
 - **Sublimation**: redirecting sexual or aggressive energy — e.g., Picasso channeling sexual energy into artwork.
 - **Displacement**: refocusing shameful desires onto a more acceptable target — a boy bullied by his father may kick the dog and think he hates the dog.
 - **Projection**: attributing one's own impulses to others — Freud's classic example is a man who, ashamed of his homosexual desires, accuses other men of lusting after him. Freud suggested "perhaps not implausibly" that men obsessed with the sexuality of other men are projecting their own desires.
 - **Rationalization**: a parent who enjoys smacking his child says, "It's for the child's own good. I'm being a good parent."
 - **Regression**: returning to an earlier stage of development under stress — children cry, suck their thumb, seek a blanket.
- When these mechanisms fail, symptoms arise. Freud's era called this **hysteria**: hysterical blindness/deafness, paralysis, trembling, panic attacks, gaps of memory including amnesia.
- The therapeutic story (often dramatized in movies): a patient with blackouts or amnesia goes to an analyst, who provides an interpretation; the patient achieves insight and the symptom remits.
- Freud first used **hypnosis** to access repressed memories, then moved to **free association**. Patients **resist**; the analyst's job is to overcome resistance and help the patient achieve **insight**.

- Core claim of psychoanalysis: your problems reflect deeper hidden phenomena; once you see them, the problems will go away.
- Bloom shows a clip from *The Sopranos* (a character with panic attacks) — not a pure Freudian analysis (which would have the patient on a couch, not seeing a nondirective therapist), but it illustrates Freudian themes of dreams, repression, and hidden meaning.
- **Dreams** (Freud):
 - **Manifest content**: what is experienced.
 - **Latent content**: hidden implication.
 - All dreams are **wish fulfillment** — sometimes forbidden wishes.
 - Dreams have **symbolism** — the idea long predates Freud.
 - **Literature, fairy tales, stories**: Freud believed they carry universal themes — aspects of unconscious struggles and preoccupations.
 - **Religion**: Freud viewed the search for a singular, all-powerful God as seeking out a father figure that some of us never had during development.

Chapter 5. Question and Answer on Freud's Theories

- **Q (non-traditional family structures)**: Freud's stages assume a child raised by a mother and a father; what about a child raised by a single parent, or never breast-fed? Bloom: Freudians have had problems with this; Freud focused on the upper-class European families he interacted with. A Freudian would presumably expect such children to be psychologically damaged by failing to go through the normal psychosexual stages.
- **Q (women and superegos)**: Do modern psychoanalysts still believe women lack superegos? Freud was notorious for claiming women were morally immature relative to men. Bloom thinks Freud would say women have superegos, just not the sturdy ones men have. Modern psychoanalytic scholars are mixed — some maintain deep sex differences, others want to jettison this aspect.
- **Q (sublimation vs displacement)**: A student asks if sublimation is a subgroup of displacement. Bloom: displacement takes anger from one target to another (you → her); projection puts your desire onto someone else; sublimation "gives up the details and keeps the energy." Example: telling your roommate they're studying so much because they haven't had sex.
- **Q (cross-cultural evidence)**: Bloom defers to the scientific-assessment section.

Chapter 6. Controversies and Criticisms on Freud's Theories

- Bloom references *The Memory Wars* by **Frederick Crews**, "one of the strongest and most passionate critics of Freud."

- Two ways to reject a theory:

1. It is wrong. (Example: the theory that autism is caused by mothers not loving children enough — once popular, just wrong.)

2. It is so vague and all-encompassing it cannot be tested.

- The second is the main critique of Freud, captured by physicist **Wolfgang Pauli's** line about another physicist: "That guy's work is crap. He's not right. He's not even wrong."
- **Karl Popper** introduced **falsifiability**: scientific predictions must make strong claims that could be proven wrong.
- Examples of falsifiable psychological claims (to be revisited in the course):

- Damage to the hippocampus causes failures of certain sorts of memory.

- Everywhere in the world, men on average want more sexual partners than women.

- Exposure to violent television tends to make children more violent.

- **Astrology** as the model non-science: predictions are not wrong, they cannot be wrong. Bloom reads his web horoscope:

- "A couple of negative aspects could make you a little finicky for the next few days."

- "The presence of both Mars and Venus suggests you want to box everything into a neat, ordered, structured way but keeping a piece of jade or carnelian close will help you keep in touch with your fun side." (His wife gave him a piece of jade.)

- A funnier, falsifiable horoscope from *The Onion*: "Riding in a golf cart with snow cone in hand, you'll be tackled by two police officers this week after matching a composite caricature of a suspected murderer."

- Freudian theory often fails the falsifiability test because it's too flexible.

- The "self-validation" problem in psychoanalysis:

- Freud tells a patient, "You hate your mother." Patient agrees → Freud is right.

- Patient denies → Freud says: "Your anger shows this idea is painful to you. You have repressed it from consciousness. I am right."

- The same dynamic plays out in scientific debate: scientists say there is little evidence for claims like "adult personality traits are shaped by psychosexual development," "all dreams are disguised wish fulfillment," "psychoanalysis is the best treatment for mental disorders." Freudian response: "Your rejection of my ideas shows they are distressing to you. This is because I am right." Often followed by: "You have deep psychological problems."
- Where Freudian-aligned researchers have made falsifiable predictions, results are bad:

- No evidence that oral/anal characteristics (needy vs. stingy) relate in any interesting way to weaning or toilet training.
- Cross-cultural studies of toilet training and weaning differences show no good evidence of corresponding personality differences.
- Freud's strong claims about sexuality (why people are straight or gay) have met with very little empirical support.
- The claim that psychoanalysis is uniquely effective at curing mental illness is "almost certainly not true" — quicker and more reliable treatments exist for most disorders. There is considerable controversy about whether the "Tony Soprano method" of insight makes any real difference for symptoms like anxiety disorders or depression.
 - As a result, **Freudian psychoanalysis is almost never studied inside psychology departments** — neither cognitive, developmental, nor clinical sides — with rare exceptions, and even there usually critically. At Yale, Freud is more likely to be found in the history or literature department than in psychology.
 - Despite all this, **the big idea — the importance of the dynamic unconscious — remains intact.**

Chapter 7. Examples of the Unconscious in Modern Psychology

- **Language understanding:**
 - "John thinks that Bill likes him" → understood in a fraction of a second to mean John thinks Bill likes John.
 - "John thinks that Bill likes himself" → understood to mean John thinks Bill likes Bill.
 - This is not conscious; you don't know how you do it or that you are doing it.
- **Automatic action:**
 - Driving, chewing gum, tying shoelaces — when you are expert enough, you don't know you're doing them.
 - Anecdote: a friend hosting a party ran out of food and went to pick some up "around the corner"; an hour later they called his cell phone — he had gotten on the highway and driven to work (an hour away).
- **Social psychology — likes and dislikes due to factors we are unaware of:**
 - Harsh initiation into a club makes people like the club *more*. Hazing is illegal but "a remarkably successful tool."
 - The more you pay for something and the more pain you go through to get it, the more you like it.

- Political campaign implication: "If you want loyal people in a political campaign, do not pay them. If you pay them, they'll like you less. If they volunteer, they'll like you more."

- **Terror management — subliminal death primes:**

- Participants sit before a computer screen and answer questions like, "How much do you love your country?" "What do you think of Asians/Jews/blacks/vegetarians/people with different political views?"

- Words like "corpse," "dead," "dying" are flashed too fast to consciously see (**subliminal**).

- Effect: people become more nationalistic, more patriotic, less forgiving of others, less liking of other races and people from other countries.

- **In-class demonstration — Norbert Schwarz's research:**

- Bloom splits the class in half. Each half is asked to think of someone they love.

- One half lists **three** positive features of that person; the other half lists **ten**.

- Result (not subtle): liking goes *up* in the three group and *down* in the ten group.

- Mechanism: three features is easy, so the person seems great; ten is hard because "nobody has ten positive features" — the difficulty leads subjects to infer they don't like the person very much.

- Subjects do not know this is why: asked why they rated the person lower, they don't say "because you told me to list ten."

- This is "a substantial and an important part of the study of psychology, and particularly... the study of racial and sexual prejudice." People have strong views about other races that they don't know about and can't control.
- Closing point: the particulars of Freud have largely been rejected, but the general idea of the dynamic unconscious has been "so successful" that Freud has become "a victim of his own success" — we underestimate his importance because his worldview is now hard to think outside of.

Chapter 8. Further Question and Answer on Freud

- **Q (medication):** Did Freud believe in medication? Freud began as a neuroscientist, intensely interested in the neural basis of thought and behavior, but the answer is **no** — he was convinced disorders like depression and anxiety must be cured by talk therapy and insight, not drugs. Modern therapists, including some who are not psychoanalytically defined, say drugs "mask the symptoms" — panic attacks "are actually not your real problem" — so making them go away with drugs doesn't get to the root. Both Freud and modern psychoanalysts think medications are substantially overused.

- **Q (dreams research):** Bloom defers — will devote half a class to sleep and dreams. Some evidence that dreams relate to what you're thinking about and worrying about during the day, but the strong Freudian view of symbolism and wish fulfillment "has not been supported by the study of dreams."
- **Q (Electra complex):** It is the penis envy story. Bloom: "Freud developed the Oedipal complex... and then it's as if somebody reminded him, 'Sigmund, there are also women.'" The Electra complex was an "add-on" to the main interest of the Oedipal complex.
- **Q (does penis envy fixate like the oral or anal stage?):** Not a fixation in the same sense, but the woman's discovery that she lacks a penis "plays a fundamental role later on determining her allegiances in life and in fact her own sexual preferences and interests." Not just a short-period effect.

Notable Quotes

"Don't give the prize to Freud. He doesn't deserve a Nobel Prize. He's just a psychologist." — Albert Einstein (as Bloom reports the letter)

"The Gestapo has not harmed me in any way. In fact, I highly recommend the Gestapo to everybody." — Freud (added to the Gestapo's letter)

"The id, according to Freud, is present at birth. It's the animal part of the self. It wants to eat, drink, pee, poop, get warm, and have sexual satisfaction. It is outrageously stupid."

"In between these two screaming creatures, one of you; one of them telling you to seek out your desires, the other one telling you, 'you should be ashamed of yourself,' is you, is the ego."

"That guy's work is crap. He's not right. He's not even wrong." — Wolfgang Pauli (used to summarize the critique of Freud)

"The problem with astrological predictions is not that they're wrong. It's that they can't be wrong. They're not even wrong."

"Your rejection of my ideas shows that they are distressing to you. This is because I am right." — Bloom's caricature of the Freudian response to scientific criticism

"If you want loyal people in a political campaign, do not pay them. If you pay them, they'll like you less. If they volunteer, they'll like you more."

"Freud's been a victim of his own success. We tend to underestimate the importance of Freudian thought in everyday life because he's transformed our world view to such an extent that it's difficult for us to remember if there's any other way to think about it."

Assigned Readings / References

- **Frederick Crews**, *The Memory Wars* — referenced as "an excellent book" and as the work of "one of the strongest and most passionate critics of Freud."
- **Karl Popper** — concept of falsifiability.
- **Wolfgang Pauli** — quoted for the "not even wrong" formulation.
- **Norbert Schwarz** — research on the three-versus-ten positive features demonstration.
- *The Sopranos* — television episode shown in class to illustrate Freudian themes (dreams, repression, hidden meaning).
- *The Onion* — source of the parody horoscope used to illustrate falsifiability.

PSYC 110 - Lecture 4 - Foundations: Skinner

Overview

This lecture, delivered by Professor Paul Bloom, completes loose ends from the Freud lecture and then introduces B. F. Skinner and behaviorism. Bloom first revisits the question of why an unconscious would evolve, offering a deception-based hypothesis: the best lies are ones we tell ourselves. He then turns to behaviorism, presenting it as a movement championed by figures like John Watson and later packaged and publicized by Skinner, organized around three radical claims — strong emphasis on learning, anti-mentalism, and no interesting differences across species. The bulk of the lecture walks through the three learning principles behaviorists used to explain all of mental life: habituation, classical conditioning, and operant (instrumental) conditioning. Bloom closes with the criticisms — innate knowledge, the legitimacy of mental notions, animal-specific constraints (e.g., the Garcia effect), and Chomsky's argument that Skinner's terms are too vague to be scientific — concluding that while behaviorism faded, its mechanisms and training techniques remain an important legacy.

Key Concepts & Definitions

- **Deception (broad sense):** "to act or be in some way that fools others into believing or thinking or responding to something that's false."
- **Behaviorism:** A school of thought, predating Skinner, championed by psychologists like John Watson, built on three radical views (strong emphasis on learning, anti-mentalism, no interesting cross-species differences).
- **Habituation:** "a decline in the tendency to respond to stimuli that are familiar due to repeated exposure." The very simplest form of learning.
- **Classical conditioning:** "the learning of an association between one stimulus and another stimulus," where stimulus is "a technical term meaning events in the environment like a certain smell or sound or sight." Thought up by Pavlov.
- **Unconditioned stimulus (UCS):** Stimulus that gives rise to an unconditioned response without learning (e.g., poking with a stick; food powder in the dog's mouth).
- **Unconditioned response:** The natural, unlearned response to the unconditioned stimulus (e.g., saying "ouch"; salivation).
- **Conditioned stimulus (CS):** A previously neutral stimulus that, through pairing with the UCS, comes to elicit a response (e.g., the bell).

- **Conditioned response:** The learned response to the conditioned stimulus.
- **Reinforced trial:** When the CS and UCS go together — "you're teaching the dog that the bell goes with the food."
- **Unreinforced trial:** When the CS occurs without the UCS — "you get the food without the bell."
- **Extinction:** When teaching stops, the response goes away.
- **Spontaneous recovery:** After extinction, if you wait a while and try the CS again after a couple of hours, the response (e.g., saliva) comes back.
- **Stimulus generalization:** A conditioned response extends to similar stimuli (e.g., Little Albert's fear extending from a rat to a white rabbit).
- **Systematic desensitization:** A cure for phobias that exposes you to what causes fear while relaxing you at the same time, so you replace the aversive classically conditioned fear with something positive.
- **Operant (instrumental) conditioning:** "learning the relationships between what you do and how successful or unsuccessful they are, learning what works and what doesn't." Voluntary; unlike classical conditioning, the organism chooses to act.
- **Law of effect:** "a tendency to perform – an action's increased if rewarded, weakened if it's not."
- **Positive reinforcement:** You do something (give a reward) to increase a behavior.
- **Negative reinforcement:** A type of reward in which you "take away something aversive" (e.g., remove a heavy collar). Different from punishment.
- **Shaping:** Working an animal or child gradually toward a complex behavior by rewarding successive approximations.
- **Primary reinforcers:** Things organisms naturally like (food) or naturally don't like (being hit, shocked).
- **Token economies:** Trained systems where "seemingly arbitrary rewards like poker chips" function as reinforcers via classical conditioning.
- **Schedules of reinforcement:** Fixed vs. variable, ratio vs. interval. Ratio = reward every certain number of times; interval = over a period of time.
- **Partial reinforcement effect:** "Behaviors last longer if they're reinforced intermittently."
- **Garcia effect:** Conditioned taste aversion — a connection between a new food's taste and getting sick is "natural" and "hard wired," but is special to taste and nausea; the same association does not form with painful shock.

Detailed Notes

Chapter 1. A Brief Review on the Unconscious

- Bloom returns to Freud to address material skipped on Wednesday — specifically why an unconscious would exist.
 - He notes that the right question may not be "Why are some things unconscious?" but rather "why is this tiny subset of mental life conscious?" because "by far the vast majority of what our brains do, the vast majority of what our minds do, is unconscious and we're unaware of it."
 - **Evolutionary hypothesis: deception.** Most animals deceive.
- Physical examples:
- Chimpanzees: when threatened, "their hair stands up on end and that makes them look bigger to fool others to thinking they're more dangerous than they are."
 - Angler fish at the bottom of the ocean has "a rod sticking up from the top of its head with a lure to capture other fish."
 - Humans, especially, use minds and behavior continually to trick others into thinking we're "tougher, smarter, sexier, more reliable, more trustworthy."
 - Social psychology in part concerns self-presentation to make "the maximally positive impression even when that impression isn't true."
- We've also evolved good lie-detection mechanisms — so there's evolutionary pressure both to lie and to detect lies.
 - **Key hypothesis:** "the best lies are lies we tell ourselves. You're a better liar, more generally, if you believe the lie that you're telling."
 - **Hitchcock anecdote:** Hitchcock hated working with child actors. A child actor couldn't cry, so Hitchcock leaned over and whispered, "Your parents have left you and they're never coming back." The kid burst into tears, Hitchcock said "Roll 'em," and the child looked genuinely sad because he was.
 - Other illustration: if there was a \$100,000 prize for looking most in pain, jamming a pen into your groin works because you really will be in pain.
 - Implication: "certain motivations and goals, particularly sinister ones, are better made to be unconscious because if a person doesn't know they have them they will not give them away."
 - **Anecdote on falsification of Freud:** Bloom's young son was asked who he would marry — instead of naming his mother (Oedipal prediction), he answered, "I would marry a donkey and a big bag of peanuts." Both parents are psychologists, "he hates these questions and at times he just screws around with us."

Chapter 2. B. F. Skinner and Behaviorism

- Skinner vs. Freud:

- Freud's psychoanalysis is "as close as you could find in science to a solitary invention."
- Behaviorism predates Skinner and was championed by figures like **John Watson**.
- Skinner came later but packaged, expanded, publicized, and developed behaviorism — both scientifically and popularly — through the 1960s and 1970s. He appeared on talk shows; his books were bestsellers.

- **Three radical views at the core of behaviorism:**

1. **Strong emphasis on learning:** "everything you know, everything you are, is the result of experience. There's no real human nature." People are infinitely malleable.

- Watson's famous boast, paraphrasing the Jesuits ("Give me a child until the age of seven and I'll show you the man"):

> Give me a dozen healthy infants, well-formed and my own specified world to bring them up and I'll guarantee to take any one at random and train them to become any type of specialist I might select — doctor, lawyer, artist, merchant, chief, and yes, even beggar-man and thief, regardless of his talents, penchants, tendencies, abilities, vocations and race of his ancestors.

- This view has "egalitarian" appeal: if there's no human nature, no group can be superior by race or sex.

2. **Anti-mentalism:** A reaction to Freud — behaviorists felt that "claims about internal mental states like desires, wishes, goals, emotions and so on, are unscientific." The manifesto: develop a science using only **stimulus, response, reinforcement, punishment, environment** — real-world tangible events.

3. **No interesting differences across species:** A behaviorist might say differences are "just general associative powers that differ," or even that "humans and rats aren't different at all. It's just humans tend to live in a richer environment than rats." Methodologically: study human learning by studying nonhumans.

Chapter 3. Habituation: The Very Simplest Form of Learning

- Definition: "a decline in the tendency to respond to stimuli that are familiar due to repeated exposure."
- Demonstrated by Bloom shouting "Hey!" — second time startles less.
- Common examples: getting used to a ticking clock, traffic noise.
- Imagine life without habituation: you'd be startled every time someone waved their hand.

- It's "critically important to get used to things because it's a useful adaptive mechanism to keep track on new events and objects." New things may harm you, so you must notice them; but once they show themselves to be part of the environment, you stop noticing.
- Counts as learning because it happens through experience and changes how you think.
- **Use in developmental psychology:** habituation has been used by clever developmental psychologists to study creatures who can't talk — nonhuman animals and young babies (to be discussed Wednesday).

Chapter 4. Classical Conditioning: Associating Stimulus

- Definition: learning of an association between one stimulus and another. Thought up by **Pavlov**.
- **Origin (scientific serendipity):** Pavlov originally had no interest in learning — he was interested in saliva. He attached apparatus to dogs and put food powder in their mouths to make them salivate. He noticed that when a person who typically gave the food entered the room, salivation would start. Pairing a bell with food meant the bell came to cause salivation.
- **Distinction between two kinds of stimulus-response relationships:**
 - **Unconditioned stimulus** → **unconditioned response** — innate, unlearned. Examples: poke with a stick → "ouch"; food powder → saliva.
 - **Conditioned stimulus** → **conditioned response** — learned via repeated pairing. The bell starts as a neutral stimulus; paired with food repeatedly, it eventually generates saliva on its own.
- **Reinforced vs. unreinforced trials:**
 - Reinforced = CS + UCS together (teaching the dog).
 - Unreinforced = UCS without CS (not teaching).
- **Extinction:** stop the teaching → the response goes away. Pattern: "bell, food, bell, food, bell [pant]," then just "bell, bell, bell" → give up.
- **Spontaneous recovery:** After extinction, waiting a couple of hours and trying again — the saliva comes back.
- **Stimulus generalization — Little Albert (John Watson):**
 - In *The Norton Reader*, Watson reported a bizarre experiment with a baby named Little Albert.
 - Little Albert originally liked rats (shown via video).
 - Watson banged a metal bar behind the baby while Albert played with the rat. Albert screamed/sobbed.
 - **Unconditioned stimulus:** loud noise (bar bang). **Unconditioned response:** crying/sadness/misery.
 - **Conditioned stimulus:** the rat. **Conditioned response:** fear.

- The fear generalized to other things, including a white rabbit.
- Bloom: "This is one of a long list of experiments that we can't do anymore."
 - **Generality:** Classical conditioning has been replicated in many species — crabs, fish, cockroaches — and applied to human responses.
 - **Applications to human responses:**
 - **Fear / phobias:** Plausible story: a dog (neutral) bites you (UCS → UCR pain/fear) and becomes a CS → CR fear of dogs.
 - Treatment by extinction: present the CS without the UCS.
 - Cartoon joke: a man trapped in a small booth filled with snakes, dangling from a building — simultaneously confronting heights, snakes, and the dark. The logic isn't bad (nothing bad happens), but the panic attack often makes the fear worse.
 - **Systematic desensitization:** expose person to the feared stimulus while relaxed. Traditionally via relaxation exercises; "nowadays they just pump you full of some drug to get you really happy," replacing aversive conditioning with positive.
 - **Hunger:** Environmental cues associated with eating, smoking, or drinking can generate desire — "those of you who are trying to quit smoking... certain activities... really make you want to smoke."
 - **Sexual desire / fetishes:** Speculative behaviorist story — "your lover's caress brings you to orgasm, your eyes happen to fall upon a shoe," and the shoe becomes a CS → sexual pleasure. "This almost certainly is not the right story" but may play a role.
 - Some treatment for pedophiles and rapists has involved "controlled fantasies during masturbation to shift the association from domination and violence... to develop more positive associations with sexual pleasure."
- **What is classical conditioning for?**
 - Traditional view: "It's just association" — CS and UCS happen together, so the response transfers. Predicts that strongest learning is when CS and UCS are simultaneous, and CR = UCR.
 - **Current mainstream view: preparation.** "you become sensitive to a cue that an event is about to happen and that allows you to prepare for the event."
 - Predictions: best timing is CS *before* UCS. CR may differ from UCR.
 - Example: A child beaten by his father flinches when the father raises his hand. Flinching is *preparation* for being hit, not the same response as being hit.
- **Final exercise — A Clockwork Orange:**
 - A violent murderer/rapist is given a drug that makes him violently ill, eyes propped open, shown scenes of violence. Afterward, encountering real-world violence triggers nausea.

- UCS = drug; UCR = nausea; CS = violence; CR = nausea.

Chapter 5. Operant Conditioning: Operating on the Environment

- Also called **instrumental conditioning** — Skinner's signature theory.
- Definition: "learning the relationships between what you do and how successful or unsuccessful they are, learning what works and what doesn't."
- Key contrast with classical conditioning: classical can occur even if you're strapped down and immobile; operant requires voluntary choices.
- **Thorndike's puzzle box:**
 - A cat must pull a special lever to escape.
 - Cats "do not solve this problem through insight." They bounce around scratching bars, pushing the ceiling, digging the floor, howling, etc.
 - Pressing the lever happens to get them out. Over trials, they stop other behaviors and just press the lever. Gradual improvement.
- **Analogy with Darwin:** Behaviors are like random variation; reinforced ones survive, others go extinct. Skinner "explicitly made the analogy from the natural selection of species to the natural selection of behavior."
- **Law of effect:** "a tendency to perform – an action's increased if rewarded, weakened if it's not."
- **Example — training a pig:**
 - Reinforce walking forward; punish walking backward → over time, pig walks forward.
- **Positive vs. negative reinforcement:**
 - Both increase behavior (both are rewards).
 - **Positive:** you give something.
 - **Negative:** you take away something aversive (e.g., remove a heavy collar).
 - Negative reinforcement is **not** punishment.
- **Shaping** (for complex behaviors):
 - You can't just wait for the pig to dance. Reward closer and closer approximations: stumbling → more dance-like motion → dancing.
 - Child-as-beer-fetcher example: reward the child for walking to the kitchen, then for opening the fridge, then for retrieving and opening the beer.
- **Skinner's examples:**
 - Skinner developed a **pigeon-guided missile** in WWII (never used).

- Military history includes attempts to train pigeons and dolphins for deadly tasks.
- Skinner taught animals to play **Ping-Pong** by rewarding approximations.
- **Primary reinforcers vs. learned reinforcers:**
 - Primary: food (positive), being hit/shocked (negative — aversive).
 - Learned via classical conditioning: e.g., saying "Good dog" — not innately pleasurable, but pair it with a treat and the phrase acquires reward value.
 - **Token economies:** Use seemingly arbitrary rewards (poker chips) — increases utility and ease of training.
 - **Schedules of reinforcement:** Fixed vs. variable, ratio vs. interval.
 - **Ratio:** reward after a certain number of behaviors (e.g., every tenth newspaper retrieval).
 - **Interval:** reward after a period of time (e.g., dog dances for an hour straight).
 - **Fixed:** rigid schedule (every 5th time).
 - **Variable:** changing (sometimes 3rd, sometimes 7th).
 - **Slot machine** = variable ratio (depends on number of pulls, but the number is unpredictable).
 - **Partial reinforcement effect:**
 - "Behaviors last longer if they're reinforced intermittently."
 - Slot-machine analogy: continuous reward then sudden stop → you give up quickly. Intermittent reward → you persist much longer.
 - Parenting example: telling a kid "No, no, no, no... Okay" inadvertently exploits the partial reinforcement effect — "the evolution of nagging."
 - **Extension to humans / society (Skinner):**
 - Skinner argued the prison system should be reformed — focus on reinforcing good and punishing bad behavior instead of justice/retribution.
 - Argued operant conditioning should be extended to everyday life; people's lives would be fuller "if they were controlled in a properly behaviorist way."

Chapter 6. Question and Answer on Behaviorism

- **Q (student):** Wouldn't extinction occur for things like poker chips?

- **Bloom:** Yes — the connection with poker chips is established via classical conditioning, so eventually they lose reinforcing power. You'd need to retrain. Similarly, "Good dog" without occasional treats eventually won't cut it.

• **Q (student):** [inaudible — about Skinnerian involvement in prisons]

- **Bloom:** Skinner and Skinnerian psychologists were never directly involved in creating prisons. But behaviorism shaped criminal justice thinking. Some mental institutions and prisons have installed token economies (poker-chip style rewards for good behavior) influenced by behaviorist principles.

Chapter 7. Controversies and Criticisms on Behaviorism

• **Three behaviorist claims, all now believed mistaken:**

1. No innate knowledge; everything is learned.
2. You can explain psychology without mental notions like desires/goals.
3. Mechanisms apply across all domains and species.

• **(1) Innate knowledge:** Considerable evidence exists for innate knowledge and innate desires — discussed later via language learning, sexual preference development, understanding of material objects. "There's nobody who doubts nowadays that a considerable amount for humans and other animals is built-in."

• **(2) Mental states / unobservables:** Mature sciences like physics and chemistry are full of unobservables. "It makes sense to explain complex and intelligent behavior in terms of internal mechanisms and internal representations." The computer revolution illustrates this — you can't explain a chess-playing computer without talking about its programs and mechanisms.

• **(3) Cross-domain / cross-species generality fails in multiple ways:**

- **Tolman's maze study:** Rats learned to run a maze faster whether rewarded or not. Reinforcement helps but is not necessary. Tolman's work suggests rats have **mental maps** — internal mechanisms — contrary to behaviorist doctrine. "A huge embarrassment for the Skinnerian theory."

- **Species-specific constraints on responses:**

- You can train a pigeon to peck for food (natural response) but it's "very difficult to train it to peck to escape a situation."

- You can train it to flap its wings to escape (natural) but it's very difficult to train wing-flapping for food.

- Animals have natural response repertoires that learning channels and exploits.

- **The Garcia effect** (discussed in the Gray textbook):

- Food aversions in humans and other animals can form through association — getting nauseous after eating a new food (even if the food wasn't the cause) generates strong aversion.

- Bloom's anecdote: as a teenager, he drank the Greek liqueur **ouzo** with beer, became violently ill, and now cannot abide the smell of ouzo. (Thankfully no aversion to beer developed.)

- **The aversion is special to taste-nausea pairings.** New food + nausea-inducing drug → aversion. New food + painful shock → no aversion. We're hard-wired to connect tastes with sickness, but the effect doesn't generalize.

- **Phobias and evolution:** The claim that phobias form via classical conditioning "is almost always wrong." Some phobias are evolved — humans and chimpanzees are particularly prone to fear snakes. The phobias you're likely to have depend more on evolutionary than personal history.

- **Chomsky's *Review of Verbal Behavior*:**

- Other assigned reading. Chomsky is "one of the most prominent intellectuals alive" — still a professor at MIT.

- This excerpt is "one of the most influential intellectual documents ever written in psychology" — more than anything else, it ended behaviorism's dominance.

- **Chomsky's core argument:** When applied to humans, Skinner's notions of reward and punishment are so vague they're no longer science. This raises the issue of **unfalsifiability** (paralleling Freud).

- Example: Skinner's *Verbal Behavior* tries to explain why we talk to ourselves, imitate sounds, create art, give bad news to an enemy, fantasize — all as "reinforced behaviors." But Skinner doesn't mean food pellets or pats on the head; he means things like "self-reinforcing" or "reinforcing because it makes your enemy feel bad."

- Chomsky: This is not wrong — it's "just so vague as to be useless." Saying "giving bad news to an enemy is reinforcing because it makes the enemy feel bad" says nothing different from "giving bad news to an enemy feels good because we like to give bad news to an enemy."

- Chomsky's dilemma: The law of effect, applied to humans, is **either trivially true or scientifically robust and obviously false.**

- Example: "Why did you come to Yale for a second semester?" "Because the first semester was rewarding." That's true, but you don't mean pellets — you just mean you chose to come back. "There's nothing wrong with saying that but we shouldn't confuse it with science."

- **Legacy of behaviorism:**

- Behaviorism has faded as a dominant intellectual field but remains a major 20th-century contribution.

- It produced a "richer understanding of certain learning mechanisms" — habituation, classical conditioning, operant conditioning — real, scientifically studiable, important in animal life and probably human life. "They just don't explain everything."
- Behaviorists provided **powerful training techniques**, especially for nonverbal creatures — useful to animal trainers, teachers of young children/babies, and clinicians working with severely autistic or severely retarded populations.

Notable Quotes

"By far the vast majority of what our brains do, the vast majority of what our minds do, is unconscious and we're unaware of it. So the right question to ask may not be, 'Why are some things unconscious?' but rather, why is this tiny subset of mental life — why is this conscious?"

"The best lies are lies we tell ourselves. You're a better liar, more generally, if you believe the lie that you're telling."

"Give me a dozen healthy infants, well-formed and my own specified world to bring them up and I'll guarantee to take any one at random and train them to become any type of specialist I might select — doctor, lawyer, artist, merchant, chief, and yes, even beggar-man and thief, regardless of his talents, penchants, tendencies, abilities, vocations and race of his ancestors." — John Watson

"This is one of a long list of experiments that we can't do anymore."

"Behaviors last longer if they're reinforced intermittently and this is known as 'the partial reinforcement effect.'"

"Skinner explicitly made the analogy from the natural selection of species to the natural selection of behavior."

"There's nothing wrong with saying that but we shouldn't confuse it with science."

"Behaviorism as a dominant intellectual field has faded, but it still leaves behind an important legacy and it still stands as one of the major contributions of twentieth century psychology."

Assigned Readings / References

- John Watson's article in *The Norton Reader* (covering the **Little Albert** experiment and stimulus generalization).
- Gray textbook discussion of the **Garcia effect**.
- Noam Chomsky, *Review of Verbal Behavior* (excerpt) — described as one of the most influential intellectual documents ever written in psychology, credited more than any single event with ending behaviorism's dominance.
- Skinner, *Verbal Behavior* (referenced via Chomsky's critique).

PSYC 110 - Lecture 5 - What Is It Like to Be a Baby: The Development of Thought

Overview

This lecture, delivered by Professor Paul Bloom, surveys the development of thought from infancy through childhood. It begins with Jean Piaget's foundational stage theory and its mechanisms (assimilation and accommodation), then moves to the modern science of infant cognition, which uses clever non-verbal methods (looking-time, habituation, surprise, sucking) to test what babies actually know. Bloom argues that, contrary to Piaget's claims, babies appear to possess significant knowledge of the physical world (e.g., object permanence, rudimentary arithmetic) and possibly the social world (helping/hindering, mimicry) from very early on. The lecture also explores explanations of developmental change — neural maturation, inhibition problems, and learning — and considers the modular view of cognition (Chomsky, Fodor), illustrated by autism as a disorder selectively affecting social understanding. It closes with the broader idea that studying development can illuminate questions about adult cognition.

Key Concepts & Definitions

- **Genetic epistemology:** Piaget's discipline — the origins of knowledge; he studied individual children to learn about the development of knowledge more generally.
- **"Ontogeny recapitulates phylogeny":** The idea that the development of an individual mimics or repeats the development of the species. Bloom notes it is "entirely not true, but it's a beautiful phrase," and Piaget was committed to it.
- **Assimilation:** "The act of expanding the range of things that you respond to." Example: a baby used to sucking on a breast may come to suck on a bottle or a rattle.
- **Accommodation:** "Changing how you do it." Example: a baby forms his mouth differently depending on what he is sucking on. More psychologically, changing one's system of knowledge itself.
- **Sensorimotor stage / period:** Piaget's first stage. The child is "purely a physical creature" with no real understanding of the external world — no past, no future, no stability, no differentiation. The child touches and sees but doesn't yet reason. Object permanence is gradually acquired during this stage.
- **Object permanence:** The understanding that things exist when you no longer see them. Piaget claimed children before six months lack it — when an object is hidden, "it's like it's gone."

- **A-not-B task** (Peter Gray's "changing hiding places" problem): An object is repeatedly hidden at location A (a cup) and the nine-month-old reaches for it. When moved to a new location B in plain view, the child still reaches at A.
- **Preoperational stage**: The capacity to represent the world internally emerges, but with striking limits — egocentrism and failure of conservation.
- **Egocentrism** (Piaget's technical sense, not "selfish"): Children "literally can't understand that others can see the world differently from them."
- **Three mountains task**: Demonstration of egocentrism. A child can draw three mountains from his own viewpoint but finds it "extraordinarily difficult" to draw them as they appear from the other side.
- **Conservation**: The notion that "there's ways to transform things such that some aspects of them change but others remain the same" (e.g., pouring water into a different-shaped glass, spreading out pennies). Preoperational children fail to recognize this.
- **Concrete operations (age 7–12)**: Children can solve conservation but are still limited in abstract reasoning (e.g., mathematical infinity, logical entailment are beyond them).
- **Abstract / scientific reasoning stage** (around age 12): Capacity for abstract and scientific reasoning emerges.
- **Habituation**: "A fancy word for boredom." Showing a baby something repeatedly until the baby stops attending; then a new stimulus is presented to see if the baby perks up.
- **Surprise method**: Showing babies something that shouldn't happen; longer looking is taken as evidence the baby also finds it surprising. "Scientists do magic tricks to explore this very thing."
- **Neural maturation**: Most neurons exist already in utero; development largely involves pruning excess neurons, growth of synaptic connections (peaking around age 2), and myelination (continuing through teenage years, especially in frontal lobes).
- **Inhibition**: Capacity to stop a prepotent response; tied to the frontal lobes, not fully active in babies. Offered as an alternative explanation for A-not-B errors.
- **Modules** (Chomsky, Jerry Fodor): Separate pre-wired systems for reasoning, each with built-in knowledge and its own learning pattern. Opposed to a single across-the-board developmental story.
- **Autism**: A disorder striking about one in a thousand people, mostly boys; characterized by lack of social connectedness, problems with language, problems dealing with people — what Simon Baron-Cohen calls "mind blindness." Autistic individuals show no impairments on the physical world, math, or spatial skills.
- **False-belief task**: A test of understanding others' minds. A doll named Maxie hides a ball; a second doll moves it; the child is asked where Maxie will look. The correct answer is the original

(cupboard) location because Maxie has a false belief. Three-year-olds fail; four- and five-year-olds pass; autistic individuals often fail even when otherwise high-functioning.

- **"Extreme male brain" theory** (Baron-Cohen): Social abilities are "more sequestered" in females; autistic individuals have extreme male brains. "To be a man is to suffer from a very mild form of autism."
- **Autistic savants:** A very small minority of autistic individuals who have extraordinary mathematical, artistic, or musical abilities (e.g., the Rain Man character).

Detailed Notes

Chapter 1. Jean Piaget, Stage Theory and Its Limits

- Piaget's influence shapes most current methods, theories, and ideas in cognitive development.
- Piaget viewed children as "active thinkers... little scientists" who construct schemas — "miniature theories of the world."
- His ambition was not children per se but **genetic epistemology** — the origins of knowledge. He believed studying the individual child would reveal the development of knowledge more generally.
- He endorsed the phrase "**Ontogeny recapitulates phylogeny**" — development of an individual mimics development of the species. Bloom: "It's entirely not true, but it's a beautiful phrase."

Two mechanisms of learning:

- **Assimilation** — expanding the range of things you respond to (baby sucks on breast → bottle → rattle).
- **Accommodation** — changing how you respond (baby shapes mouth differently for different objects). Psychologically: changing your system of knowledge.

Method: ask children to solve problems and ask them questions. Different answers at different ages led to the Stage Theory.

Stages:

Stage	Approx. Age	Key Features
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Sensorimotor	Before ~2	Purely physical; no past/future/stability; gradually acquires object permanence
Preoperational	~2 to 7	Internal representation present but limited by egocentrism and failure of conservation

| Concrete operations | 7 to 12 | Solves conservation, but limited in abstract reasoning (infinity, logical entailment) |

| Abstract / scientific | ~12+ | Abstract and scientific reasoning |

Object permanence demonstrations:

- Adult example: ducking behind the lectern doesn't make the audience think he's gone.
- Before six months, hide a rattle and "it's like it's gone." Babies surprised by peek-a-boo because they think you actually disappear.
- **A-not-B task** ("changing hiding places," Peter Gray's name): Nine-month-old reaches at A repeatedly; when the object is moved to B in plain view, the child still reaches at A. Piaget took this as evidence babies don't fully understand objects.

Preoperational limits:

- **Egocentrism** — Three mountains task: child can draw the mountains from his own side but fails when asked to draw them from the other side.
- **Conservation failures:**
 - Two equal rows of checkers; one is spread out — child says the spread row has more.
 - Pouring water into a taller/shallower glass — child says the amount changed.
 - Bloom: "Any of you who have access to a four- or five-year-old... Do not take one without permission... you can do this yourself."

Why Piaget fared better than Freud or Skinner:

- Interesting and falsifiable claims.
- Rich theoretical framework with many books and articles.
- "Really striking findings" — before Piaget, no one noticed conservation failures or babies' object-tracking problems.

Limitations:

- *Theoretical*: Does he really explain how a child moves from concrete to abstract thought, or from lacking to having object permanence?
- *Methodological*: Heavy reliance on question and answer; children aren't good with language, especially when young, which may underestimate what they know.
- *Factual*: Methodological limits may have led Piaget to systematically underestimate infant/child knowledge.

Aside on methodology: "Often 90% of the game is discovering a clever method through which to test your hypotheses." Bloom cites Henry Gleitman at University of Pennsylvania, who built a tickle

machine — "this box with these two giant hands that went 'r-r-r-r.'" It failed because people convulsed in laughter before reaching it.

Chapter 2. The Modern Science of Infant Cognition

- Long-standing consensus (captured by an Onion headline): "babies are stupid... don't know much about the world."
- Modern research suggests "babies might be smarter than you think" — but discovering this requires clever methods.

Methods (babies can't talk or move skillfully):

- **Brain waves** (with EEG-style electrodes — Bloom jokes about a child "crushed by the weights of the electrodes").
- **Pacifier sucking**: machines deliver music or language contingent on sucking; sucking rate measures preference.
- **Looking times** (most influential): pioneered by **Elizabeth Spelke**, a developmental psychologist.

Two looking-time paradigms:

- **Preference**: Place two images side by side (e.g., dog vs. cat; pretty vs. ugly face) and see which the baby looks at more.
- **Habituation and surprise**: Show the same stimulus repeatedly until the baby is bored, then show a different one. Longer looking = perceived difference. Used, e.g., for color vision: green, green, green, then red.
- **Surprise**: Show something that shouldn't happen; longer looking suggests the baby finds it surprising. "Scientists do magic tricks to explore this very thing."

Studies revisiting object permanence:

- **Spelke and Baillargeon — bar/block study**: Babies see a bar moving behind a block. Adults assume a complete bar in the middle. When the screen drops, babies look longer at a broken bar than a complete one — i.e., they expected a whole bar.
- **Rene Baillargeon — screen and block**: A block on a stage is occluded by a rising screen. In one condition the screen stops where the block should be; in the other (via a trap door) the screen rotates through. Babies look longer at the impossible event.
- **Karen Wynn's lab at Yale — addition/subtraction**: With real objects and animated versions. Babies expect $2 - 1 = 1$; when shown $2 - 1 = 2, 3$, or 0 , they look longer. Even six-month-olds are sensitive to "rudimentary facts of arithmetic" and to object continuity out of sight.

Things babies don't know:

- Show a block on a surface and another block floating in midair. Six-month-olds find floating in midair surprising (violates gravity).
- A block stuck protruding off the edge: six-month-olds aren't surprised; even 12-month-olds aren't surprised; adults recognize it as an unstable configuration that should fall.
- "Although some things are built in, some things develop."

Explanations of development:

1. **Neural maturation:** Most neurons exist already in utero. Development largely involves **pruning** (getting rid of excess neurons), **synaptic growth** (peaks at about two years), and **myelination** (continues through development — even teenagers' frontal lobes aren't fully myelinated; frontal lobes underlie restraint and willpower).
2. **Inhibition problems:** A-not-B may reflect not lack of knowledge but inability to stop a prepotent response. **Adele Diamond's** evidence: kids reach for A but *look* at B, "as if they know it's there but they can't stop themselves from reaching." Bloom illustrates with a Simpsons clip.
3. **Learning:** Some things must be learned — true across social, economic, political, and physical domains.
 - **Michael McCloskey** study with college students: A ball is shot through a curved transparent tube; subjects asked whether it exits in a straight line (A) or continues to curve (B). At Johns Hopkins, 50/50; at Yale, students get it right. With *water* squirted through the tube, nobody chooses B — everyone knows water goes straight. "When you have experience that helps you out, but in absence of experience you're kind of lost."

Chapter 3. Babies in the Social World

- Newborns have social preferences. Newborn research is hard "because of the consent procedure and everything, so most of this work is done in France." Newborns prefer face-like configurations over scrambled ones. Humans and other primates favor faces.
- Babies are natural mimics. **Andrew Meltzoff:** stick out your tongue at a newborn, and the newborn more often than not sticks its tongue back out — "some sort of social connection from one person to another."
- The nature of face preference and mimicry is debated.

Helping/hindering study (Bloom, Valerie Kuhlmeier, Karen Wynn):

- Tested nine-month-olds and twelve-month-olds with movies of a ball being helped or hindered by other characters.
- Controls: shapes, positions varied across versions.

- Result: Babies look statistically longer when the ball approaches the *hinderer* than the *helper*, suggesting a social interpretation of the events.

Follow-up (Kiley Hamlin, graduate student):

- 3-D scenes; baby is offered the characters to reach for.
- "The person giving a choice is **blind** to the study" — she didn't know what the baby had seen, to avoid intentional or unintentional bias.
- Suggests some social understanding may be present from very early.
- Bloom notes the evidence is "tentative, very controversial."

Developmental puzzle: In some respects, young children are "very clueless" about people. Two film clips were shown demonstrating young children's ignorance of others' minds.

Chapter 4. Question and Answer on Learning and Development

- **Q: ages of children in failure/success videos?** Bloom: Typically three-year-olds and young four-year-olds fail; around age four to five children make the transition.
- **Q: has discriminant conditioning been used with babies?** A graduate student (Koleen) answers: it's "not as effective as other methods." Operant conditioning is hard with babies because they don't behave systematically; looking-time measures are more subtle.
- **Q: why did the kids choose the rocket ship sticker over the Raphael one?** Bloom: not the experiment's focus; researchers assumed boys' sticker preferences to get the study off the ground. Sex differences in preferences would be revisited later.

Chapter 5. Review of Studies Presented in Class; Autism

Two interpretations of children's failures on deception/false-belief tasks:

- The psychologist in the video argued children "need to know more about minds" — they don't understand intent to deceive.
- **Alternative:** They have the right knowledge but suffer from **inhibition** problems.

Examples:

- **Big shoes / little shoes deception task:** The child, asked who the mother would think stole the food, says the real thief. Possible "pull towards the right answer" overwhelms the deception logic.
- **Mean monkey / rocket ship task:** The boy keeps pointing to the rocket ship he wants, even though the mean monkey then takes it. Could be Homer-Simpson-like failure of inhibition — "the extra work it takes to lie was beyond him." Supporting this, even adults are slower and more error-prone on tasks involving lying and deception.

Relationship between different domains of development:

- Piaget (and Freud) believed in across-the-board general changes.
- **Alternative: modularity** (Noam Chomsky, Jerry Fodor) — separate pre-wired systems with built-in knowledge and distinct learning patterns.
- Reason to take modularity seriously: developmental disorders affecting one system but not others.

Autism:

- Strikes about 1 in 1,000 people, mostly boys.
- Symptoms: lack of social connectedness, language problems, problems with people, "mind blindness" (**Simon Baron-Cohen**).
- No impairments on physical, mathematical, or spatial tasks necessarily.
- Many autistic children have no language; even high-functioning ones suffer severe social impairment.
- Bloom personally entered psychology working with autistic children.

Baron-Cohen — Charlie and chocolates task: Four candies are arranged; "Charlie" is in the middle looking toward one. Asked "Which chocolate will Charlie take?" most three- and four-year-olds answer correctly (the one he is looking at). Autistic children often "just shrug, say, 'How could I know?'" — they don't appreciate that desires track gaze.

False-belief task (Maxie):

- Maxie puts a ball in the cupboard, leaves. A second doll moves the ball under the bed. Maxie returns.
- Question: Where will Maxie look?
- Correct: the cupboard (Maxie has a false belief).
- Three- and two-year-olds fail. Four- and five-year-olds and normal adults pass. Autistic individuals — even high-functioning — often fail, saying "He's going to check under the bed."

Chapter 6. Question and Answer on Autism

- **Q: Is the false-belief task different from the deception task?** No — both tap appreciation of false belief; the Maxie task is a "stripped-down version" of the deception study.
- **Q: Why mostly boys?** "Nobody knows," but Baron-Cohen's theory: certain abilities are more sequestered by sex; social abilities are more female. Provocatively: **"To be a man is to suffer from a very mild form of autism."** Autistic individuals have "extreme male brains." To be revisited under sex differences.

- **Q: Severity of autism vs. developmental stages?** Surprisingly, autism can't really be thought of as a developmental delay. "Somebody with autism isn't like any child at all, any normally developing child at all." However, severity is partly indexed by how much language the person has.
- **Q: Can someone with autism overcome their deficiencies?** Media tends to showcase exceptional cases like **Temple Grandin**, who wrote books on her experience but is very unusual. Depends partly on whether **Asperger syndrome** (milder) is included. Majority of people with autism have severe problems and will not lead a normal life.
- **Q: Rain Man / autistic savants?** Rain Man (Dustin Hoffman) had extraordinary mathematical abilities. Some autistic individuals have extraordinary artistic, mathematical, or musical abilities — "amazing" but a very small minority.

Candidate modules raised:

- **Physics** (object module) and **people** (social module) — covered in the developmental data.
- **Artifacts** — for tables, chairs, cars, forks.
- **Sociology** — for human groups, races, classes.
- **Intuitive biology** — common-sense biological understanding; defended by **Frank Keil**, Master of Morse College at Yale.

Beyond modules — general differences:

- A "very, very big difference between a creature that doesn't have language and a creature that does." Learning to speak may reconfigure the human brain in a way unparalleled in other species. To be revisited in the language lecture.

Closing — Stephen Jay Gould's example:

- Even if you hate babies and development, developmental data can inform questions about adults.
- Gould asked: "Is a zebra a black animal with white stripes or a white animal with black stripes?" Looking at adult zebras won't answer it — but watching embryological development of a zebra will.
- Final quote from biologist **D'Arcy Thompson**, author of *On Growth and Form*: "**Everything is the way it is because it got that way.**"

Notable Quotes

"Ontogeny recapitulates phylogeny." (Endorsed by Piaget; Bloom: "It's entirely not true, but it's a beautiful phrase.")

"The child essentially — he does A, A, A. It's moved. You go, 'doh!' and he keeps going for it. And there's some evidence that's true."

"Habituation is a fancy word for boredom."

"Often 90% of the game is discovering a clever method through which to test your hypotheses."

"Most of this work is done in France, where they have no laws at all. They just rush in to — Women give birth and they rush in and they say, 'We are psychologists,' and then we do experiments on the babies, and it's terrific."

"To be a man is to suffer from a very mild form of autism."

"Somebody with autism isn't like any child at all, any normally developing child at all."

"Everything is the way it is because it got that way." — D'Arcy Thompson

Assigned Readings / References

- Peter Gray, psychology textbook (refers to the A-not-B task as the "changing hiding places" problem; further reading on the deception/inhibition debate).
- The Norton readings on development (further reading on the deception/inhibition debate).
- D'Arcy Thompson, *On Growth and Form* (referenced in closing quote).

PSYC 110 - Lecture 6 - How Do We Communicate?: Language in the Brain, Mouth and the Hands

Overview

This lecture by Professor Paul Bloom introduces the scientific study of language, restricting "language" to systems like English, Dutch, Warlpiri, Italian, Turkish, Urdu, and sign languages, in order to make scientific generalizations possible. Bloom argues that any adequate theory of language must account for both deep universals shared across languages and striking differences between them. He frames language as plausibly part of human nature, citing Darwin's claim, the universal presence of language in human societies, the creation of creoles from pidgins within a single generation, neurological dedication, and possible genetic bases. He walks through the three core structural levels of language — phonology, morphology, and syntax — and then turns to Chomsky's claim that language acquisition is more like growth than learning, surveying the developmental time course from prenatal melody preferences through the post-puberty decline in language-learning ability.

Key Concepts & Definitions

- **Language (scientific/narrow sense):** Systems like English, Dutch, Warlpiri, Italian, Turkish, Urdu, American Sign Language, Navajo. Bloom restricts the term in order to find "interesting generalizations" and "do good science."
- **Linguistics:** An entire field devoted to studying the nuances and structures of different languages.
- **Pidgin:** A makeshift communication system developed when speakers of different languages (e.g., enslaved people from different language backgrounds, mixed deliberately in the slave trade to avoid revolt) need to talk to one another. "Strings of words borrowed from the different languages around them and put together in sort of haphazard ways." Not a language.
- **Creole:** A full language with rich syntax, morphology, and phonology, developed within a single generation by children raised in a pidgin-speaking society.
- **Aphasia:** Language deficit resulting from damage to dedicated language parts of the brain; may include losing the ability to understand or create language.
- **Phonology:** The system of sounds (or signs) of a language. English has about forty phonemes.
- **Phoneme:** A basic sound unit. Example: English has phonemes "l" and "r," so English speakers hear "lip" vs. "rip" as different words.

- **Morphology:** The system of words/morphemes.
- **Morpheme:** "The smallest meaningful unit in a language." Often the same as a word ("dog"), but words can contain multiple morphemes ("dogs" = "dog" + plural 's'; "complained" = two morphemes).
- **Arbitrariness of the sign (Ferdinand de Saussure):** Languages map arbitrary sounds/signs to ideas. Most concepts (e.g., "country," "idea") cannot be iconically represented.
- **Syntax:** Rules and principles that combine words into phrases and phrases into sentences. Same as "grammar" (syntax is the more technical term).
- **Infinite use of finite media (Wilhelm von Humboldt):** Going from a finite vocabulary to an infinite number of sentences via a combinatorial system.
- **Recursion:** The infinity mechanism in language — one rule invoking another rule (e.g., a sentence expanding to noun + verb + sentence), allowing potentially infinite embedding.
- **Top-down processing:** "When you know what something is you hear it that way." Fills in gaps in sound (e.g., hearing "sentence" through a cough).
- **Segmentation:** Determining word boundaries from a continuous sound stream. The "pauses" between words in your own language are a psychological illusion inserted by the mind.
- **Specific language impairment:** Genetic disorder where children, though smart, socially motivated, and wanting to communicate, cannot learn language normally.
- **Subject-verb-object (SVO) vs. subject-object-verb (SOV):** English is SVO ("Bill hit John"). The majority of languages are SOV ("Bill John hit").
- **Linguistic isolate:** A person (e.g., a deaf person in a society where no one signs to them) who grows up without language input.

Detailed Notes

Chapter 1. The Scientific Notion of Language and Structure

- Language is a battleground for theories of human nature. Every major thinker about people has had to make claims about it: Aristotle, Plato, Hume, Locke, Freud, Skinner; modern computational theory, cognitive neuroscience, evolutionary theory, cultural psychology.
- Bloom restricts "language" to systems like English, Dutch, Warlpiri, Italian, Turkish, Urdu, etc. Using "language" too broadly (e.g., for traffic signals, music, dog communication) makes scientific generalization useless.
- After defining narrowly, we can later ask whether animal communication systems share the same properties.

- Framing questions for lecture: basic facts about language; what languages share; how language develops; language/communication in nonhumans.
- Pre-class demonstration with bilingual students illustrated two facts:
 1. Languages share deep, intricate universals — every language can convey abstract notions about thoughts, propositions, and spatial relations in objects.
 2. Languages differ greatly — sounds, words, look unusual (for sign language); knowing one does not mean knowing another.

Darwin's claim

"Man has an instinctive tendency to speak, as we see in the babble of our young children, while no child has an instinctive tendency to bake, brew or write."

- Claim: language is special — a propensity, capacity, or instinct.

Evidence that language is part of human nature

- Every human society has language. No group encountered without one in human history.
- This alone doesn't prove it's built in — could be a useful cultural innovation (like utensils for eating: knife/fork, chopsticks, spoon).
- **Pidgin → Creole case:** Slave trade (tobacco, cotton, coffee, sugar) deliberately mixed laborers from different language backgrounds to prevent revolt. Adults developed pidgins. Children raised in this society did not speak pidgin; within a single generation they produced a creole — a real language with rich syntax, morphology, phonology.
- **Nicaragua sign language case:** Children acquired sign language from adults who were themselves second-language learners. Rather than copying the makeshift adult system, the children "creolized" it into a full-blown language.
- Every normal human has language; not everyone can ride a bicycle or play chess. Exceptions are due to brain damage.
- Neurological evidence: dedicated brain parts for language; damage produces aphasia.
- Genetic evidence: some people have point mutations in genes implicated in language and cannot learn/use language normally.

Creativity of language (Descartes)

- Descartes argued language is the best evidence we are more than machines — language is unbounded and free; we have free will.

- We can produce a virtual infinity of sentences. Grammatical English sentences under twenty words: "a lot."
- Comprehension cannot be a list — we understand sentences never said before. Bloom's invented examples:
 - "It's surprisingly easy to get a purple tie on eBay if you don't care much about quality."
 - "I am upset that one cannot easily download 'Buffy the Vampire Slayer' through iTunes."
- Rules in our heads are abstract, automatic, unconscious — similar to rules of visual perception.

Examples of implicit syntactic knowledge

- "The pig is eager to eat" — pig is the eater.
- "The pig is easy to eat" — pig is being eaten.
- "Bill knew that John liked him" — "him" can refer to Bill or to a third party (Fred), but not to John. Natural reading: him = Bill.
- "Bill knew that John liked himself" — only means John liked John.
- Bloom credits Steven Pinker's *The Language Instinct* for examples.

Three core levels of language

| Level | Definition |

| ---| ---|

| Phonology | System of sounds or signs |

| Morphology | System of words/morphemes (basic units of meaning) |

| Syntax | Rules and principles combining words/phrases into meaningful utterances |

Chapter 2. Phonology: A System of Sounds

- English has about forty phonemes; native monolingual English listeners categorize sounds into these.
- "l" vs. "r" example: English speakers hear "lip" vs. "rip"; speakers of languages lacking the distinction struggle to learn it.
- **Segmentation problem:** must figure out word boundaries from sound signal.
- The pause between words is a psychological illusion — an oscilloscope shows no pauses. Your mind inserts pauses where you already know the boundaries are.
- Demonstrations of segmentation:
 - French: "Je ne sais pas" — non-French listeners hear no pauses.

- Hebrew: "Sleecha, eypho ha-sheeruteem" (a request for the bathroom).
- When bilingual students gave demos, they sounded like "blublublublub" because the listener didn't know the language.
- Imagined properly-segmented English from another's perspective: "Glorp [pause] fendel [pause] smug [pause] wuggle."
 - Children make segmentation mistakes — visible in misheard song lyrics:
- "I'll never be your pizza burnin'" → "Beast of burden."
- "A girl with colitis goes by" → "A girl with kaleidoscope eyes."
- "The ants are my friends; they're blowin' in the wind."
- Religious example: "Our father with Bart in heaven; Harold be they name... Lead us not into Penn Station..."
 - **Top-down processing:** what you know shapes what you hear.
- Rick James "Super Freak" — many hear "the kind of girl you read about in Newsweek magazine"; actual lyric: "in new wave magazines." Once told, you hear it that way.
- Coughed-over "s[cough]entence" is perceived as "sentence" — gap is filled.
- "Get Crunk" example: even the clean (censored) version still sounds obscene because of top-down processing.
 - Theme to be revisited with vision: prior knowledge clears up confusing perceptual input.

Chapter 3. Morphology: A System of Words

- Saussure's **arbitrariness of the sign**: any arbitrary idea can be linked to a sound/sign.
- "Dog" could be "woof woof" (somewhat iconic), but "country" cannot sound like a country.
- In sign language, "drink" can resemble drinking; "country" or "idea" cannot be iconic.
 - Vocabulary = the set of these arbitrary mappings (technically: morphemes).
 - A morpheme is the smallest meaningful unit. "Dog" = one morpheme; "dogs" and "complained" = two morphemes each. Knowing "dog" + plural 's' lets you form "dogs" without ever having heard it.
 - **Vocabulary size:** Average speaker knows ~60,000 words (low-ball); Bloom's estimate closer to 80,000–100,000. Starting around age 1, children learn roughly **nine new words per day** on average (varies by age).
 - Multilinguals may carry 200,000–300,000 words, accessed in fractions of a second — "one of the most astonishing things that people do."

Chapter 4. Syntax: Communicating Complicated Ideas

- Syntax combines words into phrases into sentences.
- **Humboldt**: "infinite use of finite media." Achieved via a **combinatorial system**.
- Other combinatorial systems: music (finite notes, infinite compositions); DNA (finite bases/ amino acids, possibly infinite strings).
- **Recursion** as the infinity mechanism.

Toy grammar example

- Vocabulary: 3 nouns (Fred, Barney, Wilma) + 2 verbs (thinks, likes).
 - Rule: sentence = noun + verb + noun.
 - Total sentences: **18** ($3 \times 2 \times 3$). E.g., "Fred likes Wilma," "Fred thinks Barney," etc.
 - Add recursion rule: sentence $\rightarrow$ noun + verb + sentence.
 - Yields e.g. "Fred thinks Barney likes Wilma" — and a potential infinity of sentences.
 - Real-life recursion example (escalating embedding):
- "John hates cheese."
 - "My roommate heard a rumor that John hates cheese."
 - "It disturbed Mary when I told her that my roommate heard a rumor that John hates cheese."
 - ...and so on. "You could keep producing a sentence deeper and deeper embedded until you die."

Ambiguity

- Different rules can produce the same surface sentence.
 - Groucho Marx: "I once shot an elephant in my pajamas. How it got into my pajamas I'll never know."
 - Newspaper headline ambiguities:
- "Complaints about NBA referees growing ugly."
 - "Kids make nutritious snacks."
 - "No one was injured in a blast which was attributed to the buildup of gas by one town official."
 - "General arrested for fondling privates." (Seoul/Korea University)
- There is a legal sub-field using linguistic theory to disambiguate sentences in the Constitution, legislation, and criminal cases.

- **Criminal case:** Two brothers (one with cognitive disability) committed a robbery. A police officer pointed his gun and said, "Give me the gun." The brother holding the gun was told by the other: "Let him have it." Ambiguity:
 - "Shoot him, let him have it" (= fire), or
 - "Give him the gun, let him have it."
 - The brother fired and killed the officer. Bloom's understanding: the speaker was found guilty.

Chapter 5. Question and Answer on Language Structure

- **Q: How does syntax differ from grammar?** A: They are the same; "syntax" is the more technical term.
- **Q: Neurologically normal humans without language?** A: Bloom corrects: not every neurologically normal person ends up with language. Cases of children raised by wolves/dogs (likely apocryphal); 20th-century children locked away by insane/evil parents; deaf "linguistic isolates" in societies where no one signs to them. These show: a brain alone is not enough — someone must use language with you.
 - **Susan Goldin-Meadow:** studied deaf children with no signing input but who had deaf siblings. They create their own language — not a full system like ASL or *langue des signes québécoise*, but a language with words, syntax, phonology.
 - **Q: Are there inherent limits on grammars?** A: Yes. Most linguists agree languages are highly constrained.
 - No language constructs a question by reversing word order.
 - No language has a rule "the fifth word must be a verb."
 - Constraints reveal what counts as a humanly natural language; but a constrained grammar still yields infinite sentences (analogy: restricting to odd numbers still gives infinitely many).

Chapter 6. Noam Chomsky and Language Acquisition

- Chomsky (author of *A Review of Verbal Behavior*) makes a radical claim: language is not learned but **grown**.

"No one would take seriously the proposal that a human organism learns through experience to have arms rather than wings, or that the basic structure of particular organs results from accidental experience. [Language] proves to be no less marvelous and intricate than these physical structures."

Why, then, should we not study the acquisition of a cognitive structure like language more or less as we study some complex bodily organ?"

- So you didn't "learn" English the way you learned baseball or the American Civil War — exposure triggered growth.
- Environment does shape language: English requires English input; specific contrasts (l/r) and words ("dog" vs. *chien* vs. Greek term) are language-specific historical accidents across ~6,000 languages (Vietnamese, Urdu, Czech, etc.).
- Word order varies: English is SVO ("Bill hit John"); the majority of languages are SOV ("Bill John hit").
- Evidence that development is like growth:
 - All normal children acquire language.
 - Specific impairments exist: aphasias from trauma/stroke; genetic disorders (**specific language impairment**) in which smart, social, motivated children cannot learn language. Suggests language ability is separable from general intelligence, social motivation, and communicative drive.
 - Language is learned without explicit feedback/training.
 - American "teach your baby language" industry (DVDs, flash cards) — not necessary.
 - In some communities adults don't speak to babies (saying "It'd be ridiculous to speak to a baby. The baby has nothing to say. You might as well just speak to your dog"). When American linguists answered "Yeah. We speak to our dogs," they revealed a cultural difference. Lack of baby-directed speech does not prevent language learning.
 - **Brown and Hanlon (1970s)**: parents respond to affect/cuteness/sociability, not to grammaticality. A parent will not correct "I loves you, Mommy" for verb agreement; will not praise "I hate your guts, Mother" for clean subject-verb-object structure. Grammatical errors disappear without correction.

Chapter 7. The Time Course of Language Acquisition

- **Prenatal/newborn melody preference:**
 - Studies in France with four-day-old babies using a sucking method on a pacifier — babies sucked harder to hear French than Russian.
 - Reviewers (mostly French) said maybe French just sounds better. Replicated in Russia: Russian babies sucked harder for Russian than French.
 - Babies are responding to the **rhythm** of the language, not words or syntax.
- **Early phonemic sensitivity:**

- English-speaking babies distinguish English phonemes (e.g., "lip" vs. "rip") AND non-English contrasts (e.g., Czech, Hindi).

- You as a baby were a "multilingual fool"; this sensitivity **narrows down to the language you hear by about twelve months of age.**

- **Q&A about baby talk:** Speaking in perfect adult English to babies is unusual; most people use "Oh. You're such a cute little baby."-style talk. Evolutionary psychologists debate function — aiding language learning vs. calming. Whether you do it or not doesn't make a big difference. It is very difficult to find effects of how parents talk to babies on language outcomes.

- **~7 months: babbling.**

- **Sign language acquisition:** A surprising finding of the last 10–20 years — acquisition of sign languages is, as far as known, **exactly the same** as spoken languages. Same developmental milestones: babbling, first words, first sentences, first complicated constructions, at the same points. No advantage for speech over sign.

- **~12 months:** First words — objects and actions ("dog," "up," "milk"). Sensitivity to word order ("dog bites cat" ≠ "cat bites dog").

- **~18 months:** Faster word-learning. Mini-sentences ("Want cookie," "Milk spill"). Function morphemes ("in," "of," "a," "the") gradually appear.

- **~7 years through puberty:** Language-learning ability declines.

- **Elissa Newport and Sam Supalla:** studied immigrants in the US for 30–40 years. Best predictor of English proficiency is **not** intelligence, family size, or motivation — it is **age at first exposure**.

- Learning a second language in the first few years: speak like a native.

- Past puberty: huge variation; rare to lose an accent; phonology, syntax, morphology all affected.

- "It's like the part of the brain that's responsible for language learning is only around early in development and if you don't get your language by then it'll just run out."

- Next class: animals; vision, attention, memory. Second reading response will be posted.

Notable Quotes

"Man has an instinctive tendency to speak, as we see in the babble of our young children, while no child has an instinctive tendency to bake, brew or write." — Darwin

"No one would take seriously the proposal that a human organism learns through experience to have arms rather than wings... Why, then, should we not study the acquisition of a cognitive structure like language more or less as we study some complex bodily organ?" — Chomsky

"The infinite use of finite media." — Wilhelm von Humboldt

"The arbitrariness of the sign." — Ferdinand de Saussure

"You could keep producing a sentence deeper and deeper embedded until you die."

"When you were a baby you were a multilingual fool. You could understand the sound differences of every language on earth. Now, if you're like me, you could barely understand English."

"It's like the part of the brain that's responsible for language learning is only around early in development and if you don't get your language by then it'll just run out."

"It'd be ridiculous to speak to a baby. The baby has nothing to say. You might as well just speak to your dog." — adults in some communities, as quoted by Bloom

Assigned Readings / References

- Steven Pinker, *The Language Instinct* — Bloom credits it for the discussion of phonology/morphology/syntax examples.
- Noam Chomsky, *A Review of Verbal Behavior* — referenced as the source of Chomsky's critique of Skinner.
- Brown and Hanlon (1970s) — classic study on parental responses to children's utterances.
- Susan Goldin-Meadow — work on deaf children with deaf siblings creating their own language.
- Elissa Newport and Sam Supalla — studies of age effects in second-language acquisition.
- Chapters on the brain (assigned earlier in the course) — referenced for neurological language deficits/aphasias.
- A "second reading response" was to be posted by Wednesday.

PSYC 110 - Lecture 7 - Conscious of the Present; Conscious of the Past: Language (cont.); Vision and Memory

Overview

This lecture wraps up the topic of language and transitions into perception, attention, and memory. Bloom argues that nonhuman animals possess communication systems but not language in the technical sense (phonology, morphology, syntax, recursion), and that attempts to teach apes human languages have largely failed and may be misguided. He then introduces perception as a hard problem: the retina is a 2D surface from which the mind must infer a 3D world, and it does so through unconscious assumptions and Gestalt cues, illustrated by color/brightness, object segmentation, and depth illusions. Finally, he frames attention and memory together, demonstrating that attention is a narrow spotlight (change blindness, the gorilla demo, the Stroop effect) and that memory comes in many kinds (sensory, short-term/working, long-term; implicit/explicit; semantic/episodic) with encoding, storage, and retrieval stages. The overarching moral: "We are often wrong about our experiences, both of the present and of right now."

Key Concepts & Definitions

- **Language (technical sense):** a communication system with phonology, morphology, syntax, combinatorial systems including recursion, and arbitrary names.
- **Animal communication categories:** (1) a finite list of calls (e.g., vervet monkey warnings for snake vs. leopard), (2) a continuous analog signal (e.g., bee dance where intensity corresponds to richness of the food source), (3) random variation on a theme (e.g., birdsong).
- **Linguistic relativity / Sapir-Whorf hypothesis:** the claim that the language you know affects how you think.
- **Reading:** a cultural invention, not an instinct; acquired with tremendous difficulty over many years; distinguished by Darwin from things that come naturally.
- **Perception (problem):** getting from a 2D array of neural firings on the retina to objects, people, actions, and events in a 3D world; "from a mathematical point of view, impossible" without unconscious assumptions.
- **Gestalt principles** (cues to object segmentation): proximity, similarity, closure, good continuation, common movement, good form.

- **Kanizsa Triangle / Kanizsa Square:** illusory shapes induced by incomplete circles ("Pacmen") or regularities of form.
- **Depth cues:** binocular disparity, interposition, relative size, texture gradient, linear perspective.
- **Binocular disparity:** the only depth cue that involves two eyes; the images differ more for close objects and become almost identical for far ones.
- **Mueller-Lyer illusion:** arrow-end lines that appear different lengths but are equal.
- **Ponzo illusion:** two converging lines (like train tracks) make a horizontal line at the top appear longer than an equal one at the bottom.
- **Shepard tables:** two tables that look very different in proportion but are the same shape and size (developed by Roger Shepard).
- **Sensory memory:** a residue in your senses (e.g., afterimage of lightning; echoic memory for sounds for a few seconds).
- **Short-term memory / working memory:** spans for a few minutes.
- **Long-term memory:** the store you walk around with (e.g., knowing Elvis, your name, where you live).
- **Implicit vs. explicit memory:** explicit = conscious access (what you had for dinner); implicit = unconscious (word meanings, how to walk, how to ride a bicycle).
- **Semantic vs. episodic memory:** semantic = facts (capital of Canada, "Yale is in New Haven"); episodic = autobiography (you went on vacation away from New Haven last week).
- **Encoding / storage / retrieval:** getting the memory in, holding it, getting it out.
- **Recall vs. recognition:** recall = pulling it out; recognition = recognizing a match to something in the past.
- **Attention:** "a flashlight, a spotlight on experience that willingly zooms in on something and makes it memorable."
- **Stroop effect:** expert reading subverts the task of naming font color because attention to word meaning cannot be blocked.
- **Change blindness:** a narrow focus of attention causes obliviousness to huge changes in the environment.
- **Two types of amnesia:** (1) losing memory of the past (Matt Damon amnesia); (2) losing the ability to form new memories.

Detailed Notes

Chapter 1. Non-human Communication [00:00:00]

- Two follow-ups from Monday's lecture:

- **Preference for own language in development:** studies of newborns show that "pretty much the moment they pop out they favor their own language over other languages," suggesting they listen in utero to the rhythms of the language.

- **"Let him have it!" court case:** the defendant was tried and found guilty of murder, executed in 1957, then pardoned in 1988. A movie was made about it.

- Roadmap recap: covered universals of language, phonology, morphology, syntax, arbitrary symbols, combinatorial systems including recursion, and development from babies (good at language) to adults (whose "language capacities are dead").

- **Animals and language:**

- Not controversial that animals have communication systems; if "language" = communication, then dogs, bees, monkeys have it.

- In the technical/narrow sense, the answer is almost certainly "no."

- **Three categories of animal communication systems:**

1. Finite list of calls — vervet monkeys have warnings like "attack from a snake" / "attack from a leopard."

2. Continuous analog signal — bee dance: intensity corresponds to the richness of the food source (no syntactic structure).

3. Random variation on a theme — birdsong.

- None show phonology, morphology, syntax, combinatorial systems, or arbitrary names.

- **Trained primates (Kanzi, Nim Chimpsky, etc.):** this is controversial. Bloom thinks Gray's textbook is "a little bit too credulous."

- Even if they learn words, they learn very few, over years of extensive training.

- A normally developing child can learn "a word in a day or a word in an hour."

- Utterances have limited order and lack recursion (not controversial).

- Trained chimpanzee utterances are extremely repetitive. Example sample: "Nim eat, Nim eat. Drink, eat, me Nim. Me gum, me gum. Tickle me, Nim play. Me eat, me eat. Me banana, you banana, me you give. Banana me, me eat. Give orange, me give, eat orange, me eat orange."

- **Lila Gleitman:** "if any normally developing child spoke like this, his parents would rush him screaming to a neurologist."

- **Why expect chimps to learn human language at all?**

- "Chimps are smart" — but smartness isn't enough (smart children with language deficits exist).

- "Chimps are our nearest evolutionary relatives" — but we split a long time ago; five million years either way is more than enough for language to evolve.

- Bloom's personal view: attempts to teach apes ASL are "misguided" — "as if a team of monkeys kidnapped a human child and tried to train him how to hoot like a monkey." Studying animal communication systems in the wild (vervet cries, bee dance) using the linguistics program would be more interesting.

Chapter 2. The Relationship between Language and Thought [00:09:16]

- Intro Psych skips topics that fuller courses cover:

- Language in the brain.

- Language disorders: aphasias, specific language impairment, dyslexia.

- Language perception and production (understanding and producing words in a fraction of a second).

- Reading: a cultural invention, not universal; Darwin distinguished language as instinct from reading; reading "intersects with language" — moving from speech to writing.

- Bilingualism and multilingualism.

- **Language and thought** — Bloom once taught a full seminar on this. Two general questions:

1. **Is language necessary for abstract thought?** Look at creatures without language (babies, chimpanzees) and see how smart they are.

2. **Does the structural properties of language affect the way you think?** = linguistic relativity / Sapir-Whorf hypothesis. Research compares speakers of e.g. English vs. Korean.

- The Gray textbook and selections from The Norton Anthology address this; it is the basis of a reading response.

Chapter 3. Question and Answer on Language [00:12:39]

- **Why do some people learn languages more easily?**

- Applies to both first and second language learning.

- Einstein anecdote (didn't speak till four, then "The soup is too hot," "Well, up to now everything's been fine") — "It's not a true story."

- Slight advantage for females, but small (need ~100 people to see it statistically).

- Big genetic factor: if your parents learned languages quickly, you're more likely to.

- Brain, cognitive, and social bases largely an open question.

- **Bilingual parents:** more the norm worldwide than the US single-language situation. Children learn both languages, distinguishing them by sound system and rhythms, and don't typically confuse them.
- **Hemispheric specialization:** if right-handed, language is probably in the left side. For left-handers, it varies — left side, right side, or diffuse. Why people are right- or left-handed is also an open question.
- **Does learning more than one language slow each down?** Common sense says yes (finite resources), but data say no — bilingual children reach milestones in each language at the same time as monolinguals. No cognitive deficit either (cited the Quebec debate). "There appears to be, as far as we know, no down side to learning many languages when you're young."

Chapter 4. Introduction to the Complexity of Perception and Expectation [00:18:06]

- Perception, attention, and memory treated as one connected story: you perceive a scene, hold it in memory, and a week later recall it.
 - **Claims to be argued:**
- Perception is hard; successful perception involves educated and unconscious guesses.
- We attend to some things and miss a surprising amount.
- There are many types of memory; the key is organization and understanding; some memories cannot be trusted.
- **9/11 memory study:** psychologists on Sept 12 asked people where they were when they heard the news; later checks (a year, two years later) often showed the recalled story was wrong.
 - **Big moral:** "We are often wrong about our experiences, both of the present and of right now."
 - **Marvin Minsky / MIT story:** A.I. guru; thought building a robot that could see and recognize objects would "take a graduate student a whole summer." Computer vision has been a "profound failure" — no machine recognizes objects at the level of "a really dumb one-year-old."
 - **Why perception is hard:**
- The "little man in the head watching a TV monitor" model just pushes the question back.
- The Terminator's gauges/numbers view doesn't solve it either (he still has to "read" them).
- Real picture: retina = nerve cells firing or not firing; you must infer the world from an array of firings.
- Retina is 2D; the world is 3D. Mathematically, for any 2D image there is an indefinite number of 3D images that correspond to it (e.g., a square shape could be a square head-on or a tilted square).
- The mind has **unconscious assumptions** that yield educated guesses.

- Three examples to be covered: **color** (conflated with brightness), **objects**, **depth**.

Color / brightness

- Coal vs. snowball: not simply read off from retina. Object color depends on illumination as well.
- As Bloom walks across the stage into shadow and light, no one screams that he is changing colors — we automatically factor out the change in illumination.
- **Two-blocks demo**: a block under a table appears lighter than one out in the open, but they are the same. Removing the surrounding cues (covering the background) reveals they are identical.
- Logic: "We know shadows make surfaces darker... wired up in our brains." A surface seen as in shadow is assumed lighter than it looks, and seen as lighter.

Objects

- A scene is intuitively segmented into man, house, birds, trees. Programming a computer to do this is "hugely difficult."
- **Gestalt cues**:
 - **Proximity**: things close together belong together.
 - **Similarity**: same-texture items group as one object.
 - **Closure**: a closed square reads as one object overlapping a circle.
 - **Good continuation**: overlapping lines AB and CD are seen as A-to-B and C-to-D, not A-to-C and D-to-B.
 - **Common movement**: things moving together are one object.
 - **Good form**: two perpendicular overlapping rectangles read as a plus sign; non-perpendicular ones read as two things lying on each other.
- **Illusions**: Kanizsa Triangle (illusory triangle from three Pacmen — cover the Pacmen and the triangle disappears); Kanizsa Square (no square there; "very Matrix").

Depth

- Looking at a flat picture, you still perceive depth (man in front of house; podium with Bloom walking behind it). If neurologically abnormal, you'd see a "strange creature" merging Bloom and the podium.
- **Depth cues**:
 - **Binocular disparity**: only depth cue using both eyes; close objects yield very different left/right images, far objects yield nearly identical images.

- **Interposition:** object that blocks another is in front (Bloom walking behind a podium; the man blocking part of the house).
- **Relative size:** judgments about size dictate judgments about distance. If Bloom looked 50 feet tall, you'd assume he's farther. Empire State Building / tower example.
- **Texture gradient:** textures get smaller with distance; receding dots cue receding depth.
- **Linear perspective:** parallel lines converge with distance.
 - **Mueller-Lyer illusion:** lines with inward vs. outward arrowheads look different lengths but are equal.
 - **Ponzo illusion:** two converging "train track" lines crossed by horizontal segments — the upper line looks longer; it isn't.
 - **Explanation:** arrow/track cues suggest one line is farther away; for equal retinal size the "farther" one must be physically bigger, so the brain sees it bigger.
 - **Shepard tables** (Roger Shepard): two tables — one looks longer/leaner, one thicker — but the rectangles are identical. Bloom spent ~seven hours on the computer demo proving it; a paper cut to one shape fits the other perfectly. Same explanation: depth cues and the visual system's correction for depth.

Chapter 5. Linking Attention and Memory [00:39:15]

- Memory fascinates especially when it goes wrong (amnesia).
- **Volunteer demo:** a "pretend amnesia" patient says "I don't know." Bloom: wrong answer — you still remember English. A second volunteer "still maintains bowel and bladder control so he hasn't forgotten everything."
- **Memory is broad:** autobiographical memory (the movie kind, "Where am I?") but also language, standing, chewing, swallowing — all learned and shaped by experience.
- **Two types of amnesia:**
 1. Losing memory of the past (Matt Damon amnesia).
 2. Losing the ability to form new memories.
 - **Choir director case** (film): suffered viral encephalitis; brain damage destroyed most of his temporal lobes, hippocampus, and a lot of his left frontal lobe. Retains speech, not intellectually impaired, but can't form new memories — lives in a perpetual "now," "continually reborn at every moment."
 - **Basic distinctions in memory:**
 - **Sensory memory** — residue in senses (lightning afterimage; echoic memory of last few seconds of speech, as when "you're not listening" but can replay).

- **Short-term / working memory** — spans for a few minutes.
- **Long-term memory** — knowing Elvis, your name, your address.
- **Implicit vs. explicit** — conscious recollection vs. unconscious know-how (word meaning "had," walking, riding a bicycle).
- **Semantic vs. episodic** — facts (capital of Canada; Yale in New Haven) vs. autobiography (your vacation last week).
- **Encoding, storage, retrieval** — getting it in, holding it, getting it out.
- **Recall vs. recognition** — pulling out vs. matching. ("What color tie I had on two days ago?" — impossible to recall; "purple or orange?" much easier.)
 - **Stage theory:** sensory → short-term → long-term.
 - **Attention as gatekeeper:** not everything gets into memory (the seat pressing your butt, your neighbor's smell). Attention "is a flashlight, a spotlight on experience."

Attention demos

- **Automatic visual search:** finding the one green X in black Xs, or the one O among Xs, is effortless and fast. Finding the red O among black Os and red Xs is harder.
- **Stroop effect:**
 - Naming colors of colored rectangles — easy.
 - Naming colors of color-words written in matching color — easy.
 - Naming font color of color-words written in mismatching color — slow and error-prone.
 - "Being an expert reader... your attention to what the words meant, subverted your desire to do the task. You couldn't make that go away even if you wanted to."
- **The gorilla demo** (basketball passing): white vs. black T-shirt teams pass balls; subjects count white-team passes. A gorilla-suited person walks through. "About 50% of people when counting, who have never seen this before don't notice anything." This shows a "very small window of attention."
- **Change blindness demos:**
 - Conversation movie where camera cuts change the scarf, plate color, food, etc. — most people miss the changes.
 - Two pictures flickering back and forth — subjects must clap when they see the difference; many never do.

- Work by **Dan Simons**.

- **Cornell campus experiment (Dan Simons):** a confederate stops a passerby asking for directions; two construction workers carry a door between them; while passing, the original confederate is swapped for a different person. Subjects typically don't notice. They do notice sex changes ("Didn't you used to be a woman?") and race changes; most other changes go unnoticed.
- **Consent form experiment (Brian Scholl or Dan Simons):** a subject signs a human-subjects form; the experimenter goes "down here" and a different person pops up. People don't notice.
- "What's weird is we don't see — we don't think we are. We think we see the world as it is and we don't know — notice that when we're attending to something; everything else gets blanked out."

Chapter 6. Question and Answer on Attention and Memory [00:58:04]

- **Why do practiced things become automatic and involuntary?** Bloom doesn't know. Reading, listening to speech, and seeing all become hard to shut off ("You could put your fingers in [your ears] but you can't shut off your [hearing]"). Over-practice → involuntary.
- **Did the choir director remember things before his accident?** Yes — he had some amnesia of events before the accident but knew his name and other things about his life.

Notable Quotes

"Animal communication systems fall into sort of one of three categories. Either there is a finite list of calls... There is a continuous analog signal... And then, you get things like random variation on a theme such as birdsong."

"Nim eat, Nim eat. Drink, eat, me Nim. Me gum, me gum. Tickle me, Nim play. Me eat, me eat. Me banana, you banana, me you give. Banana me, me eat. Give orange, me give, eat orange, me eat orange."

"If any normally developing child spoke like this, his parents would rush him screaming to a neurologist." — Lila Gleitman

"It would be as if a team of monkeys kidnapped a human child and tried to train him how to hoot like a monkey."

"We are often wrong about our experiences, both of the present and of right now."

"There is, at this point, no machine on earth that could recognize people and objects and things at the level of a really dumb one-year-old."

"We know shadows make surfaces darker. We don't know this like 'Here's something I know.' Rather, we know this in that it's wired up in our brains."

"Attention... could be crudely viewed as a flashlight, a spotlight on experience that willingly zooms in on something and makes it memorable."

"Being an expert reader, as you are, your knowledge of reading, your attention to what the words meant, subverted your desire to do the task. You couldn't make that go away even if you wanted to."

Assigned Readings / References

- Gray textbook — sections on nonhuman communication and on language and thought; characterized by Bloom as "a little bit too credulous" about ape language claims.
- Selections from The Norton Anthology — relevant to the language-and-thought reading response.
- Wikipedia — used by a TA to research the "Let him have it!" court case.
- Roger Shepard — Shepard tables illusion.
- Dan Simons — change-blindness work, including the Cornell campus door experiment.
- Brian Scholl — possibly the consent-form swap experiment.
- Lila Gleitman — quoted on chimpanzee utterances.
- Marvin Minsky — A.I. anecdote about robot vision.

PSYC 110 - Lecture 8 - Conscious of the Present; Conscious of the Past: Vision and Memory (cont.)

Overview

This lecture continues the discussion of vision and memory, beginning with change-blindness as evidence that our perception of the world is far sparser than it feels. Professor Bloom then distinguishes short-term (working) memory from long-term memory, arguing that storage capacity, "chunking," and depth of processing determine what is retained. He covers retrieval (cues, context- and state-dependent memory, elaborative rehearsal/retrieval) and several types of memory failure: ordinary forgetting, brain damage (retrograde and anterograde amnesia, including implicit memory in amnesics), and reconstructive failures including suggestion-induced false memories. The lecture closes with hypnosis, the repressed-memory debate, flashbulb memories, and a Q&A on motor/skill memory and photographic memory. A central theme is that confidence in a memory is no guarantee of accuracy.

Key Concepts & Definitions

- **Change blindness:** A phenomenon in which large changes in a scene go unnoticed, demonstrating that perception and memory of the world are limited and sparse.
- **Sensory memory:** "The sort of fraction of a second of sensory residue of what you're hearing and what you're seeing."
- **Working memory / Short-term memory (STM):** Very limited memory held in consciousness; capacity is "seven, plus or minus two" (George Miller), i.e., roughly five to nine items.
- **Long-term memory (LTM):** The "major storage system"; "like the hard drive of your computer"; huge but finite capacity (e.g., 60–80,000 English words, faces, songs, etc.).
- **Chunk** (George Miller): "A basic memory unit, something you think of as a single, individual entity." How material is chunked depends on knowledge.
- **Maintenance rehearsal:** Repeating information over and over to hold it in STM; typically does NOT get information into LTM.
- **Depth of processing:** The more deeply (meaningfully) you process something, the better it is entrenched in LTM.
- **Elaborative rehearsal:** The more you think about something (connecting it to images, jokes, explanations, counterfactuals), the easier it is to remember.

- **Elaborative retrieval:** People give up too soon when searching memory; persistent, cued searching recovers more.
- **Retrieval cues:** Things associated with what you are trying to remember that bring it back.
- **Compatibility principle:** You remember something better in the context in which you learned it. Also known as context-dependent memory and state-dependent memory.
- **Déjà vu:** A feeling that an event has happened before; worse with frontal lobe damage; one theory (per Marcia Johnson) is a "glitch" where an event is stored without a proper time/date stamp, so it feels familiar half a second later.
- **Retrograde amnesia** ("retro" for past): Loss of memory of the past, often a blackout period prior to a blow to the head because memories did not get consolidated.
- **Anterograde amnesia:** Loss of the ability to form new memories; appears in Korsakoff's syndrome, in patient H.M., and in Clive Wearing.
- **Implicit memory:** Memory that persists without conscious awareness, e.g., amnesic patients improving on the mirror-star task even though they don't remember doing it before.
- **Explicit memory:** Conscious memories you are aware of and understand.
- **Childhood amnesia:** Difficulty recovering memories from before about age three.
- **Flashbulb memories:** Vivid memories of socially relevant events (e.g., September 11, 2001), which cannot fully be trusted because they get reshaped through retelling.
- **Repressed memories:** Allegedly forgotten traumatic memories recovered later (often via therapy/hypnosis); their veracity is heavily debated.

Detailed Notes

Chapter 1. Distinctions between Short-Term and Long-Term Memory

- **Change-blindness demonstration (Dan Simons, at Cornell; advised by Frank Keil, now at Yale):** A student on a college campus asks an older man for directions. Two construction workers carrying a door pass between them; the student swaps places with one of them. The conversation resumes and the older man never notices he is speaking to a different person.
- **Big moral:** Your perception of reality is "a lot more sparse, a lot more limited, than you might think it is." You think you have a full sense of the world around you, but if classmates changed positions, or the lecturer changed clothes, you wouldn't notice unless you had consciously encoded it.
- **Memory and the visual system** "exploit a basic fact about the universe, which is that most things stay the same most of the time." You don't need precise representations.

- Two ways the limitation becomes visible: (1) psychologists changing reality (change-blindness studies); (2) movies — continuity errors are pervasive and hard to catch.
- Review from last class: sensory memory → working memory → long-term memory; attention determines what passes through.
- **Storage differences:**
 - LTM: huge but finite; includes ~60–80,000 English words, people met, languages, faces, stories, locations, nursery rhymes, songs, TV programs. It is **not** true that you remember everything that has ever happened to you.
 - STM/working memory: very limited.
 - **Number-recall demonstration:** Bloom reads 14, 59, 11, 109, 43, 58, 98, 487, 25, 389, 54. Most students get only a few. **George Miller:** STM $\approx$ "seven, plus or minus two" (roughly 5–9).
 - **Chunks:** The unit is the chunk. Examples:
 - "L, A, M, A, I, S, O, N" = 8 chunks; broken into 4 words = 4 chunks; as French "la maison" = 1–2 chunks.
 - A binary string ("1,1,0,1,1,0,0,1,0,1,1,0") is harder for non-experts but compresses to a single number for those who know binary.
 - **Chess study:** Non-players can hardly remember a chessboard after 5 minutes of study; chess masters remember a legal/meaningful position at a glance ("Oh. It's the Fibonacci defense").
 - **Football coaches** have photographic memory for football diagrams; **architects** for floor plans — because they understand the structure.
- Storage in memory depends greatly on how much you understand the material (expertise effects).
- **Maintenance rehearsal:** Example phone number "514-688-9057." Repeating to yourself holds it as long as you keep rehearsing, but this does NOT typically get things into LTM. Like a character on 24 memorizing a message by repetition.

Chapter 2. How Structure and Organization Affects Long-Term Memory

- **Depth-of-processing experiment:** 48 words shown to subjects (not told to memorize). Three groups:
 1. Press button for capitals vs. non-capitals (surface/visual).
 2. Press button if it rhymes with "train" (sound).

3. Press button if it fits the sentence "The girl placed the blank on the table" (meaning).

- Result on surprise recall: looks (worst) < sound < **meaning (best)**. "If you want to remember something, the best way to remember it is to give it meaning, to give it sense."

- **Mnemonic techniques** (vivid imagery, songs, poetry):

- The hippocampus is involved in spatial memory ("finding your way around"). Mnemonic: "The way I find my way around campus is through the hippocampus."

- Name-memory books pair a face with a vivid image (e.g., Mr. Fish with spiky hair → fish impaled on hair).

- To remember 10 letters, turn them into a song or "dirty poem."

- Generate a "grand and obscene image" for arbitrary material.

- **Understanding > tricks:** At a deep level, understanding is the way to get things into memory.
- **Kite paragraph demonstration:** Bloom reads a passage ("A newspaper is better than a magazine... Finally, a rock will serve as an anchor..."). It is murderously hard to remember. Revealing that "This paragraph is about flying a kite" enables comprehension and recall. Context aids memory.

Chapter 3. Memory Retrieval

- Getting information out: exam recall, eyewitness testimony, recalling someone's name.
- **Retrieval cues:** Associations that bring memories back (e.g., seeing a cracked window reminds you to replace it; seeing a person reminds you of the lunch date).
- **Compatibility principle / context-dependent / state-dependent memory:**

- **Scuba diver study:** Subjects taught material either on a boat or underwater; tested later. Recall is better when tested in the same context as learning.

- Implication: You'll do better on the final exam in the same room you learned the material in.

- **Drug state:** People who learn while mildly stoned remember better when stoned again than when sober. ("So, if you study while you drink you should tinkle a little bit before coming in to the Final exam.")

- **Mood:** Material learned while depressed is recovered slightly better when depressed than when elated.

- **Elaborative rehearsal:** Connect a target memory to many things — images, jokes, explanations, counterfactuals — to make multiple retrieval paths.
- **Elaborative retrieval:** People give up too soon. Study with older adults trying to recall high-school classmates (verified via yearbooks): initial recall is poor, but persistent cuing ("Who was

your teacher? What clubs? Sports? How did you get to school? Friends whose last name began with B, C, D?") brings more memories back over time. Some forgetting is real, but some "forgetting" is just insufficient searching.

- **Déjà vu:** Worse with frontal lobe damage. Per **Marcia Johnson** (chair of the Yale department), one theory is that a "glitch" causes an event to be stored in memory without a time/date stamp, so when the same event continues half a second later it feels strangely familiar. Not evidence of psychic powers.

Chapter 4. Memory Failure

- **Why forgetting?** Three explanations:

1. **Decay:** The brain is a physical thing; memory traces decay over time.
2. **Interference:** New similar information (e.g., 114, 81, 66, 42) blocks recovery of the originals; related memories get confused.
3. **Changes in retrieval cues:** As the world changes around you, the cues that originally accessed the memory are no longer present.

- **Childhood amnesia:** People have trouble recovering memories from before about age three.

Hard to test because people fabricate plausible-sounding early memories. Trauma in children at age 1–2 is typically not later remembered accurately. Possible explanations:

- Retrieval cues change radically (a clinical psychologist's idea: build giant tables/chairs to recreate baby's-eye view).
- Language reformats memory; pre-linguistic memories can't be re-accessed afterward.
- Neural maturation: memory parts of the brain develop around age 2–3.

- **Brain damage:**

- **Retrograde amnesia:** After head trauma/loss of consciousness, memories from immediately prior do not get consolidated and are lost.
- **Anterograde amnesia:** Inability to form new memories. Seen in **Korsakoff's syndrome**, patient **H.M.** (lives in Hartford, Connecticut), and **Clive Wearing** (film shown last class). Damage typically to the temporal lobe and hippocampus.

- **Implicit memory in amnesics:** Mirror-star tracing task. Amnesic patients each time say they've never seen the task before, yet they get steadily better at it. Implicit (procedural) memory is preserved even when explicit conscious memory is impaired.

- **Films illustrating amnesia:**

- A "very bad" Drew Barrymore / Adam Sandler movie ("Definitely don't watch that").

- *Memento* — highly recommended; the main story is told backwards, but a parallel story about "Sammy" told forwards (modeled after real experiments) dramatically illustrates what is and isn't impaired in severe memory damage.

- **Two general lessons:** (1) Normal everyday forgetting is expected. (2) Exotic brain-damage cases illuminate normal memory mechanisms.

Chapter 5. The Power of Suggestion on Memory

- **Sam Stone / Steve Ceci children's study:**

- Three children describe an event involving a man named Sam Stone. Students vote on whom to believe — opinions roughly evenly split.
- 2,300 experts shown the films thought the man ripped the book, messed up the bear, tossed the book in the air — the majority believed it happened.
- **Truth:** Nothing happened. A man named Sam Stone walked in, said "Hi," walked around, and left. The second girl was right.
- Memories were **implanted** in the first and third children through:
 - **Stereotype:** Children were pre-told Sam Stone is clumsy, trips, breaks things, spills things. (The third child says "He always does that.")
 - **Suggestive/leading questions:** "Sam Stone came in? Did he rip a book while he was there?"
 - **Combination** over several months (the children shown were in this group).
- The interviewer was naïve to the truth. Children weren't lying — they genuinely believed it. Police officers, child caseworkers, judges, and lawyers find such children highly believable precisely because they aren't lying.
 - **False memories in adults:** Similar experiments succeed. Dramatic cases of people confessing to crimes they didn't commit — not because lying, deranged, schizophrenic, or delusional, but because they've been persuaded.
 - **Filling in blanks:** If a story about a restaurant omits paying the bill, listeners later "remember" that it was paid — schemas fill gaps.
 - **Elizabeth Loftus eyewitness studies:**
 - "Did you see **the** broken headlight?" vs. "Did you see **a** broken headlight?" — "the" presupposes a headlight existed; subjects later more likely to remember one.
 - Film with no school bus; asking "Did you see the children getting on the school bus?" leads people later to report seeing a school bus.

- "How fast were the cars going when they **smashed** into each other?" vs. "**hit** each other?" — a week later, "smash" subjects more often report seeing broken glass that wasn't there.

Chapter 6. Hypnosis, Repressed Memory and Flashbulb Memories

- **Hypnosis:** Some people are readily hypnotizable. Hypnosis does NOT magically retrieve stored memories (no "zoom in on the license plate" as in movies). It makes people eager to please the hypnotist, so they make stuff up — enthusiastically and believably.
- **Hypnotic regression:** Asked to go back to one's 4th birthday party, a subject acts like an adult's idea of a 4-year-old, not a real 4-year-old.
- Some subjects claim to speak languages like "ancient Egyptian"; linguists find these are just nonsense sounds from a North American imagining ancient Egyptian.
- Hypnotism is an extreme form of what normally happens in eyewitness testimony — it "brings out the actor in you."
- **Repressed memories:** Heated debate, especially in the U.S., about adults recovering memories of childhood sexual abuse during therapy (often using hypnotic techniques).
- Some psychologists: memories are real and Freudian-style repressed; therapy brings them out.
- Most psychologists: such memories are untrustworthy and may be created by the therapist's actions.
- Real legal consequences: criminal charges by women against their fathers based on possibly false memories; counter-suits against therapists for implanting memories.
- Whether repression is ever real is controversial; that false memories *can* be implanted (without sinister intent) is not.
- **Flashbulb memories:** e.g., where you were on September 11, 2001. They feel vivid but cannot be fully trusted: people repeatedly tell and hear the story, and "stories change." Bloom notes his own memory of 9/11 may now be partly his wife's experience because he spent so much time listening to her tell it.
- **Resist the temptation:** "Being sure is no guarantee that a memory isn't false, reconstructed or even implanted."
- **Three main morals of the memory unit:**
 1. There are many types of memories (STM/LTM, implicit/explicit; possibly dedicated systems for faces, objects, spatial locations); they are dissociable.
 2. The key to remembering is organization and understanding. Intro courses (including Intro Psych) are among the hardest at Yale because the material is diverse; "highfalutin seminars" are often easier

because background context makes everything understandable. "The more you understand something, the easier you'll remember it."

3. You can't trust some of your memories.

- **Reading response:** Describe, based on lecture and readings, how to implant a false memory.

Chapter 7. Question and Answer on Memory

- **Q: What kind of memory is knowing how to play the piano?**

- It is **long-term memory** (you carry it around for years) **and also implicit memory** (you can do it unconsciously). "It's not sensory but it's as if, put it crudely, that your fingers know and not your mind."

- **Q: Photographic memory?**

- Claims of photographic memory tend not to be substantiated. Sometimes linked with savant-like skills in people with severe impairments (came up earlier in the autism discussion). Bloom is "not convinced" that true picture-taking photographic memory exists; there may be one or two case studies, but it's controversial.

- Closing note: A guest lecturer, Dean Peter Salovey, will talk about love on Wednesday.

Notable Quotes

"Your perception of reality is a lot more sparse, a lot more limited, than you might think it is."

"The standard memory storage of short-term memory is seven, plus or minus two."

"How much you know depends — affects how much you memorize — how much you could store in memory because it affects what counts as a basic unit of memory."

"If you want to remember something, the best way to remember it is to give it meaning, to give it sense."

"The way I find my way around campus is through the hippocampus."

"There is honest to God forgetting but sometimes you think you forget and it's because you haven't looked long enough."

"It is not true that you remember everything that has ever happened to you."

"Being sure is no guarantee that a memory isn't false, reconstructed or even implanted."

"It's not sensory but it's as if, put it crudely, that your fingers know and not your mind."

Assigned Readings / References

- Referenced by name in lecture (not formal reading assignments):
- **George Miller** — "seven, plus or minus two"; chunks.
- **Dan Simons** (at Cornell), advised by **Frank Keil** — change-blindness door study.
- **Marcia Johnson** (chair of the Yale Psychology department) — déjà vu theory.
- **Steve Ceci** — Sam Stone false-memory studies in children.
- **Elizabeth Loftus** — eyewitness-testimony / leading-question studies (some discussed in the text-book).
- Patient **H.M.** (Hartford, Connecticut); **Clive Wearing**; **Korsakoff's syndrome**.
- Films cited: *Memento* (recommended); a Drew Barrymore / Adam Sandler amnesia film (not recommended); the TV show 24.
- **Reading response assignment:** Describe, based on lecture materials and readings, how to implant a false memory.
- **Next class:** Guest lecture by Dean **Peter Salovey** on love.

PSYC 110 - Lecture 9 - Evolution, Emotion, and Reason: Love (Guest Lecture by Professor Peter Salovey)

Overview

This guest lecture, delivered on Valentine's Day by Dean Peter Salovey, examines love and, more centrally, the social psychology of attraction. Salovey opens by defining love through Robert Sternberg's three-component theory (intimacy, passion, commitment) and walks through the permutations that arise when only some components are present. He then surveys the social psychology of attraction, organized into the "big three" (proximity, similarity, familiarity) and the "more interesting four" (competence/Pratfall Effect, physical attractiveness, gain-loss effects, and misattribution of arousal). Salovey acknowledges that much of the research is older, sometimes ethically impossible to replicate today, and skewed toward androcentric, heterosexual dyads. He closes with the serious implications of misattributed arousal, including its possible role in why people remain in abusive relationships.

Key Concepts & Definitions

- **Love (Sternberg's theory):** Made up of three components — intimacy, passion, and commitment (decision/commitment). All three must be present for love.
- **Intimacy:** "The feeling of closeness, of connectedness with someone, of bonding." Operationally, sharing secrets/information not shared with others.
- **Passion:** "The drive that leads to romance"; physical attraction or sex.
- **Decision/Commitment:** "The decision that one is in a love relationship, the willingness to label it as such, and a commitment to maintain that relationship at least for some period of time."
- **Non-love:** No intimacy, no passion, no commitment.
- **Liking:** Intimacy only — typical casual friendships.
- **Infatuation / Infatuated love:** Passion only — "love at first sight."
- **Empty love:** Commitment only; often the final stage of long-term relationships gone bad, or the first stage in arranged marriages.
- **Romantic love:** Intimacy + passion, no commitment ("Romeo and Juliet when they first met").
- **Companionate love:** Intimacy + commitment, no passion — "your best friend."

- **Fatuous love:** Passion + commitment, no intimacy — "whirlwind courtship," Hollywood romance, shotgun wedding.
- **Consummate love:** All three components — Sternberg's complete love.
- **Pratfall Effect:** Our liking for a competent person grows when they make a mistake/blunder. Conversely, blunders make mediocre people seem even more mediocre.
- **Confederate ("stooge"):** A person in the employ of the experimenter who plays a role in the experiment.
- **Gain Effect (Aronson):** We are attracted to people whose regard for us is gaining momentum over time — "the first derivative of their regard for us is positive."
- **Loss Effect:** People who really hurt us are those who were once positive and become negative — "you can only hurt the one you love."
- **Misattribution of arousal:** Feeling physiologically aroused without being sure why, and inventing an explanation — sometimes mistakenly labeling the arousal as love/attraction.

Detailed Notes

Chapter 1. Introduction to Dr. Peter Salovey

- Professor Paul Bloom introduces Dean Peter Salovey as the first guest lecturer.
- Salovey is described as Dean of Yale College, an active social psychologist studying health psychology, framing of health messages, and the founder/developer of the idea of emotional intelligence.
- Bloom notes Salovey once taught the largest course ever in Yale College — a course on Psychology in Law — and was a legendary Introduction to Psychology teacher.

Chapter 2. Defining Love and Its Permutations

- Lecture is given on Valentine's Day; Salovey's main research area is human emotion (he doesn't personally study love in the lab).
- Caveats up front:
 - Some experiments described could not be done today (1950s–early 1970s ethics were different).
 - Much of the social psychological research on romantic love is androcentric and heterosexual — usually male participants, female targets.
 - Question to keep in mind: do these results generalize to other kinds of dyadic relationships?
 - Most of the experiments are about **attraction**, not love.

- **Sternberg's theory of love** (Robert Sternberg, formerly Yale faculty ~30 years, then dean at Tufts): love = intimacy + passion + commitment.

- Sternberg: not love if you don't call it love and don't desire to maintain the relationship.

- **Permutations** (Sternberg's terminology):

| Intimacy | Passion | Commitment | Type |

| --- | --- | --- | --- |

| - | - | - | Non-love |

| ✓ | - | - | Liking |

| - | ✓ | - | Infatuation / Infatuated love |

| - | - | ✓ | Empty love |

| ✓ | ✓ | - | Romantic love |

| ✓ | - | ✓ | Companionate love |

| - | ✓ | ✓ | Fatuous love |

| ✓ | ✓ | ✓ | Consummate love |

- **Empty love** is often the final stage of long-term relationships gone bad ("we'd better stay together for the kids...appearance's sake...financially"). It's also the first stage in arranged marriages.
- Arranged marriages do not seem to have a greater chance of ending in divorce than love marriages, but there is a major **confound**: arranged marriages tend to occur in societies that frown on divorce, making divorce socially costly.
- Reference: Meat Loaf's song "Two Out of Three Ain't Bad" used to introduce the two-element permutations.
- **Fatuous love** illustrated with: "Maybe you find yourself in Las Vegas and you get married for a day and a half and then realize that this wasn't such a good idea. And maybe your name is Britney and you're a singer."
- Suggested (humorous) homework: tally everyone you know on the three elements.

Chapter 3. The Social Psychology of Love and Attraction

- The social psychology of love has essentially been a social psychology of attraction.
- **Seven variables**, grouped into the "**big three**" and the "**more interesting four**".

The Big Three (all other things being equal)

1. Proximity (close spatial proximity)

- Tested in New York City using city-block metrics: people who live more closely together are more likely to end up in romantic relationships.
- On college campuses: negative correlation between feet from one's door to another's door and likelihood of a romantic relationship.
- "It isn't the stranger you fall in love with."

1. Similarity — "Birds of a feather flock together."

- On any measured dimension (height, age, attitudes toward capital punishment, Red Sox vs. Yankees preference), more similar = more attracted.
- "Opposites attract" — not much evidence; "similarity breeds contempt" — not much evidence.
- When students claim their partner is their opposite, Salovey responds: "Good luck."

1. Familiarity

- We tend to fall in love with people in our environment with whom we're already familiar.
- The cultural myth of "some enchanted evening you will see a stranger across a crowded room" (from "South Pacific") is just that — a myth.
- Familiarity effect even works with **words in unknown languages**: briefly flashed unfamiliar characters are later rated as more likable than new ones, even when participants don't remember seeing them.

The More Interesting Four

1. Competence and the Pratfall Effect

- We prefer competent people over incompetent ones, but **super-competent people are threatening** and make us feel diminished.
- We are most attracted to the **competent individual who occasionally blunders** — the **Pratfall Effect**.
- Example: **Bill Clinton's** popularity at the end of his term did not suffer much despite the Monica Lewinsky mistake — described in the media as "just shows he's human."
- **Classic Pratfall Experiment** (Elliot Aronson, UC Santa Cruz):

- Subjects listen to audio interviews with possible "College Bowl" quiz-show representatives.
- **Nearly perfect person:** answers 92% of questions correctly, modestly admits being in campus honor society, was editor of the yearbook, ran varsity track.
- **Mediocre person:** answers 30% correctly, average grades, yearbook proofreader, tried out for track but didn't make it.
- On attractiveness, the competent person is rated much more attractive.
- **Blunder condition:** tape continues with clattering dishes and "Oh, my goodness. I've spilled coffee all over my new suit."
- Result: the **competent person who blunders** is rated as *even more* attractive; the **mediocre person who blunders** is rated as *even more* mediocre.
- **Salovey's personal anecdote** — arriving at Yale in 1981 as a grad student, trying to persuade **Judy Rodin** (later President of the University of Pennsylvania, now President of the Rockefeller Foundation) to be his adviser. He spilled coffee on her desk and tried to invoke the Aronson experiment. Rodin replied: "Peter, Peter, Peter. You know that effect only works if I think you're competent first."

2. Physical Attractiveness

- College students surveyed say warmth, sensitivity, intelligence, compassion, sense of humor matter — and that looks don't.
- Empirically, **looks predict who gets a second date** after a blind date, not warmth, sensitivity, intelligence, compassion, or sense of humor.
- Long-term relationships likely depend on more than looks; looks specifically predict the second-date outcome.
- **University of Minnesota study:** a computer algorithm (essentially random) paired students for a dance; those who rated their randomly assigned partner as attractive were more likely to continue the relationship.
- **Aronson's "Frizzy Wig" experiment:**
 - A confederate (a naturally attractive woman) was made to look either attractive or unattractive (frumpy clothes, bad make-up, frizzy wig).
 - She posed as a clinical psychology grad student interviewing male participants and giving each a personal clinical evaluation.
 - Conditions: attractive vs. unattractive × favorable vs. unfavorable feedback.
 - Results:
 - Attractive + favorable feedback → delighted.

- Attractive + unfavorable feedback → very upset; often sought future interaction to disprove her evaluation.
- Unattractive (either feedback) → didn't really care.
- Interpretation: feedback from attractive people has more impact on us.

3. Gain and Loss

- General principle: we are wired to be more sensitive to **change** than to steady states (change signals danger or opportunity — survival/reproductive value).
- **Gain Effect (Aronson)**: more powerful than steady positive regard is regard that **starts lower and rises** over time. "The first derivative of their regard for us is positive."
- Even if the average regard is lower than someone who was always positive, the rising trajectory captures attention.
 - **Loss Effect**: people who have always been negative cannot hurt us much; the **person who was positive and becomes negative** is who hurts us most.
 - Salovey references a Ricky Skaggs country song line: "Nothing can hurt you like the person you love."
 - "You can only hurt the one you love because you are expecting positive feedback from the one you love."

Chapter 4. Misattribution for the Causes of Arousal

- Connects to Schachter & Singer's emotion work (referenced in passing).
- **Misattribution of arousal**: feeling physiologically aroused, being unsure why, and constructing an explanation — sometimes wrongly labeling it love.
- **Thought experiment (Yale/Starbucks)**: A Pierson student takes a friend to a New Blue concert and then to Starbucks. The friend orders a "decaf double espresso mocha skinny frothed," but the barista mistakenly uses caffeinated espresso. Walking back, her heart pounds, palms sweat, breath shortens — she misattributes the caffeine arousal to love. Quote: "What a day this has been. What a rare mood I'm in. Why, it's almost like being in love."
- She turns to "the most salient object in her immediate social environment" — the narrator — and concludes she's in love.

The "Rickety Bridge" Field Experiment (University of British Columbia)

- Two bridges across a river on campus: a **rickety rope bridge** hundreds of feet above the water (sways in the breeze, ~3 feet wide) and a **low solid wood bridge** near the water.
- An attractive female confederate intercepted male students crossing each bridge, asked questions, requested a short story, and gave her phone number in case of questions.
- Results:
 - Men on the **rickety bridge** wrote sexier, more ribald stories and were more likely to call her later.
 - Men on the **low bridge** wrote boring stories and were less likely to call.
 - Interpretation: misattributed arousal — fear/heart pounding from the rickety bridge gets attributed to attraction.
 - **Methodological flaw:** no random assignment to bridges → not a true experiment. People who choose the rickety bridge may be more adventure-seeking by disposition.

Lab Experiments on Misattributed Arousal

- **Painful shock instruction:** Subjects told "we'll begin the painful shock experiment in a moment" rate the experimenter as **more attractive** than those given a neutral instruction — even though the source of the arousal (fear of shock) is obvious.
- **Treadmill experiment:** Subjects asked to run 10 minutes on a treadmill before rating the experimenter find the experimenter **more attractive** than the control group who just sat — even though the arousal source (exercise) is obvious. Salovey jokes the "fourth floor of Payne Whitney Gym" is dangerous because of "that combination of aerobic exercise and spandex."
- **Playboy centerfold / fake heartbeat experiment:**
 - Male subjects view ~10 Playboy centerfolds while wearing headphones that supposedly amplify their heartbeat.
 - In reality, they hear a **tape recording** of a heartbeat that the experimenter speeds up at random intervals.
 - Subjects rate the slide that coincided with the sped-up heartbeat as the most attractive.
 - Shows that even **phony arousal** can be misattributed.

Serious Implications: Domestic Violence

- **Number one reason** people stay in abusive relationships: economic dependence or perceived lack of alternatives (homelessness, starvation, danger to kids).
- Additional factor: sometimes people misattribute the arousal of fights, screaming, name-calling, or violence as love — "Well, he wouldn't be yelling and screaming at me if he didn't love me."
- Salovey relates a story from a social psychologist friend: 30 years ago, while dating her now-husband, during a parking-lot argument he punched out her car window. She told Salovey, "That's when I knew he really loved me." Salovey calls this misattributed arousal — misattributing his aggression as love and her own fear as mutual attraction.

Chapter 5. Question and Answer

- **Q: If subjects are told in advance about painful shock (via consent form), does the misattribution effect still occur?**

- A: The more salient the actual source of arousal, the less likely the misattribution. But the effect can still occur even when the cause is somewhat obvious.

- **Q: Do the big three (proximity, familiarity, similarity) affect maintenance of relationships or just initial attraction?**

- A: Probably both, but the literature focuses on initial attraction; following couples over time creates "Heisenberg-esque" problems (observation interferes with the relationship). Most of these experiments are really about initial attraction.

- **Q: Can someone feel consummate love for more than one person?**

- A: Debated in the literature. From an **evolutionary perspective**, feeling that kind of love for more than one person could facilitate passing genes to more of the next generation. From a **socially constructed perspective**, most societies (except very unusual polygamist societies) hold that you cannot love more than one. Salovey's guess: yes, you could, but you would be conflicted because the evolutionary impulse and social constraints clash.

- **Q: Wouldn't natural selection favor people who learn this material and apply it?**

- A: This feels like passing on an acquired characteristic (counter to Darwinian theory), but if there were a heritable proclivity for learning such material, evolution might favor it. It would "make the several thousand social psychologists in this world very happy."

Notable Quotes

"It is not, however, a required component of taking a shower in Calhoun College."

"Birds of a feather may flock together but opposites don't really attract each other."

"Some enchanted evening you will see a stranger across a crowded room... Of course it happens, but much more common is somebody you already know."

"Peter, Peter, Peter. You know that effect only works if I think you're competent first."
— Judy Rodin

"What a day this has been. What a rare mood I'm in. Why, it's almost like being in love."

"You can only hurt the one you love because you are expecting positive feedback from the one you love. And when that turns negative, it's a blow."

"The first derivative of their regard for us is positive."

"That's when I knew he really loved me."

Assigned Readings / References

No formal reading list is given in the transcript. Named researchers, studies, and cultural references mentioned:

- **Robert Sternberg** — triangular theory of love (intimacy, passion, commitment); formerly Yale faculty, dean at Tufts University.
- **Elliot Aronson** (UC Santa Cruz, retired) — Pratfall ("College Bowl") experiment; "Frizzy Wig" attractiveness experiment; Gain Effect.
- **Judy Rodin** — Yale faculty when Salovey arrived in 1981; later President of University of Pennsylvania; now President of the Rockefeller Foundation.

- **Schachter & Singer** — emotions/misattribution work (referenced in passing).
- **Rickety Bridge** field experiment at the University of British Columbia.
- **University of Minnesota** computer-paired dance study.
- Cultural references: Meat Loaf's "Two Out of Three Ain't Bad"; "South Pacific" ("Some Enchanted Evening"); Ricky Skaggs ("Nothing can hurt you like the person you love"); Bill Clinton / Monica Lewinsky; Britney (Spears); The New Blue (Yale a cappella group); The Rumpus (Yale satirical newspaper); Payne Whitney Gym; Pierson College / Davenport Theater; Calhoun College.

PSYC 110 - Lecture 10 - Evolution, Emotion, and Reason: Evolution and Rationality

Overview

This lecture introduces a second "astonishing hypothesis" of modern psychology — what Daniel Dennett called Darwin's dangerous idea — the modern biological account of the origin of biological (and psychological) phenomena. Professor Bloom contrasts William Paley's creationist argument from design with Darwin's account of natural selection, showing how variation, differential survival and reproduction, and adaptation can produce complex structures like the eye without a designer. He then explores how evolutionary theory does and doesn't apply to psychology, warning against confusing ultimate and proximate causation and against assuming everything is adaptive. The lecture closes with an introduction to human rationality through the heuristics-and-biases tradition of Kahneman and Tversky, including framing effects, base rates, availability bias, and confirmation bias, ending with the Wason selection task and Leda Cosmides's ecological reformulation.

Key Concepts & Definitions

- **The Astonishing Hypothesis (Crick):** The idea that our mental life, consciousness, morality, and capacity to make decisions and judgments is the product of a material physical brain.
- **Darwin's dangerous idea (Dennett):** The modern biological account of the origin of biological phenomena, including psychological phenomena.
- **Creationism:** The view that biological structures were created by an intelligent being.
- **Argument from design (Paley):** A complicated artifact like a watch (or an eye) implies an intelligent designer; the eye's complexity, like a camera's, suggests a designer "far smarter than any human engineer."
- **Vestigial characteristics:** Features of the human body (e.g., tailbone, goose-bumps) that are inexplicable unless the body is viewed as modifications from a previous form.
- **Natural selection:** Has three components — variation, differential survival and reproduction, and inheritance — giving rise to adaptations ("that perfection of structure that justly excites our imagination").
- **Adaptation:** A trait shaped by natural selection to solve a problem of survival and reproduction.
- **Byproduct / accident:** Traits that are not adaptations but accidental consequences of how the body/mind is built (e.g., back pain, hiccups, self-pity, bloating).
- **Ultimate causation:** Why something is there in the first place, over millions of years of history.

- **Proximate causation:** Why you are doing something now (the day-to-day psychological reason).
- **Auditory cheesecake (Pinker):** Music as an evolutionary accident with no adaptive advantage — something we like to gorge ourselves on.
- **Heuristics (Kahneman & Tversky):** Rough and ready rules of thumb that served us during evolutionary history but can lead us astray.
- **Framing effect:** Responding differently to a situation depending on how options are framed.
- **Loss aversion:** People hate a certain loss; framing in terms of losses changes decisions.
- **Endowment effect:** Once you own something, its value shoots up.
- **Base rates:** Background statistical frequencies, which people tend to ignore in favor of individuating information.
- **Availability bias:** Judging frequency by how easily examples come to mind.
- **Confirmation bias:** When holding a hypothesis, we look for confirmations.
- **Wason selection task:** A logical card task that is hard in abstract form but easier when framed as an ecologically valid social rule (Leda Cosmides).

Detailed Notes

Chapter 1. The Modern Biological Account of the Origin of Psychological Phenomena

- Course opened with Francis Crick's "Astonishing Hypothesis": mental life, consciousness, morality, and decision-making are products of a material physical brain.
- A second, equally or more shocking idea concerns the **origin** of mental life — Daniel Dennett's "Darwin's dangerous idea."
- **Paley's argument from design:**
 - Walking on a beach: a rock needs no special explanation, but a watch does.
 - Paley (a medical doctor) compared the watch to the eye: both have many parts interacting in complex ways.
 - Updated analogy: the eye is like a camera.
 - Both have lenses that bend light and project an image onto a light-sensitive surface (retina vs. film).
 - Both have a focusing mechanism (muscles changing the lens vs. a diaphragm).
 - Both are encased in black.

- The eye is far more complicated than a camera. Bloom's "Six Million Dollar Man" (test pilot Steve Austin, 1974) anecdote: even 30+ years later, building an artificial eye that matches the human eye is fantasy, not even science fiction.

- Paley's conclusion: complicated biological things must be created by an intelligent designer — God.

- Bloom showed a Google Images search for "God," yielding "an old-bearded man wearing a crown."

- **Problems with the creationist view:**

- It pushes back the question: where did the intelligent being come from? Particularly serious for psychological structures.

- Evidence for evolution (broadly, not a specific mechanism):

- **Fossil evidence** of rudimentary forms of body parts.

- **Vestigial characteristics** like the human tailbone and goose-bumps.

- **Parallels with other animals:** human, rat, cat, monkey brains follow a common plan — suggesting evolutionary descent.

- **Occasional poor design:** the eye's blind spot; the male urethra running through the prostate gland — God is either malicious or incompetent.

- **Richard Dawkins** (*The Blind Watchmaker*): 100–150 years ago, anyone who didn't believe God created humans was a "moron" because the argument from design is "a damn good argument" in the absence of an alternative.

- **Darwin's accomplishment:** showed how complex biological structures like the eye emerge through a purely non-intentional, purely physical process — possibly the most important idea in the sciences.

- **Three components of natural selection:** variation → differential survival/reproduction → inheritance → adaptations ("that perfection of structure that justly excites our imagination").

- Examples:

- **Camouflage:** animals better hidden survive and reproduce more.

- **The eye:** parallels in other animals show rudimentary forms; computer simulations have generated eyes from plausible starting points under selective pressure.

- Why evolution matters for psychology: combine Darwin's idea with the rejection of Descartes (minds are physical). Then mental life is no less the product of natural selection than the eye or camouflage. Cognitive mechanisms evolved for survival and reproduction, shaped to solve specific problems (perception, communication, nutrition, rest, etc.).

Chapter 2. Avoiding Misconceptions When Applying Evolutionary Theory to Psychology

- **Misconception 1:** Natural selection means animals consciously want to spread their genes.

- Confuses **ultimate** and **proximate causation**.

- A cockroach knows nothing about genes; mechanisms differ from mental states.

- **William James** on why we eat:

> Not one man in a billion when taking his dinner ever thinks of utility. He eats because the food tastes good and makes him want more. If you asked him why you should want to eat more of what tastes like that, instead of revering you as a philosopher, he will probably laugh at you for a fool.

- "You eat because you're hungry" (proximate) and "you eat to spread your genes" (ultimate) are different levels of explanation, not competing alternatives.

- Parents protect children because offspring carry 50% of their genes (ultimate); psychologically, no one but "a deranged psychologist" answers this way.

- **Misconception 2:** Everything is adaptive.

- Distinguish **adaptations** vs. **byproducts/accidents**.

- Back pain is not an adaptation — it's an accidental byproduct of back shape. Same for hiccups, self-pity, bloating.

- Three big modern preoccupations — **pornography, television, and chocolate** — are largely evolutionary accidents:

- Heterosexual male attraction to women is plausibly adaptive; porn substitutes are "evolutionary adaptive dead ends."

- Liking chocolate stems from an evolved taste for sweet things (e.g., fruit in our natural environment); chocolate bars are recent and not good for us.

- **Clear cases:**

- **Color vision** — adaptation (visual distinctions).

- **Fear of snakes** — adaptation (ancestors not afraid didn't reproduce as much).

- **Liking chocolate bars and NASCAR** — cannot be adaptations; too recent.

- **Hard cases (debated):**

- **Music:** Steven Pinker (*The Language Instinct*) called music "auditory cheesecake" — an evolutionary accident. Others argue music has an adaptive advantage.

- **Male sexual violence:** adaptation or accident?

- **Multiple languages:** accidental byproduct or group-selective advantage?
- **Visual art, fiction, love of stories:** all subject to heated debate.

Chapter 3. Claims against the Evolutionary Psychology

- **Objection 1 (metaphysical / dualist):** If mental life is not the product of the brain, evolution is irrelevant.
- **Lisa Simpson** said the Pope favored evolution — half right.
- **John Paul II** said Darwinian theory is not incompatible with truth about man, but drew a line:
 - > If the human body takes its origin from preexisting living matter, the spiritual soul is immediately created by God. Consequently, theories of evolution which consider the mind as emerging from the forces of living matter or as a mere epiphenomenon of this matter are incompatible to the truth about man.
- **Objection 2:** Even if the mind is physical, humans have no instincts.
- **Ashley Montagu (1973):**
 - > With the exception of the reactions of infants to sudden withdrawals of support and to sudden loud noises, the human being is entirely instinctless. Man is man because he has no instincts, because everything he is and has become he has learned from his culture...
- **Louis Menand** (New Yorker): "Every aspect of life has a biological foundation in exactly the same sense, which is that unless it was biologically possible it wouldn't exist. After that it's up for grabs." Despite work by **Spelke** and **Baillargeon** on infant hard-wiring, Menand thinks the interesting aspects of human nature are cultural.
- **Objection 3:** Even if there's a human nature, you can study it directly without evolution — what could evolution add?

Chapter 4. Ways in Which Evolution Helps Describe the Mind

- Evolution of consciousness, morality, etc., is intuitively interesting in its own right.
- **What can be innate and what cannot:**
 - Plausibly innate adaptations: talking, mate selection, childcare.
 - Cannot be innate (too recent): written communication, interacting with strangers, driving a car, playing chess.
- **Group differences:**
 - **Children vs. adults:** face different evolutionary problems — predictable differences in brains.

- **Racial / ethnic differences:** evolutionary theory makes no predictions about profound mental differences across humans evolved in different parts of the world.
- **Sex differences:** evolutionary biology gives tools to discuss what to expect; Bloom shows a clip from the movie "**Roger Dodger**" as an example of "barroom evolutionary psychology" — a mix of reasonable claims and "total bull crap."
- Map-reading sex difference claims: implausible biological root.
- Trait developing to attract females: less crazy.
- Telepathic claims: crazy.
 - **Design perspective gives insights into current nature.** Two examples:
- **Margie Profet on pregnancy sickness** (won the MacArthur Genius Award):
 - Traditional view: hormonal breakdown.
 - Profet: morning sickness is designed to protect the developing fetus from **teratogens** (poisons).
 - Predictions:
 - Onset/offset matches the period of maximal embryonic/fetal vulnerability.
 - Foods avoided correspond to those most deadly for the fetus **during the period when humans evolved.**
 - No aversion to **alcohol** — wasn't around in evolutionary history.
 - Relationship between **miscarriage and birth defects**: more morning sickness → more protection. Evidence supports this.
- **Peter Salovey's "big three"** of attraction: **proximity, similarity, familiarity.**
 - Evolutionary problem: siblings are extremely close, similar, familiar — yet not "hot."
 - Reason: **incest avoidance.** Inter-mating with kin produces bad offspring.
 - No parenting guides address keeping siblings from having sex — they don't want to.
 - Illustrates proximate vs. ultimate causation: gut-level "eew," not reasoning about non-viable offspring.
- **How do you tell who your kin are?**
 - Earlier evidence — **kibbutz studies**: people raised communally know they're unrelated but show diminished sexual desire toward each other.
 - **New paper in *Nature* (five days ago)**: an even bigger cue is whether you observed your **mother taking care of that person** — this diminishes desire to the level of disgust.

Chapter 5. Heuristics: Framing Effects, Base Rates, Availability Bias and Confirmation Bias

- Psychology has produced Nobel Prize winners — most recently **Danny Kahneman**, in collaboration with **Amos Tversky**.
- Kahneman & Tversky shifted views away from purely logical decision-making toward **heuristics** — useful during evolutionary history, but error-prone.

Framing Effects

- Classic disease problem: 600 people will die.
- **Program A:** 200 saved.
- **Program B:** 1/3 chance all saved, 2/3 chance nobody saved.
- Most pick A.
- Reframe in terms of deaths and chance of dying: preferences flip — a **framing effect**, combined with **loss aversion**.
- Practical example: hamburger advertised as "80% fat free" vs. "20% fat."
- **Laurie Santos** capuchin monkey studies:
- Monkeys taught to use discs as money to buy banana or apple pieces.
- Framing test: two experimenters both average 1.5 pieces.
- One shows 1 piece, sometimes gives 2 (gain frame).
- The other (pink experimenter) shows 2 pieces, sometimes gives 1 (loss frame).
- Capuchins prefer the experimenter who shows 1 and sometimes gives 2 — framing/reference-point effects in nonhuman primates.
- **Endowment effect:**
- "How much will you give me for this chocolate bar?" → ~\$2.
- "How much will you sell me this chocolate bar for?" → ~\$2.50.
- Owning something raises its value; selling is treated as a loss.

Base Rates

- 70 engineers, 30 lawyers; John is "40, married, three children, conservative, cautious, no interest in politics, awkward around people. His hobbies include carpentry, sailing, and solving mathematical puzzles, like online dating."

- Most say engineer; switching the base rates (30 engineers, 70 lawyers) doesn't change responses — people **ignore base rates**.
- **Disease test problem:**
 - Disease hits 1 in 1,000.
 - Test never misses (100% sensitivity), 5% false positive rate.
 - Given a positive test, probability of disease?
 - Medical students typically guess 50–95%.
 - Correct answer $\approx$ 2%:
 - Of 1,000 people: 1 has disease, tests positive.
 - 999 don't have disease; $\sim$ 5% ($\approx$ 50) false positives.
 - So $\sim$ 1 in 51 positive tests is a true case $\rightarrow$ $\sim$ 2%.
- People panic because minds aren't evolved for base-rate computation.

Availability Bias

- Frequency judged by ease of mental retrieval.
- Classic Kahneman & Tversky: more "ing" words estimated than "ng" words, though every "ing" is "ng."
- **Shark attacks:**
 - Injury risk: 1 in 6 million.
 - Death: 1 in 500 million.
 - In Florida ("Shark Central"): injury risk $\sim$ 1 in half a million.
 - Overestimated because shark attacks are salient/newsworthy.
- **Potato salad / food poisoning:**
 - Injury $\sim$ 1 in 800; death $\sim$ 1 in 55,000.
 - Potato salad is $\sim$ 1,000 times more dangerous than shark attacks.
- **Estimating proportion of Jews in the US:**
 - Actual: between 1.9% and 2.1%.
 - Average American thinks 20%.
 - New York City: $\sim$ 9% (most dense); New Haven: $\sim$ 3.5%.
 - Cause: Jews prominent in highly visible positions (entertainment, academia) $\rightarrow$ easily come to mind.

Confirmation Bias

- **Child custody scenario** (jury must give sole custody):
 - **Parent A:** average income, average health, average work hours, reasonable rapport, relatively stable social life.
 - **Parent B:** above-average income, minor health problems, lots of work travel, very close relationship with child, extremely active social life.
 - When asked **who to award custody to** → slight preference for B.
 - When asked **who to deny custody to** → slight preference for B.
 - When asked to award, salient good-parent traits of B jump out; when asked to deny, salient bad-parent traits of B jump out.
- General principle: when we have a hypothesis, we look for confirmations.

Wason Selection Task

- Four cards: D, G, 3, 8 (letter on one side, number on the other).
- Rule: "If a card has a 'D' on one side, it has a '3' on the other side."
- Most people answer "D" or "D and 3" — but the correct answer is **"D" and "8"** (since you must check for a D on the back of 8 to falsify the rule).
- People find the abstract version very hard.
- **Leda Cosmides** (evolutionary psychologist, Santa Barbara): when reframed as an ecologically valid **social rule** (e.g., bartender checking that no one under 21 drinks beer — flip cards showing "beer" and "under 21"), the same logical problem becomes easy.
- Implies our minds are tuned to solve logically equivalent problems differently depending on **ecological/social content**.
- Closing assignment: reading response involves doing "your own bit of reverse engineering and evolutionary psychology."

Notable Quotes

"The Astonishing Hypothesis... the idea that our mental life, our consciousness, our morality, our capacity to make decisions and judgments is the product of a material physical brain."

"Not one man in a billion when taking his dinner ever thinks of utility. He eats because the food tastes good and makes him want more." — William James

"If the human body takes its origin from preexisting living matter, the spiritual soul is immediately created by God. Consequently, theories of evolution which consider the mind as emerging from the forces of living matter or as a mere epiphenomenon of this matter are incompatible to the truth about man." — John Paul II

"With the exception of the reactions of infants to sudden withdrawals of support and to sudden loud noises, the human being is entirely instinctless. Man is man because he has no instincts, because everything he is and has become he has learned from his culture, from the man-made part of the environment, from other human beings." — Ashley Montagu, 1973

"Every aspect of life has a biological foundation in exactly the same sense, which is that unless it was biologically possible it wouldn't exist. After that it's up for grabs." — Louis Menand

"[Music is] auditory cheesecake, something we like to gorge ourselves on that have no — has no adaptive advantage." — Steven Pinker (paraphrased by Bloom)

"That perfection of structure that justly excites our imagination." — Darwin (on adaptations)

"Are those the humans that you find the hottest? No. They're your siblings and they are not hot."

Assigned Readings / References

- **Gray textbook and Norton readings** — recommended for background on evolution.
- **Francis Crick** — *The Astonishing Hypothesis*.
- **Daniel Dennett** — "Darwin's dangerous idea."
- **William Paley** — argument from design (watch and eye analogy).
- **Charles Darwin** — *The Origin of Species*.
- **Richard Dawkins** — *The Blind Watchmaker*.
- **Steven Pinker** — *The Language Instinct*.

- **William James** — quote on eating.
- **Ashley Montagu** (1973) — quoted on human instincts.
- **Louis Menand** — *New Yorker* article.
- **Margie Profet** — MacArthur Genius Award work on pregnancy sickness.
- **Peter Salovey** — prior lecture on sex and love ("big three").
- **Laurie Santos** — capuchin monkey studies on framing effects.
- **Daniel Kahneman & Amos Tversky** — heuristics and biases; Nobel Prize work.
- **Leda Cosmides** — ecologically valid Wason selection task (UC Santa Barbara).
- **Spelke and Baillargeon** — referenced for infant hard-wiring research.
- Recent paper in **Nature** (five days before lecture) on cues for kin detection.
- Reading response: an exercise in "reverse engineering and evolutionary psychology."

PSYC 110 - Lecture 11 - Evolution, Emotion, and Reason: Emotions, Part I

Overview

This lecture, delivered by Professor Paul Bloom, applies the evolutionary perspective introduced previously to the study of emotion. Bloom argues that emotions, which seem natural and self-evident, must be examined scientifically because they emerge from contingent aspects of our biological nature, not from logical necessity. He attacks the "Star Trek" view that emotions are detrimental to rational functioning, showing through cases like Phineas Gage and Damasio's patient Elliot that without emotions people cannot set goals or prioritize. The lecture surveys facial expressions (with a focus on smiles), the non-social emotion of fear, and begins an analysis of social emotions and altruism grounded in evolutionary biology and the "selfish gene" perspective. Bloom previews how kin-directed feelings (cuteness responses, distress calls, attachment) can be explained by genetic relatedness, ending with an introduction to Harlow's "wire mother" and "cloth mother" experiments.

Key Concepts & Definitions

- **Emotions:** Mechanisms that set goals and establish priorities; without them no action would occur.
- **Phineas Gage:** 1848 construction foreman whose frontal cortex was damaged by an iron rod, illustrating loss of emotional/character function.
- **Duchenne smile:** A real happiness smile, named after a neurophysiologist; involves movement around the eyes; only about one in ten people can fake it.
- **Pan Am smile (greeting smile):** A smile to communicate hello; mouth moves but eyes do not.
- **Coy smile (appeasement smile):** A specialized smile of embarrassment/stress involving gaze aversion; appears in other primates and human infants.
- **Emotional contagion:** The phenomenon by which one's face molds in accord with the emotional facial expressions of another.
- **Pre-potent stimuli:** Things (e.g., snakes, spiders) that naturally elicit fear without learning.
- **Altruism:** The biologists' term for kindness/generosity.
- **Selfish gene theory** (Richard Dawkins): Animals are the vehicles through which genes reproduce; "an animal is just the gene's way of creating another animal."

- **Cupboard Theory** (B.F. Skinner): Babies attach to parents because parents provide food (operant conditioning).
- **Bowlby's alternative**: Babies are drawn to mother for comfort and social interaction, plus fear of strangers.
- **Wire mothers / Cloth mothers** (Harlow): Experimental surrogates—wire mothers provide food via nipple; cloth mothers provide warmth and comfort.
- **Baby-face bias**: Tendency to perceive adults with baby-like features as naive, helpless, kind, and warm.

Detailed Notes

Chapter 1. The Different Functions of Emotions

- Evolutionary psychology forces scientific scrutiny of things we take for granted.
- William James quotation (cited verbatim in transcript): asks why we smile when pleased, why we can't talk to crowds as to a single friend, why a particular maiden turns our wits upside down; notes the lion finds the lioness made to be loved, the broody hen finds eggs fascinating.
- Bloom notes the passage is (a) "incredibly sexist," (b) "beautifully written" and (c) makes the terrific point that these natural-seeming facts emerge from contingent biological nature.
- Examples of questions psychology must ask:
 - Why does poop smell bad? "To a dung beetle poop smells just fine."
 - Why does chocolate taste good?
 - Why do we love our children?
 - Why do we get angry when slapped (rather than sleepy, nostalgic, or craving cold soup)?
 - Why do we feel good when someone does us a favor?
- Parallel to other sciences: Why do things fall down? Why is our flesh warm? Why does water turn solid when cold?
- **Wrong theory of emotions**: Illustrated by Star Trek — Mr. Spock (half Vulcan) and Data (android lacking emotion chip) are portrayed as competent because they lack emotions.
- **Steven Pinker** in *How the Mind Works* refutes this: Spock must have been driven by motives — intellectual curiosity, solidarity with allies, fear (to flee predators/Klingons), modesty (not walking naked), ambition (getting out of bed), sociability. Without emotions to drive us, "we would do nothing at all."

Chapter 2. Phineas Gage and the Loss of Emotional Capacity

- **Phineas Gage**, 1848, construction foreman; iron rod (13 pounds) propelled by explosion went under his eye, popped out the top of his head, landed about 100 feet away covered in blood and brains.
- Gage was briefly knocked unconscious, got up, was taken to hospital (stopped at a tavern for cider on the way), had an infection, had surgery.
- Outcome: not blind, not deaf, not aphasic, no paralysis, no retardation — but **lost his character**.
- Description from Damasio's *Descartes' Error*:

> He used to be a really responsible guy, a family man, very reliable, very trustworthy. But after the accident he was fitful, irreverent, indulging at times in the grossest profanity, manifesting but little deference for his fellows, impatient of restraint or advice, a child in his intellectual capacities and manifestations. He had the animal pleasures of a strong man. His foul language is so debased that women are advised not to stay long in his presence.

- Gage couldn't hold a job, lost his family, ended up in the circus traveling with his iron rod.
 - **Elliot** (Damasio's pseudonymous patient): tumor in frontal lobe removed along with part of frontal lobe.
- Not blind, deaf, retarded, or profane.
 - Lost ability to prioritize and set goals.
 - Quoted description: would read and fully understand significance of work material but suddenly turn to something else and spend an entire afternoon deliberating which principle of categorization to apply to files (date, size, pertinence, etc.).
 - Couldn't keep a job or deal with people.
 - Take-home: "Emotions set goals and establish priorities. And without them you wouldn't do anything, you couldn't do anything."
 - Themes for the unit:
 - Emotions as mechanisms for goal-setting/prioritization.
 - Universals vs. cultural differences (America vs. Japan, American South vs. North differ in emotional triggers/baselines).
 - Darwin's point: emotions have universal roots shared across humans and many animals.
 - Agenda: (1) facial expressions, especially smiling; (2) fear as a nonsocial emotion; (3) feelings toward kin; (4) feelings toward non-kin (next class).

Chapter 3. Facial Expressions and Smiles in Particular

- **Paul Ekman:** leading scholar of facial expressions; trains police and Secret Service to detect honesty/dishonesty; profiled by Malcolm Gladwell in *The New Yorker*.
 - **Anger face demo:** lower brows and draw them together; tense lower and upper eyelids; stare (eyes can bulge); press lips together with corners straight or down.
 - Universal facial-expression databases now exist across genders and national origins (example shown: Japanese women's facial expressions). Subtle cross-cultural differences exist alongside deep universals.
 - **Smile demo:** raise corners of lips back and up; raise cheeks; raise lower eyelids.
 - Yale experts on smiling:
- **Angus Trumble**, curator at the British Art Gallery, wrote *A Brief History of the Smile* (smiles in art).
 - **Marianne LaFrance**, Bloom's colleague, studies smiling in adults, children, across cultures, and its social uses.

Findings about smiles:

- Smiles are universal — young children smile; even children blind from birth smile appropriately (so smiling is not learned by watching faces).
 - Smiles are not uniquely human; nonhuman primates smile.
 - Smiles are social signals: people smile when they want to **communicate** happiness, not merely when happy.
- **Bowler studies:** bowlers who get a strike don't smile after knocking down pins; they turn around to friends and grin.
 - **Olympic gold medal winners:** don't grin on the podium; smile when they face the crowd.
 - Bloom's aside: ask whether people smile during sex — discoverable with a partner or a mirror.
- **Types of smiles:**
- **Pan Am smile (greeting smile):** named for the now-defunct airline whose stewardesses produced big fake smiles; communicates "hello."
 - **Duchenne smile (real happiness):** involves the eyes; only ~1 in 10 people can fake one.
 - Politicians often use Pan Am smiles, especially when attacked/ridiculed.
 - Anecdote: **John Boehner** vs. **Roy Blunt** — a few years ago, Republican John Boehner won the House majority leader race; a photo of Roy Blunt stepping out after losing showed an unhappy expression vs. a real smile.
- Real psychological validity of the distinction:

- **Ten-month-olds** give a real happiness smile when mother approaches, a greeting smile for strangers.
- **John Gottman** studies married couples by film clip:
 - Death knell of marriage is **contempt** — not fighting, screaming, or hate.
 - True happiness smile when partner enters lab bodes better than a Pan Am smile.
- **College yearbook photo studies**: a reliable correlation between current happiness and the type of smile in a yearbook photo from 30 years ago.
 - **Coy smile / appeasement smile**: third type; smile of embarrassment or stress; gaze averted; head turned; appears when wanting to be liked under high-stress/risky social situations.
- Shown in other primates: a rhesus monkey bites her own infant; the infant screams then gives a submissive coy smile.
- Shown in human infant clip: gaze locked, brows raise, smile starts, smile widens, head turns and averts.

Chapter 4. Question and Answer on Smiles

- **Q**: Do nonhuman primates give the same kinds of smiles humans do (Pan Am vs. genuine)? **A**: Bloom doesn't know; will check. There is evidence the coy smile appears in nonhuman primates.
- **Q**: Why are some people's smiles "better" than others? **A**: Trivially, some people are better looking; more deeply, some people are better at producing cues of genuine happiness.
- **Emotional contagion demo**: students paired off, designated smiler smiles at partner; difficult not to smile back. Holds for virtually every other emotion — facing someone in rage molds your own face.
- **Q**: Difference between teeth-showing vs. closed-lip smiles? **A**: Probably exists; not a main smile difference but likely shows regional/cross-country variation (e.g., England vs. US). Every culture has Pan Am, happiness, and coy smiles, but variants depend on upbringing/culture.

Chapter 5. Non-social Emotions: Fear

- Why study fear:
 - It is basic and universal in humans and most nonhuman species.
 - Connects back to behaviorism/classical conditioning.
 - Nonsocial (you can fear a cliff fall, not just a person).

- Has a distinctive, universally recognizable facial expression — illustrated by photo of the detective standing as **Jack Ruby assassinated Lee Harvey Oswald** (mixture of fear and anger).

- **What we are afraid of** (universal fear elicitors): spiders, snakes, heights, storms, large animals, darkness, blood, strangers, humiliation, deep water, leaving home alone.
- **What we are NOT very afraid of**: guns, cars, electrical outlets.
- Explanation: these elicitors were dangerous in our ancestral environment; chimpanzees develop phobias of similar things (spiders, snakes).
- **Urban Chicago study**: inner-city children asked what they fear most — top fears were snakes and spiders, even though many had never seen them outside a zoo. "These are natural fears."
- **Judy DeLoache** (University of Virginia): studies babies' fears of spiders and snakes using indirect methods (cannot ethically attempt to construct phobias). Finds these are **pre-potent stimuli**.

Chapter 6. Social Emotions and Altruism

- Two categories of social emotions: those toward **kin** (genetic relatives) and those toward **non-kin** (people you interact with but aren't related to).
 - Focus on emotions generating kind/altruistic behavior.
 - "Altruism" = biologists' term for kindness/generosity.
 - **Wrong old view**: animals are merely survival machines; under that view altruism is mysterious.
 - **Simple counterexample**: imagine Gene A (animal cares for offspring) vs. Gene B (animal cares only for itself). Gene A wins out across generations. The animal that eats its offspring is a "biological dead end." **What matters is reproduction, not survival per se.**
 - **Selfish gene perspective** (Richard Dawkins): animals are the vehicles through which genes reproduce.
- **Cold virus**: you sneeze not just to expel germs; the cold virus uses your body as a tool to replicate. "A person is just a germ's way of making other germs."
- **Toxoplasmosis**: parasite in rats, passed when cats eat rats and excrete it; rewires rat brain to be less afraid of cats — an adaptive strategy.
- Hypothetical maximally powerful virus would skip the respiratory system and take over the brain to make people seek sex and kiss on the mouth; some evidence that **sypphilis** arouses libido for this reason.
- A virus that made hosts run out and bite other animals would be called **rabies**.
- **Implication for altruism**: selfish genes lead to altruistic animals because, at the gene level, there is no hard and fast distinction between your body and someone else's.

- **Haldane:** Asked if he'd lay down his life for his brother — "No, but I would gladly give my life for three brothers or five nephews or nine first cousins."

- Joke that captures the ultimate causation: genes propagate at higher rate if a body sacrifices itself for sufficient genetic relatives. Three brothers = 50% more genes replicated.

- Selfish gene theory, often seen as cold-blooded, provides a scientific basis for **real, genuine altruism**.

- **Lion takeover example:** New male lion taking over a pride kills remaining cubs; lionesses undergo spontaneous abortions.

- Cubs are genetic competition; only by removing them can the new male reproduce.

- Females' spontaneous abortion is adaptive: existing cubs would not survive anyway, so best strategy is to start anew.

Mechanisms of kin care (especially in birds and mammals):

- Birds and mammals invest in **quality, not quantity** (unlike fish/reptiles); few offspring, big psychological investment.
- Offspring are vulnerable for long periods and need resources.

Distress calls (baby cry example played):

- Small animals chirp, mew, bleat, cry.
- "Delicate high-wire act" — annoying enough to elicit help but not so annoying that caretakers kill the infant.
- Cry is annoying because our brains are wired to respond, not because of volume or pitch.

Cuteness:

- Babies are cute because of how our brains are wired, not because of "metaphysical" cuteness; Martians wouldn't say "Oh, cute baby."
- Cute features (mammalian universal): big protruding foreheads, upturned little nose, chubby cheeks, big eyes.
- **Stephen Jay Gould** essay: evolution of **Mickey Mouse** — started as an ugly little rodent, became cuter as artists converged on baby-like features.
- **Baby-face bias:** same studies in US and Asia; baby-faced adults (example: **Leonardo DiCaprio**) judged as naive, helpless, kind, warm. In mock trials, baby-faced individuals more likely to be found innocent than non-baby-faced (example: **Ben Affleck**).
- Class poll (men only) on attractiveness between baby-faced vs. testosterone-faced; women's preferences are more complicated and depend on menstrual cycle phase (to be discussed in sex lecture).

Babies' responses to caregivers:

- **Cupboard Theory** (B.F. Skinner): attachment is operantly conditioned via food (breast milk or bottle).
- **Bowlby**: babies seek mother for comfort and social interaction, plus fear of strangers.
- **Harlow's experiments** with nonhuman primates:
 - **Wire mothers**: built with a nipple providing food.
 - **Cloth mothers**: provide no food but provide warmth and comfort.
- Aside about Yale psychology department: one professor was nicknamed "the cloth mother" (supportive, not warm); another, productive but loveless, was called "the wire mother."
- Bloom warns the third Harlow film is an example of why this research is not currently done but illustrates an important scientific point.
- Lecture ends with the promise to wrap up Harlow and move to altruism toward non-kin next class.

Notable Quotes

To the psychologist alone can such questions occur as: Why do we smile when pleased and not scowl? Why are we unable to talk to a crowd as we talk to a single friend? Why does a particular maiden turn our wits upside down?

The stinkiness of poop is not an irreducible fact about the universe. Rather, the stinkiness of poop is a fact about human psychology. To a dung beetle poop smells just fine.

Without emotions to drive us we would do nothing at all.

He used to be a really responsible guy, a family man, very reliable, very trustworthy. But after the accident he was fitful, irreverent, indulging at times in the grossest profanity... His foul language is so debased that women are advised not to stay long in his presence.

Emotions set goals and establish priorities. And without them you wouldn't do anything, you couldn't do anything.

No, but I would gladly give my life for three brothers or five nephews or nine first cousins.

A person is just a germ's way of making other germs.

The death knell of a marriage is contempt.

Assigned Readings / References

References mentioned within the lecture (not listed as a separate assignment block in the transcript):

- William James — quoted passage on why we smile, talk to crowds, love the maiden, etc.
- Steven Pinker — *How the Mind Works* (Star Trek / Spock example).
- Antonio Damasio — *Descartes' Error* (Phineas Gage description; case of Elliot).
- Paul Ekman — facial expression research; New Yorker profile by Malcolm Gladwell.
- Angus Trumble — *A Brief History of the Smile*.
- Marianne LaFrance — research on smiling.
- John Gottman — film-clip studies of married couples; finding on contempt.
- Judy DeLoache (University of Virginia) — babies' fears of spiders and snakes.
- Richard Dawkins — selfish gene theory; animals as vehicles for genes.
- J.B.S. Haldane — "three brothers or five nephews or nine first cousins" quote.
- Stephen Jay Gould — essay on the evolution of Mickey Mouse.
- B.F. Skinner — Cupboard Theory of attachment.
- John Bowlby — alternative attachment theory.
- Harry Harlow — wire mother / cloth mother experiments.

PSYC 110 - Lecture 12 - Evolution, Emotion, and Reason: Emotions, Part II

Overview

This lecture continues the two-part treatment of emotions, moving from emotions toward kin to social emotions toward non-kin. Professor Bloom uses the exotic (Emperor penguins, vampire bats) to motivate scientific study of things that feel "natural" (like parental love), echoing William James. He explains how reciprocal altruism evolves only when animals can detect and punish cheaters, and uses The Prisoner's Dilemma and The Ultimatum Game to demonstrate how human emotions correspond to permutations of cooperation, defection, and fairness. The lecture closes with a discussion of "cultures of honor" — arguing that mild irrationality (like a hair-trigger temper) can be adaptive, and that this plays out in cultural differences such as those between the American South and North. The overarching claim is that emotions are not noise but exquisitely complicated motivational systems shaped by evolution and modified by culture.

Key Concepts & Definitions

- **Appeasement smile** — In nonhuman primates, smiles correspond almost entirely to "don't hurt me" signals, equivalent to the human "coy smile."
- **Imprinting (debunked version)** — The discredited idea that a baby must see its mother at a "magical moment" right after birth or attachment will fail.
- **Cupboard Theory** — Skinnerian/operant conditioning account: babies love mothers because mothers provide food (law of effect).
- **Bowlby's theory** — A nativist, hard-wired account positing two things: a draw to mom for comfort and social interaction, plus fear of strangers.
- **Reciprocal altruism** — Good behavior toward another animal predicated on the expectation of reciprocation; benefits outweigh costs when animals cooperate.
- **Cheaters / Free-riders** — Animals (or agents) who take the benefits of cooperation without paying the costs.
- **Cheater detection** — The mental apparatus (recognize, remember, motivation to punish) required for reciprocal altruism to be an evolutionarily stable strategy.
- **The Prisoner's Dilemma** — A game in which the dominant strategy is to defect regardless of the other player's choice, but mutual defection leaves both worse off than mutual cooperation.

- **Tit-for-Tat** — Anatol Rappaport's winning 4-line program: cooperate on first encounter, then mirror the other player's previous move. Features: starts friendly, not a sucker, forgiving, transparent.
- **The Ultimatum Game** — Player A offers a split of money to Player B; B can accept (both keep proposed amounts) or reject (neither gets anything).
- **Adaptive irrationality** — Having a reputation for irrational responses (e.g., a temper, head-over-heels love) can give an evolutionary edge by deterring exploitation.
- **Culture of honor** — A culture where law is unreliable and resources are easily taken, making a reputation for violent retaliation important.

Detailed Notes

Chapter 1. Emotional Responses to Caregivers and Kin [00:00:00]

- **Follow-up on nonhuman primate smiles:** Bloom contacted experts. Answer: nonhuman primates (chimps, gorillas, gibbons) have smiles that correspond almost entirely to appeasement smiles ("don't hurt me"), equivalent to the human coy smile.
- No equivalent to the **greeting smile** or **Pan Am smile**.
- No equivalent to the **Duchenne smile** (genuine expression of happiness).
- **Review of course arc:** functions of emotions → smiling/facial expressions → nonsocial emotion (fear) → social emotions toward kin → evolutionary basis.
- **Quality vs. quantity in offspring:** Humans/birds/mammals produce few children and invest heavily; long period of dependence creates deep parent–child bonds.

Using the exotic to study the familiar

- Analogy from a friend who studies psychology of religion: when speaking to non-specialists, start with exotic examples (e.g., a culture where men put butter on their heads in summer — Dan Sperber's example, or cultures believing trees can talk) to motivate study of practices in our own culture.
- This connects to a William James point: things natural to us feel obvious; equally odd things are natural to other species.
- Applied to **parental love**: don't look head-on; look at other species.

"March of the Penguins" Example

- Emperor penguins; the documentary was the second most popular documentary of all time, beaten only by "Fahrenheit 9/11."
- Elaborate, precarious system: long trek from water to breeding grounds (protected from wind, firm ice).
- Responses to the film:
 - Conservatives: celebration of family values (love, trust, monogamy).
 - Liberals: "monogamous for one breeding season... if you add it up, it's pretty slutty."
 - Intelligent design / creationist take: example of God's intricate design.
 - Darwinian take: agrees it couldn't be accidental — it's an exquisite biological adaptation for parental care, shaped because children share parents' genes.

How children attach to caregivers

- Babies develop attachment to whoever is closest — typically the mother (prefer her voice, face, smell).
- **Imprinting myth**: The idea of a magical post-birth moment is silly; attachment forms "in the fullness of time."
- **Two competing theories**:
 - **Cupboard Theory (Skinnerian)**: attachment via operant conditioning — mother provides food.
 - **Bowlby's nativist theory**: hard-wired draw to mom for comfort/social interaction + fear of strangers.
 - **Harlow's experiments** (reviewed from prior lecture):
 - Two surrogate mothers: wire mother (food/Skinnerian) vs. cloth mother (warmth, cuddling).
 - Babies went to wire mother to eat ("You've got to eat to live") but loved and attached to the cloth mother — used her as a base when threatened and to explore from.

Effects of no contact

- Real-world relevance: debates about daycare.
- Short answer on daycare: nobody really knows; differences (if any) are subtle and not big/dramatic.
- **Harlow's solitary confinement monkeys**: raised in steel cages with only wire mother — got nutrition but no mothering.
- Outcomes: withdrawn, don't play, bite themselves, incompetent sexually, socially, and maternally.

- One artificially inseminated solitary-raised monkey banged her infant's head on the floor and bit it to death.

- **Human natural experiments:** children in harsh orphanages with little social contact develop severe social/emotional problems — either insatiable need for cuddling or apathetic.

- **Reversibility:**

- "Monkey therapists": a younger goofing-around monkey clings to and follows the solitary one, producing gradual improvement.

- Human anecdote: children at age 1.5 taken from a harsh orphanage to a home for mentally retarded women who cuddled them — apparently brought them back to normal.

Chapter 2. Question and Answer on Emotions towards Kin [00:15:44]

- **Q: Do children in orphanages comfort each other?**

- Bloom: probably not in practice because they tend to be very young babies not thrown together that way. TA confirms there is evidence that among the young, peer support can help both monkeys and children.

- **Q: What about the middle ground (parent comforts only a little)?**

- No good evidence that mild under-affection has dramatic effects.

- Parents who aren't affectionate tend to have kids who aren't affectionate, but it's unclear if this is genetic or environmental.

- **Take-away:** outside extreme conditions, effects are not big and dramatic.

Chapter 3. Evolutionary Explanations for Emotional Responses [00:18:16]

- Kin attraction is not a big evolutionary puzzle (gene reproduction logic).

- The puzzle: animals (including humans) have complicated, nice relationships with **non-kin**.

- **Examples of niceness to non-kin:**

- **Grooming** — picking lice off each other.

- **Warning cries** — risky to the caller, often protecting non-relatives.

- **Shared childcare.**

- **Food sharing** — vampire bats: a successful hunter returns to the cave and vomits blood into other bats' mouths.

- **Reciprocal altruism:** behavior is predicated on reciprocation; the group does better than each working alone.

The cheater problem

- Two hypothetical vampire-bat genes:
 - Gene A: accepts blood and shares blood.
 - Gene B: accepts blood but doesn't share.
 - Gene B outproduces A in the long run (healthier, more offspring) — cooperation collapses.
- **Warning-cry version:** adaptive to **respond** to warning cries; not adaptive to **give** them.
- **Bar analogy:** it's good to accept drinks but not to buy rounds — if everyone defects, "buying a round" fades.

Cheater detection as solution

- Reciprocal altruism evolves only if animals are wired to **recognize, remember, and punish cheaters**.
- Vampire bat experiment: scientists prevent a successful bat from sharing. Result: when other bats strike it big, they **starve the selfish bat**. Confirmed cheater detection in bats.
- Animals (and humans) are sensitive to cheating; this plays a powerful role in evolution of social behavior and social emotions.

Chapter 4. Cooperative Behavior and the Prisoner's Dilemma [00:25:20]

- A core construct across cognitive science, psychology, economics.
- **Setup:** You and a friend rob a bank; you're caught; police interrogate separately. Each can cooperate (stay silent) or defect (squeal).
- **Payoff logic:**
 - Best individual outcome: defect while other cooperates (walk free).
 - Worst: cooperate while other defects (life in prison).
 - Best joint outcome: both cooperate (minor sentence).
 - Worst joint outcome: both defect.
- **Key structural feature:** regardless of opponent's move, defecting is individually best — but mutual defection makes both worse off.

Other examples

- **Divorce lawyers:** both better off with mediator, but each tempted to hire a lawyer; both end up with lawyers and worse off.

- **Nuclear disarmament:** country A vs. B; mutual disarm is best, but each tempted to arm.
- **Drug deal ("reefer"):** \$1,000 for "a ton" of reefer behind the gym at 2 a.m.; each tempted to bring a gun; both show up armed.

In-class one-shot game

- Students played with cards reading "cooperate"/"defect."
- Payoff: cooperate/cooperate = \$3 each; cooperate vs. defect = defector gets \$5, cooperator gets nothing.
- Then five repeated games. Bloom joked the high scorer (~\$20–25) won by exploiting cooperator partners — "really a measure of honesty."

Robert Axelrod's tournament

- About 20 years ago; **63 competitors** submitted programs.
 - Programs ranged from "always cooperate"/"always defect" to elegant genetic algorithms.
 - **Winner: "Tit-for-Tat" by Anatol Rappaport** (who died about a month before this lecture).
 - **4 lines of basic code:**
- First encounter: cooperate.
 - Afterward: do what the other program did on the previous trial.
- **Why it won — four features:**
- Starts friendly.
 - Not a sucker (defects back when crossed).
 - Forgiving (returns to cooperation if other does).
 - Transparent (other programs can tell it's not a sucker and is forgiving).

Emotions mapped onto Prisoner's Dilemma cells

- We **like** people who cooperate with us → motivates future niceness.
- We feel **anger and distrust** toward those who betray us → motivates betrayal/avoidance.
- We feel **bad/guilt** when we betray cooperators → motivates better future behavior.

Bloom's home experiment

- With his 7- and 10-year-old sons, chocolate chips as stakes.
- Older boy tricked younger: "Okay, let's both cooperate" then defected.
- Younger boy reacted with **rage** (not guilt on the older boy's side).

Chapter 5. The Ultimatum Game, Rationality and Irrationality [00:40:44]

The game

- A has \$10. A offers B any integer amount \$1–\$10. B accepts (deal goes through) or rejects (nobody gets anything).
- Class played one round.

Rationality analysis

- **Rational play:** $\$1 > \0 , so B should always accept any positive offer; A should offer \$1.
- **Actual play:** few offered \$1 because **people are not purely rational**; even in one-shot games people **reject unfair distributions out of spite**.
- Neuro-economic evidence: people offered \$1 get really angry.

Moral: rationality is exploitable

- A rational person is easily exploited; their responses to provocation are measured and predictable.
- A person with a temper / reputation for irrationality is **treated better** — others avoid messing with them.
- Mild irrationality can be **adaptively beneficial**.

Application to love

- Rational mate ("I'll stay as long as you're the most attractive available") is less trustworthy than a partner who is **head over heels in love** — the very irrationality signals reliability.

Application to violence (Daly and Wilson)

- Most murder is **not** caused by reasonable provocation — it's caused by insult, curse, petty infraction.
- This is **adaptive irrationality**, not crazy irrationality.
- Daly and Wilson: "in chronically feuding and warring societies an essential manly virtue is the capacity for violence. To turn the other cheek is not saintly, but stupid or contemptibly weak."
- Modern New York Times example: violence often due to disrespect / dirty looks — rational in environments with repeated interaction and weak police presence.

Chapter 6. Cultures of Honor [00:49:21]

Defining properties of a culture of honor

- Cannot rely on the law.
- Resources are easily taken.
- → Reputation for **violent retaliation** becomes important.

Examples

- Scottish highlanders
- Masai warriors
- Bedouin tradesmen
- Western cowboys
- **American South** — most studied by modern psychologists; settled by herdsmen, traditionally less centralized legal control.

Documented differences (South vs. North, US)

- More permissive gun laws in the South.
- More approval of corporal and capital punishment.
- More positive attitudes toward the military.
- In questionnaires, more forgiving of honor-based violence ("Somebody insults my woman and I punch him in the face").
- Streets not generally more dangerous overall, but **higher rate of honor-related crimes** (e.g., shooting an intruder, killing someone for insults).

Nisbett and Cohen's experiment

- Conducted at University of Michigan.
 - Sample: white males, **not Hispanic and not Jewish** (homogeneous sample; culture of honor seen as male phenomenon).
 - **Provocation**: subject walks down a hallway; a male grad student holding files bumps him, calls him "Asshole," and keeps walking.
- The grad student survived hundreds of these encounters — no fights or shootings.
- **Findings**:
- Southern males showed **higher hormone response** — increases in **testosterone and cortisol**.

- Differences in later behavior and fill-in-the-blank responses: northerners completed "John went to the store and bought a ___" with neutral words (e.g., "an apple"); southerners more aggressive (Bloom's joke: "an AK-47 to kill that freaking graduate student").

- Southerners were not more violent overall — they were **more sensitive to honor provocations**.

Caveats Bloom offers

- These are **average differences**; not every northerner or southerner fits.
- A southern student previously felt the discussion was unfair to southerners.
- Nisbett (himself a southerner) found northerners "astonishingly rude" — possibly because no culture of honor restrains them.
- Culture-of-honor virtues — **honor, loyalty, courage, self-reliance** — are not necessarily bad.

Closing synthesis

- Emotions (fear, love for children, anger, gratitude) are not noise — they're complex motivational systems crafted by evolution to handle natural and social environments.
- An evolutionary approach is required to see this clearly.
- Bloom closes with D'Arcy Thompson: "everything is the way it is because it got that way."
- Reading response assigned; exam Wednesday.

Notable Quotes

"Some liberals, who hate everything that's good and true, responded by saying, 'Well, yeah, they're monogamous for one breeding season. It's a year. Then they go and find another mate. If you add it up, it's pretty slutty.'"

"You've got to eat to live."

"In one case, one of these monkeys raised in solitary confinement was artificially inseminated. When she had a child she banged its head on the floor and then bit it to death."

"If everybody fell to that solution, the idea of buying a round would fade."

"The first time you meet a new program, cooperate... After that, do on each trial what the other program did on the previous trial. This beat sixty-two others."

"A rational person is easily exploited."

"in chronically feuding and warring societies an essential manual — manly virtue is the capacity for violence. To turn the other cheek is not saintly, but stupid or contemptibly weak."

"everything is the way it is because it got that way."

Assigned Readings / References

- Reading response for the week was assigned (specific reading not named in transcript beyond the D'Arcy Thompson quote shown on a slide).
- Exam on Wednesday.

Named figures / works referenced in lecture

- Paul Bloom (lecturer)
- William James
- Dan Sperber (butter-on-heads example)
- "March of the Penguins" (documentary)
- "Fahrenheit 9/11"
- B.F. Skinner (operant conditioning / Cupboard Theory)
- John Bowlby (nativist attachment theory)
- Henry Harlow (wire vs. cloth mother experiments; solitary confinement studies)
- Robert Axelrod (Prisoner's Dilemma tournament)
- Anatol Rappaport (Tit-for-Tat)
- Daly and Wilson (homicide and adaptive irrationality)
- Richard Nisbett and Dov Cohen (culture of honor study; transcript also mentions "Nisbett and Wilson" at one point)
- D'Arcy Thompson (closing quotation)

PSYC 110 - Lecture 13 - Why Are People Different?: Differences

Overview

After many lectures on human universals (language, rationality, perception, emotions, development), Professor Paul Bloom turns to the question of why people differ from one another psychologically. He frames the lecture around two main interesting factors at the root of all human differences: personality and intelligence — first asking how to characterize them and then asking why these differences exist. He warns that this is the class likely to bother students the most because the scientific findings on psychological differences are, to many, "shocking and unbelievable." After introducing measurement concepts (reliability and validity), personality frameworks (Eysenck, Cattell, the Big Five), and the structure of IQ ("G" and "S"), he turns to behavioral genetics — concluding with two surprising findings: (1) high heritability for almost everything, and (2) almost everything that's not genetic is due to non-shared environment, meaning shared environment (i.e., parents) counts for little or nothing once genes are controlled for.

Key Concepts & Definitions

- **Personality:** A stable trait across situations and time — how someone deals with the world and with other people. Not a momentary reaction to a situation, but something carried around.
- **Reliability:** A test has no measurement error; testing the same person at different times yields the same result.
- **Validity:** The test measures what it's supposed to measure.
- **Rorschach inkblot test:** Personality test invented by Herman Rorschach using inkblots; widely used but, per Bloom, "transcendently useless."
- **The Big Five (OCEAN):** Openness, Conscientiousness, Extroversion, Agreeableness, Neuroticism — five fundamental dimensions of personality.
- **Intelligence:** Hard to define, but core components named by experts include abstract reasoning, problem solving, and the capacity to acquire knowledge.
- **"G" (general intelligence):** A general intellectual capacity people bring to all tasks in common.
- **"S" (specific intelligence):** Ability on specific tests/subtests within an IQ battery.
- **Wechsler test:** Modern IQ test (for adults and children) where 100 is set as the average score.

- **Heredity / heritability:** The proportion of variation in a trait (across people) due to genetic differences — a claim about differences, not about the origin of any specific trait in one individual.
- **Shared environment:** Environmental influences shared by people raised in the same household (e.g., the same parents).
- **Non-shared environment:** Everything else not heredity and not shared environment — a "garbage can" category.
- **Monozygotic (MZ) twins:** Genetic clones; share 100% of their genes.
- **Dizygotic (DZ) twins:** Share 50% of genes; like regular siblings.
- **Adopted siblings:** Zero percent genetic overlap beyond randomness.
- **Flynn effect:** People have been getting smarter over generations; IQ tests hide this because 100 is always reset as the average.

Detailed Notes

Chapter 1. Personality, Intelligence and Determining Difference [00:00:00]

- Prior lectures focused on universals; this lecture shifts to differences.
- Plan: (1) how people are different (theories of difference) and (2) why people are different.
- Bloom warns: "the scientific findings on human psychological differences are, to many of us, shocking and unbelievable."
- Examples of ways people differ:
 - **Sexual identity** — male/female; central to most people's being; indexed in pronouns.
 - **Sexual orientation** — who you're attracted to (deferred to post-spring-break lecture).
 - **Happiness** — rank 1–10 differs across the room; deferred to last class of semester.
 - **Success and failure in life** — relationships, job satisfaction, criminal records.
- Two main factors at the root of human differences: **personality** and **intelligence**.

Personality characterized:

- A person's style in dealing with the world and with other people.
- Example: Homer (impulsive, irresponsible, sometimes lazy, good-hearted) vs. Bloom's colleague (responsible, reliable, very kind).
- Must be a **stable trait across situations and time** — not a one-off reaction to being smacked in the head, but how you walk around all the time.

Reliability vs. validity:

- **Reliability:** no measurement error; same result across testings. Example: a bathroom scale that gives the same number is reliable.
- **Validity:** the test measures what it claims. Example: predicting intelligence from birth date (Bloom was born Christmas Eve) — perfectly reliable but absolutely not valid.
- Something can be reliable but not valid, and vice versa.

Personality tests in the wild:

- Web-based tests are common. Bloom took "which super hero are you?" and got Batman ("you are dark, love gadgets, and have vowed to help the innocent not suffer the pain you have endured"). Earlier attempts gave him The Incredible Hulk, then Wonder Woman; eventually he tailored answers to get Batman — undermining both reliability and validity.
- The same site also offers a "hot or not" test.

The Rorschach inkblot test:

- Invented by **Herman Rorschach**, whose teenage nickname was "Inkblot."
- Originally for psychiatric cases, now extremely common.
- ~80% of clinical psychologists claim to use it; most APA-accredited graduate programs teach it; Catholic seminaries use it for seminary applicants.
- Used in child custody cases.
- Right/wrong answers exist: "it is important to see the blot as two human figures, usually females or clowns." Failing to do so is "a sign you have problems relating to people." Acceptable alternatives: "a cave entrance," "butterfly," or "vagina."
- Bloom's verdict: "transcendently useless... as useless as throwing dice... as useless as tea leaves."
- Inkblots and answers are available on the web.

Reducing personality dimensions:

- **Gordon Allport** went through the dictionary, found ~18,000 personality-related traits, but many overlapped (friendly/sociable/welcoming/warm-hearted tap the same thing).
- **Cattell** and others tried to narrow it down — how many parameters of personality difference are needed?
- **Eysenck:** initially two dimensions — introverted–extroverted, neurotic–stable → four types of people. Later added **psychoticism vs. non-psychoticism** (aggressive vs. empathetic) → eight types.
- **Cattell:** 16 personality factors.

Chapter 2. Measuring Personality with "The Big Five" [00:13:53]

- Modern consensus: 2–3 too few, 16 too many. The **Big Five** dimensions:
- **Neurotic vs. stable** (natty/worrying vs. calm)
- **Extrovert vs. introvert**
- **Open to experience vs. closed to experience**
- **Agreeable** (courteous, friendly) **vs. non-agreeable** (rude, selfish)
- **Conscientious vs. not conscientious** (careful/reliable vs. careless/undependable)
- Mnemonic: **OCEAN** — Openness, Conscientiousness, Extroversion, Agreeableness, Neuroticism.

Why take the Big Five seriously:

- **Reliability over time:** Stable across years; once you pass age 30, very stable indeed. Test parents now; in 10 years they'll still be there.
- **Agreement across observers:** Roommate, professor, and mom's ratings of you tend to overlap a lot. "Your personality leaves a trail in the minds of people around you."
- **Predicts real-world behavior:**
- **Conscientiousness** → faithfulness to spouse.
- **Openness** → likelihood of changing jobs.
- **Extroverts** look people in the eye more and have more sexual partners.

Simpsons study:

- 13 subjects rated Simpsons characters on the five dimensions; considerable agreement.
- Most **agreeable**: Ned Flanders (6.27) and Marge Simpson (5.46).
- Least agreeable: Mr. Burns and Nelson. (Nelson is "the little kid that when trouble happens he goes, 'Ha ha.'")

Chapter 3. Defining and Measuring Intelligence [00:19:47]

- No easy definition. A survey of 1,000 experts:
- Core consensus: **abstract reasoning, problem solving, capacity to acquire knowledge.**
- Other commonly mentioned: memory, mental speed, language, math, knowledge, creativity.
- Gut intuition: Homer (limited intelligence), Bloom's colleague (very smart), Einstein (really, really smart), Ralph Wiggum (particularly stupid).

Spearman's **G** and **S**:

- **S**: ability on specific subtests (math, reading, spatial — different scores on each).

- **G:** general intelligence brought to all tasks in common.
- Why G is needed: performance on multiple intellectual tasks is correlated — there's a sort of general correlation people bring to tasks.
- Casual usage: "I think he is high 'G'" means he's a smart guy.

Athletic analogy: A gym with running, basketball shooting, swimming, fencing, etc. Scores aren't independent — someone good at running and swimming is probably good at climbing. Likewise IQ has a general factor.

Modern IQ tests (Wechsler, adult and child):

- **100 is average** by definition (like grading the Midterm and giving the average 100).
 - Works on the **normal curve**:
- 68% score between 85 and 115.
 - 95% score between 70 and 130.
 - Above 145 IQ = 0.13% of the population.

The Bell Curve and the controversy:

- **Herrnstein and Murray**, *The Bell Curve*: argued IQ matters immensely for everyday life; status, wealth, success follow from IQ.
- Sparked huge controversy.
- **American Psychological Association** convened 50 leading researchers to write a report; another group of IQ researchers wrote a separate report. Both reports' links were on the slide.

Broad consensus on IQ's importance:

- IQ is "strongly related — probably more so than any other single measurable human trait — to many important educational, occupational, economic, and social outcomes."
- Strong correlation: success in school, success in military training.
- Moderate but robust: social competence.
- Smaller but consistent: law-abidingness.
- Conclusion: "whatever IQ tests measure it is of great practical and social importance."
- IQ matters for **social achievement, prestigious positions, on-the-job performance**.

Caveats:

- Some role of IQ may be a **self-fulfilling prophecy**: SAT to get into Yale, GRE for grad school, etc. Society could make height matter the same way (no one under six feet gets into Yale → height "predicts" educational success).
- But IQ is not entirely a social construction: IQ scores relate to **mental speed and memory span** (how fast you think, your memory abilities).

Chapter 4. Question and Answer on Personality and Intelligence [00:30:29]

- **Q (gaming the test):** A student took a personality test and was "accepted into Slytherin" (Hogwarts reference). How do you avoid gaming?
- **A:** Test constructors use **catch questions** (e.g., "I have never done anything I am ashamed of" — endorsement suggests a liar). Same question is asked in different ways across the 100 items; correlations expose liars. Proof is in the pudding: a test easily fooled wouldn't survive.
- **Q (Emotional IQ):** Is emotional/social intelligence a better predictor?
- **A:** Evidence for predictive power not as strong as standard IQ; **Peter Salovey** has done interesting research on it. Emotional intelligence is also related to "good, old-fashioned intelligence" — they pull together.
- **Q (what makes a good test):** Reliability + validity. Replicability over time plus relationship to real-world phenomena (grades, number of girlfriends, being liked). SAT is useful because it corresponds with other things like grades.
- **Q (stability of personality):** Personality does change somewhat — a ten-year-old's test only weakly relates to the same person at 50. But "if you're an extrovert now you'll likely be an extrovert twenty years from now." Same for IQ — might go up or down but not much, justifying it as a "more or less stable trait."

Chapter 5. The Roles of Genes and Environment in Explaining Human Differences [00:36:33]

- Two factors: **genes/heredity** and **environment/experience**.
- Crucial clarification: heritability is about the role of genes in explaining **differences between people**, not the proportion of an individual's trait due to genes.

Height example:

- Height is reasonably **heritable** — population differences are largely (not entirely) due to genes.

Number of legs example:

- Whether you have legs is highly genetic, but **variation in leg count (0, 1, 2) is not very heritable** — people with fewer legs typically lost them in accidents.
- Heredity is a claim about differences, not about the origin of any specific trait.

Two kinds of environment:

- **Shared environment:** things siblings in the same household share (e.g., having the same lousy parents).

- **Non-shared environment:** everything else — a snowball to the head at age five, winning the lottery at 21, even aliens from Pluto zapping your brain.
- Heredity + shared environment + non-shared environment = 1.

Behavioral genetics tools (you can't easily separate genes from environment in everyday families):

- **Monozygotic twins:** clones, 100% shared genes.
- **Dizygotic twins:** 50% shared genes; like regular siblings.
- **Adopted siblings:** 0% genetic overlap beyond randomness.
- Two people raised in the same house = 100% shared environment by definition.

Comparisons and inferences:

- MZ much more similar than DZ → large role of genes (caveat: MZ twins look more alike and may be treated more similarly).
- MZ no more similar than DZ → low role of heredity.
- Adopted children highly similar to their non-biological siblings → high role of shared environment.

- Hypothetical: if the 7 Bloom kids all have IQ 104 and adopted kids also end up at 104, there's a household effect.

- Adopted kids' IQ unrelated to biological Bloom children's → being raised by Bloom has no effect on IQ.

Identical twins reared apart — the gold standard:

- Cartoon: "Separated at Birth, the Mallifert Twins Meet Accidentally" (twins end up at a patent office with the same device).
- Astonishing similarities in: attitudes to death penalty, religion, modern music; rates of crime, gambling, divorce.
- Bizarre similarities: a pair who both pretended to sneeze in elevators; the "**Giggle Twins**" who both burst into giggles constantly.
- Brain scans of MZ twins reared apart often indistinguishable.

First surprising finding:

- **High heritability for almost everything:** intelligence, personality, happiness, religiosity, political orientation, sexual orientation.

Within-group vs. between-group differences:

- Clear, reliable IQ score differences exist between American racial groups: whites, Asians, African Americans, Ashkenazi Jews.

- Groups are partially socially constructed (e.g., whether a child of a Jewish man and a non-Jewish woman counts as Jewish depends on reform vs. orthodox).
- But genuine genetic differences exist across groups too (e.g., Ashkenazi Jews and Tay-Sachs vulnerability).
- **High heritability within a group does NOT entail genetic differences between groups.**

Richard Lewontin's seed analogy:

- Two plots of land planted with the same seeds; one fertilized a lot, the other a little.
- Within each plot, growth differences are largely genetic (high heritability).
- The difference **between** plots has no genetic cause — it's the fertilizer.

Bloom's two-Midterms thought experiment:

- One side of the class gets a "savagely hard" Midterm; the other gets "Which is bigger, a dog or an elephant?"
- Within-group variation could be genes or study habits; between-group difference is entirely due to the test, not genes.

Chapter 6. Genes, Environment and Intelligence [00:51:16]

Two textbook reasons group differences are largely environmental, not genetic:

1. IQ differences correspond better to **socially defined groups** (how people are treated/thought about) than to genetically defined groups.
2. IQ can differ radically without genetic differences — the Flynn effect.

The Flynn effect:

- People have been getting smarter; IQ tests hide this because 100 is always set as the current average.
- A 1980 average person taking the 1950 test would score about **120**.
- If you push today's average person back 20–30 years, they would do much better than average.
- No one knows why people are getting smarter; multiple theories. (Reading response: "Explain the Flynn effect.")
- Plotted with two lines (American and Dutch; Bloom isn't sure which is which).
- Implication: dramatic IQ change over decades without genetic change → likely environmental causes for group differences too.

Second surprising finding:

- **Almost everything that's not genetic is due to non-shared environment.**

- **Shared environment counts for little or nothing** for personality or intelligence.
- An adopted child is no more similar to his siblings than to a stranger.
- IQ correlation in genetically unrelated adults raised in the same family $\approx$ **zero**.
- "Suppose the Bloom family all has an IQ of 104 and we adopt a kid... What's his IQ? Answer? We have no idea because the IQ of the Bloom family who are unrelated to him has no effect at all."

Do parents matter?

- *Newsweek* cover captured the issue.
- Parents are shared environment. If shared environment doesn't affect intelligence/personality, then **parenting does not affect intelligence/personality**.
- Parents do have a huge effect — around **0.5** — but this is at the **moment of conception**. After that, very little role in shaping you.

Judith Harris, *The Nurture Assumption*:

- Subtitle: "Why Children Turn Out the Way They Do, Parents Matter Less than You Think and Peers Matter More."
- Harris was kicked out of Harvard graduate school and told she wouldn't amount to much. The chair who wrote the dismissal letter was **George Miller**.
- In **1997**, Harris won the **George Miller award**.

Philip Larkin's poem (her starting point of disagreement):

*They fuck you up, your mum and dad.
They may not mean to but they do.
They fill you with the faults they had
And add some extra just for you.*

Ending: "Man hands on misery to man. It deepens like a coastal shelf. Get out as early as you can and don't have any kids yourself."

Harris's rebuttal:

How sharper than a serpent's tooth to hear your child make such a fuss. It isn't fair. It's not the truth. He's fucked up, yes, but not by us.

- British psychoanalyst **Oliver James**, outraged, wrote a rebuttal book called *They Fuck You Up*.

Why parent–child correlations don't prove parenting effects:

- High correlation between parent and child for everything (bookish parents → bookish kids, religious parents → religious kids, "raised by Bonnie and Clyde, you will be a young thug," poor parents → poor children, brilliant parents → brilliant children).
- Three alternative explanations:
 1. Parents and children share genes.
 2. Children affect parents, not just vice versa.
 3. Some confound makes parents and kids both alike.

The "worst study ever done" — family meals and teens:

- Headline: family meals help teens avoid smoking, alcohol, drugs.
- Phone questionnaire of teens and parents.
- Bloom: "the worst study ever done in the history of science." Possible alternative explanations:
 - **Reverse causation:** a good kid sticks around for the family meal; a kid "smoking pot and cavorting with prostitutes" doesn't come home for dinner.
 - **Confounding family type:** good families have drug-free kids and family meals; bad families have stoned kids and no family meals.
 - **Age confound:** sample was ages 12–17. **Twelve-year-olds don't use a lot of drugs and eat with family; seventeen-year-olds "are stoned all the time and they don't eat with their family."**
- Same criticisms apply to "parents read to kids" studies — bookish kids may get parents to read to them; good parents do many good things.

The cycle of violence:

- Parents who smack kids tend to have more violent kids.
- Alternative explanations:
 - Reverse causation: troublemakers get smacked more.
 - Heritability of violence: violent kids inherit propensity to violence from violent parents, regardless of upbringing.

Why this is hard to argue against personally:

- Without being adopted, you can't disentangle "I'm like my parents because of upbringing" from "I'm like my parents because we share genes."
- "Mom smacked me a lot and that's why I turned out to be such a rotten person" — "Well, maybe she smacked you because you were rotten."

Response to "this is a dangerous message":

- Concern: if parenting has no effect, why be nice to kids?
- Answers: because you love them; because you want them happy; because you want good relationships with them.

Reading response assignment: Explain the Flynn effect.

Notable Quotes

"What we've been talking about so far in the course are human universals, what everybody shares... But honestly, what a lot of us are very interested in is why we're different and the nature of these differences and the explanation for them."

"The scientific findings on human psychological differences are, to many of us, shocking and unbelievable."

"Now, the Rorschach test is transcendently useless. It has been studied and explored and it is as useless as throwing dice. It is as useless as tea leaves."

"You walk around and you leave — and your personality leaves a trail in the minds of people around you."

"Whatever IQ tests measure it is of great practical and social importance."

"Your parents had a huge effect on your intelligence and your personality, around 0.5 actually. They had this effect at the moment of conception. From then on in, they played very little role in shaping you — what you are."

"They fuck you up, your mum and dad. / They may not mean to but they do. / They fill you with the faults they had / And add some extra just for you."

"How sharper than a serpent's tooth to hear your child make such a fuss. It isn't fair. It's not the truth. He's fucked up, yes, but not by us."

Assigned Readings / References

- Textbook reviews referenced for: history of attempts to define and measure intelligence; discussion of group differences in IQ; treatment of within-group vs. between-group heritability.
- Herrnstein and Murray, *The Bell Curve*.
- American Psychological Association report by 50 leading intelligence researchers (link on PowerPoint slide).
- A second IQ researchers' report (link on PowerPoint slide).
- Judith Harris, *The Nurture Assumption: Why Children Turn Out the Way They Do, Parents Matter Less than You Think and Peers Matter More*.
- Oliver James, *They Fuck You Up*.
- Philip Larkin (poem quoted in lecture).
- Research by Peter Salovey on emotional intelligence (mentioned).
- Richard Lewontin's seed/fertilizer analogy (mentioned).
- **Reading response assignment:** Explain the Flynn effect.

PSYC 110 - Lecture 14 - What Motivates Us: Sex

Overview

Professor Paul Bloom frames sex as both an oddly minor time commitment (an average of four minutes and three seconds per day for the average American) and an extraordinarily important force from which "everything in life follows" — marriage, family, children, aggression, competition, and much of art. Before presenting biological material, he addresses two prefatory concerns: morality (nothing moral necessarily follows from the facts of biology) and inevitability (genetic origin does not entail unchangeability). He then walks through the evolutionary logic of why two sexes exist, parental investment theory and its predictions across species, and applies it to humans: sex differences in size, aggression, sexual choosiness, and mate preferences. The lecture next examines physical beauty as a marker of youth and health (including the attractiveness of average faces), discusses possible social contributions to sex differences (empathy, math/science aptitude debates), and ends with the evolutionary puzzle of exclusive homosexuality.

Key Concepts & Definitions

- **Male / Female (biological definition):** Males have a small sex cell (sperm — genes only). Females have a big sex cell (egg — genes plus food and protective cover).
- **Parental investment** (Robert Trivers): "Any investment that's going to increase the offspring's chance of survival at the cost of the parent's ability to invest in other offspring."
- **Serial monogamy:** The pattern attributed to American culture — a series of pair-bondings rather than mating for life.
- **Coolidge Effect:** A preference for sexual variety, named after the Calvin Coolidge anecdote about the rooster and different hens.
- **Sexual choosiness:** The degree to which an individual is selective about (or receptive to) sexual partners, including anonymous sex.
- **Beauty (as signal):** A marker of (a) youth (sexually mature but young) and (b) health (absence of deformities, clear eyes, unblemished skin, intact teeth, average face).
- **Average face hypothesis:** Average faces are attractive because averaging factors out deviations/deformities.
- **Gender self-segregation:** Developmental tendency (roughly age four to eleven) of males to group with males and females with females, potentially amplifying sex differences (Eleanor Maccoby).

- **Sexual orientation (vs. behavior):** Orientation concerns what one wants (whom one would be sexually/romantically involved with all things equal), not whom one has actually had sex with.

Detailed Notes

Chapter 1. Addressing Morality and Inevitability on Evolution, Sex and Gender

- Opening puzzle: People (especially college students) name sex as their favorite activity, but data summarized in James Gleick's book show the average American spends **four minutes and three seconds per day** on sex — about the same time spent filling out IRS tax forms.
- Gleick passage lists what the four minutes excludes: "flirting, dancing, ogling, cruising the boulevard, toning up in gyms, toning up in beauty parlors, rehearsing pick up lines, showering, thinking about sex, reading about sex, doodling pornographically, looking at erotic magazines, renting videos, dreaming of sex, looking at fashion magazines, cleaning up after sex, coping with the consequences of sex, building towers or otherwise repressing, transferring, and sublimating."
- Two takeaways: (1) we don't spend much brute time on sex; (2) it remains extraordinarily important — marriage, family, children, aggression, competition, art, music, creativity all follow from it.
- Some creatures reproduce by **cloning**. The existence of two sexes is itself an evolutionary mystery: each offspring carries only half a parent's DNA, so sexual reproduction "tosses away" half the genes. (Not the puzzle of today's class.)
- Today's questions: theoretical and empirical sex differences; sexual attractiveness; origins of sexual preference (straight, gay, bisexual, harder to classify).
- Sex is "dicey" emotionally and fraught with moral implications. Bloom flags two issues at the outset:

Morality — Quote from Steve Pinker, *How the Mind Works*:

Nature does not dictate what we should accept or how we should live our lives. Well into my procreating years, I am so far voluntarily childless, having squandered my biological resources reading and writing, doing research, helping friends and students, and jogging in circles — ignoring the solemn imperative to spread my genes. By Darwinian standards, I am a horrible

mistake, a pathetic loser, but I am happy to be that way, and if my genes don't like it they can go jump in the lake.

- If "right" = leads to gene reproduction, you wouldn't think much of birth control or non-procreative sex. If "wrong" = unnatural, you wouldn't think much of planes, refrigeration, or surviving infections. Bloom: "Nothing moral necessarily follows from the facts of biology."

Inevitability — Quote from Richard Dawkins:

If a child has had bad teaching in mathematics, it is accepted that a resulting deficiency can be remedied by extra-good teaching in the following year. But any suggestion that the child's deficiency might have a genetic origin is likely to be greeted with something approaching despair. If it's in the genes, it is determined and nothing can be done about it. This is pernicious nonsense on an almost astrological scale. Genetic causes and environmental causes are in principle no different from each other. Some may be harder to reverse, others may be easy. What did genes do to deserve their sinister, juggernaut-like reputation? Why are genes thought to be so much more fixed and inescapable in their effects than television, nuns or books.

- Examples:
- **Bloom's poor eyesight:** genetically determined, but easily corrected with glasses/contact lenses.
- **Society's treatment of the obese:** views about thin/fat vary cross-culturally (not hard-wired), but once in a culture they are "almost impossible to shake."
- Conclusion: "Genetic does not mean inevitable, and cultural does not mean easy to fix."

Chapter 2. Basic Sex Ed

- Don't define male/female by penis/vagina — many animals have neither. Biological definition rests on sex cell size:
- **Male:** small sex cell (sperm) — genes only.
- **Female:** big sex cell (egg) — genes plus food and protective cover.
- Puzzle: if males have the smaller cell, why are males often larger and more aggressive?

Parental investment theory (Robert Trivers):

- Definition: investment increasing offspring's survival at the cost of investing in other offspring.
- Within a species, females typically have higher parental investment: big sex cells, often internally incubated; eggs may be sat on; carrying one offspring blocks another.
- Males with small sex cells can sometimes invest only "a few moments of copulation."

- Consequences:

- A single male can fertilize multiple females; some males go mate-less → male-male competition.
- Females can always find mates, so numbers don't matter — they compete to mate with the *right* males (best chance of surviving offspring).
- Explains evolution of larger males, special weapons (for male-male combat), and male display structures (e.g., peacock plumage) to woo choosy females.
 - Causal chain: sex cell size differences → parental investment differences → differences in psychological/physiological mechanisms.

Predictions and species evidence:

| Species | Parental investment pattern | Size / behavior |

| ---|---|---|

| Pipefish | Male takes eggs into a pouch, plugs into bloodstream; females shoot off | Females larger; females fight other females; females compete for male attention |

| Penguins (March of the Penguins) | Both parents must protect egg or it dies — equal investment | Males and females about the same size |

| Elephant seals | Males compete for harems — winner-take-all | Males four times bigger than females |

| Gibbons | Pretty monogamous; raise children together | About the same size |

- Equal-investment scenarios arise when offspring need both parents and effort can't be shared with another family. Hard to find perfect cases; some primates are close.

Chapter 3. Sex Differences among Humans

- Humans are **relatively polygamous**; most cultures are polygamous. American culture = "serial monogamy" — a series of pair-bondings, not lifelong mating.
- **Size**: human males average about **15% larger** than human females, implying some male-male competition history → parental investment is not fully equal.
- **Testicle size**: smaller than chimpanzees, larger than gorillas and gibbons — intermediate sperm competition. Suggests women in evolutionary history were "not wantonly promiscuous, but were not entirely monogamous either."

Aggression:

- "Males are meaner" (not technical). Males are more physically violent:
- More violent in utero (kick more).

- More violent as children (play fighting, combat-like sports).
- More violent as adults (prisons mostly hold men; more likely to kill/harm).
 - Male hormones like **testosterone** increase aggressiveness in humans and other primates.

Sexual choosiness / anonymous sex:

- Parental investment theory predicts males more receptive to anonymous sex (pre-birth-control logic).
- Cross-cultural / behavioral evidence:
 - **Prostitution:** near-universally a male interest. Male prostitutes mostly cater to male customers.
 - **Pornography:** universal — males in every society depict naked females for arousal (sometimes carved into trees, sculptures).
 - **Monkey porn (Duke study):** monkeys "paid" in fruit juice to look at pictures of female hindquarters or socially dominant ("celebrity") monkeys — like "some sort of combination of People Magazine and Penthouse."

Coolidge Effect:

- Anecdote about President Calvin Coolidge and Mrs. Coolidge being shown around a farm; the rooster has sex many times a day — different hen each time. Mrs. Coolidge: "Well, be sure to tell that to the president." President Coolidge, on hearing the rooster mates many times a day: "Same hen every time?" — "No, different hen every time." — "Tell that to Mrs. Coolidge."

Survey data on desired sexual partners:

| Time frame | Women's typical answer | Men's typical answer |

| --- | --- | --- |

| Next month | < 1 | 2 |

| Next 2 years | 1 | 8 |

| Lifetime | 4–5 | 18 |

Would you have sex with a very desirable partner known for...

| Known for | Women | Men |

| --- | --- | --- |

| 1 year | Yes | Yes |

| 6 months | Unsure | Yes |

| 1 week or less | No | Yes (majority say yes down to 5 minutes) |

Field experiment (~10 years ago, "wouldn't do nowadays"): Attractive male and female confederates approached strangers on campus.

- Of women approached by an attractive man: over half agreed to "go out with me tonight"; very few agreed to come to his apartment; **nobody** agreed to go to bed with him.
- Of men approached by an attractive woman: "go out" ~50%, "come to apartment" ~69%, "go to bed" ~75% agreed.
- The 25% of men who said no "apologized profusely" (e.g., fiancée in town).

Behavior — same-sex couples as natural test:

- Heterosexual frequency is a compromise between male and female priorities. Gay couples provide a "pure reflection" of one-sex desire.
- Data are messy but consistently in the expected direction: lesbians tend to be more monogamous than gay men. Pre-AIDS studies found some gay men extremely promiscuous (over a hundred or over a thousand partners). Such promiscuity not found in females.
- Interpretation: gay men do what the average heterosexual man would do if presented with equally willing women.

Mate preferences / attractiveness — Cross-cultural study:

- 10,000 people, 37 countries — asked who they want to be with (marry, mate, sexual partner).
- Universals: everyone values **kindness and intelligence** (or says they do).
- Sex differences:
 - **Females** focus more on power, status, and the male's interest in investing in children. Evolutionary rationale: a man's age (15 vs. 25 vs. 35 vs. 45) matters less for sperm; what matters is his being a good father, protector against predation/murder/assault.
 - **Males** focus on those traits too, but also on **ability to have children** — large male-perceived difference between a 20-year-old and 50-year-old female.

Chapter 4. Beauty: An Average Face

- Beauty signals two things:
 1. **Youth** — sexually mature but young. Cues: large eyes, full lips, smooth tight skin.
 2. **Health** — absence of deformities, clear eyes, unblemished skin, intact teeth (very important), **average face**.
 - Why average faces? Deformities = deviations from the average; averaging factors out anything bad (distortion, asymmetry, age-related changes).
 - **Empirical evidence:**

- Researchers morph faces together. As more faces are averaged, observers rate them as better-looking — "subtle, but it's actually not so subtle that babies don't notice it."
- Same researchers showed averaged Caucasian male and female composites to **babies**, who preferred to look at average faces → preference not purely cultural; partly hard-wired.
- Composite male and female faces "don't exist" — pure computer composites with hair held constant; they look attractive.

- **Beyond average:**

- A **feminized** version of the average female face (certain prototype features exaggerated as more feminine) tends to be preferred → cues a stronger "sexual object."
- For male faces: most people prefer the less-chiseled face B (looks healthy, young, strong, good provider). However, **women who are ovulating** prefer the more chiseled, exaggerated face A (replicated in three labs). Interpretation summarized: face A "may not be a good provider... but I'm sure he has wonderful genes" — have sex with him, have face B raise the kids.

Chapter 5. Social Factors for Sex Differences

- Some sex differences (aggression, mate preference) appear universal across cultures and are good biological-adaptation candidates. Others are debated.
- **Differential treatment of babies:** Studies — swaddle a baby in blue and call it a boy vs. swaddle in pink and call it a girl, and people treat it differently.
- **Name effects on adult treatment:** Sending an email, job application, or paper to a scientific journal as **John Smith vs. Joan Smith** elicits different reactions and expectations. Anecdote: Bloom's friend Lynn is often mistaken for female.
- **Gender self-segregation** (Eleanor Maccoby): Children self-segregate by sex roughly from **age 4 to age 11**. May exaggerate small initial differences — e.g., slightly more aggressive boys become more aggressive in all-boy groups, while girls' non-aggressive behavior is reinforced in girl groups.

Empathy (Simon Baron-Cohen, *The Essential Difference*):

- Claim: men are by nature less empathetic; women more empathetic — a core sex difference.
- Evidence cited:
 - Murder = the ultimate act of lack of empathy; men more violent.
 - Testosterone ↑ → less social.
 - Boys less empathetic than girls; some (debated) evidence boys do worse on social cognition / theory of mind tasks.

- Disorders of empathy/social cognition (autism, Asperger's Syndrome, conduct disorder, psychopathy) are predominantly male.

- Baron-Cohen's slogan: "To be a man is to suffer from a particularly mild form of autism."
- Trivia: Simon Baron-Cohen is a famous developmental psychologist; his cousin Sasha Baron Cohen is more famous.

Math and science aptitude (Larry Summers controversy):

- Larry Summers, then president of Harvard, suggested under-representation of women in science/engineering was partly due to "issues of intrinsic aptitude, and particularly of the variability of aptitude."
- Summers agreed average skills are identical; claim was that males show more **variability** → more retarded males and more male geniuses.
- Highly controversial. Bloom directs students to the **Pinker–Spelke debate** at Harvard (Steve Pinker vs. Liz Spelke, a major infant cognition researcher), available on video at The Edge (http://www.edge.org/3rd_culture/debate05/debate05_index.html).

Chapter 6. Sexual Orientation

- Approximate (hard to pin down) figures: ~98% of women are sexually attracted to men; ~96% of men are sexually attracted to women.
- Some proportion is exclusively homosexual; some bisexual (varies cross-culturally).
- **Orientation ≠ behavior**: People may have same-sex sex out of boredom or experimentation. Orientation = what you want, if all things were equal.
- The "real puzzle" is **exclusive homosexuality**.

Why? What's probably NOT the explanation:

- Not a **conscious choice**: gay people often face severe discrimination and may even believe being gay is morally wrong — implausible they would choose it.
- Not **post-pubertal seduction/experience**: signs of orientation (sexual and romantic fantasies) appear long before puberty.

Genetic / built-in? — Partly.

- Behavioral genetics: compare monozygotic vs. dizygotic twins, adopted vs. biological siblings.
- Concordance: if one identical twin is gay, the other is gay about **50%** of the time — high vs. population baseline, but not 100% as it would be if purely genetic.
- So some experience (possibly **prenatal**) also contributes.

Why is exclusive homosexuality an evolutionary puzzle?

- Not why some men sometimes have sex with men (recreation, pair bonding).
- The puzzle: why are there men who don't want to have sex with women (and women who don't want to have sex with men)? Evolutionarily, the underlying genes should be weeded out because such individuals typically (pre-modern technology) don't reproduce.
- Reading response assignment for the week: "solve that puzzle." Account should bear some relationship to the facts discussed in lectures and readings.

Chapter 7. Question and Answer on Sex and Gender

- Student: "I like your leather jacket." — Bloom: "Thank you very much."
 - Student question: Is there similar data on other species?
- Bloom: Good question — cross-species data should inform any account of human sexual preference origins.
- Many animals engage in homosexual behavior (sex with members of their own sex).
- Bloom does not know the rate of **exclusive** homosexual behavior in nonhuman primates (i.e., primates that do not want to have sex with members of the opposite sex).

Notable Quotes

Americans tell pollsters their single favorite activity is sex... On the other hand, these same studies suggested the average time per day devoted to sex is four minutes and three seconds. — James Gleick (cited)

By Darwinian standards, I am a horrible mistake, a pathetic loser, but I am happy to be that way, and if my genes don't like it they can go jump in the lake. — Steve Pinker

This is pernicious nonsense on an almost astrological scale. Genetic causes and environmental causes are in principle no different from each other. — Richard Dawkins

What did genes do to deserve their sinister, juggernaut-like reputation? Why are genes thought to be so much more fixed and inescapable in their effects than television, nuns or books. — Richard Dawkins

Genetic does not mean inevitable, and cultural does not mean easy to fix. — Paul Bloom

Males are meaner.

To be a man is to suffer from a particularly mild form of autism. — Simon Baron-Cohen

...in the special case of science and engineering, there are issues of intrinsic aptitude, and particularly of the variability of aptitude... — Larry Summers

Assigned Readings / References

- James Gleick — quoted book on Americans' time use (cited in lecture; specific title not named in transcript).
- Steve Pinker — *How the Mind Works* (quoted).
- Richard Dawkins (quoted; specific work not named in transcript).
- Robert Trivers — parental investment theory.
- Eleanor Maccoby — gender self-segregation and amplification of sex differences.
- Simon Baron-Cohen — *The Essential Difference*.
- Pinker–Spelke debate on sex differences (The Edge, Harvard, ~1 year before lecture): http://www.edge.org/3rd_culture/debate05/debate05_index.html
- Reading response assignment: account for the evolutionary puzzle of exclusive homosexuality, tied to lecture/reading facts.

PSYC 110 - Lecture 15 - A Person in the World of People: Morality

Overview

Professor Paul Bloom organizes the lecture around three facets of morality: moral feelings, moral judgments, and moral action (with particular focus on why good people do bad things, leading into a review of the Milgram study). He argues that moral feelings such as empathy and in-group caring can be grounded in evolution through kin selection and cooperation, but that empathy is unevenly distributed (as illustrated by psychopaths) and group membership powerfully shapes moral concern. Moral judgments are not just feelings; they carry obligation and sanctions, and although Western culture frames morality in terms of an ethics of autonomy, Jonathan Haidt's research shows people exhibit "moral dumbfounding," suggesting moral intuitions can run ahead of conscious reasoning. The lecture closes by laying out two forces for evil (deindividuation of self and denigration of others) and two forces for good (contact/interdependence and perspective taking).

Key Concepts & Definitions

- **Moral feelings:** feelings of condemnation, shame, pride, righteous anger, affection, caring for others, and being upset at injustice.
- **Kin selection:** evolutionary explanation in which altruism is preserved because animals act to preserve others sharing their genes; caring is proportional to shared genes.
- **Cooperation:** a second evolutionary force for kindness; even unrelated animals are nice (warning cries, grooming, food exchange) because minds evolved to surmount prisoner's dilemmas, deception, and cheating, giving rise to moral emotions like guilt and anger.
- **Empathy:** "the feeling that your pain matters to me"; being hard-wired and attuned to another's pain.
- **Psychopaths:** a population in which instinctive empathy is broken and the pain of others doesn't bother them.
- **In-group / out-group:** differential moral caring based on group membership; we favor our group over others.
- **Robber's Cave study (Sherif):** study with 11–12-year-old boys that demonstrated how arbitrary group divisions produce within-group solidarity, negative stereotypes, and intergroup hostility.
- **Superordinate goal:** a shared goal (and ideally a common enemy) that brings warring groups together.

- **Minimal groups (Tajfel):** groups created with trivial criteria (e.g., Klee vs. Kandinsky lovers, coin flips) still elicit in-group favoritism.
- **Moral judgment:** an evaluation carrying a sense of obligation and sanctions (punishment for violation).
- **Ethics of autonomy:** morality framed in terms of rights, equality, and freedom.
- **Ethics of community:** morality framed in terms of duty, status, hierarchy, and interdependence.
- **Ethics of divinity:** morality framed in terms of purity, sanctity, pollution, and sin.
- **Moral dumbfounding (Haidt):** people struggle to find rational explanations for moral judgments they hold strongly, suggesting intuitions precede reasoning.
- **Deindividuation of self:** diffusion of personal responsibility in groups, through obedience, or via anonymity.
- **Diffusion of responsibility:** in groups, individuals assume others will act, decreasing willingness to help.
- **Denigration of others:** reducing the moral status of others via distance, dehumanizing language, humor, removing names, or seeing them as disgusting.
- **Disgust:** a "body and soul emotion" (Paul Rozin); a basic human universal with a characteristic facial expression, elicited universally by feces, urine, blood, vomit, rotten flesh, and most meat.
- **Contact Hypothesis:** simple contact with others (especially under equal status, common goal, and social support) increases liking.
- **Perspective taking:** imagining oneself in another's place; expands moral concern.

Detailed Notes

Chapter 1. Introduction [00:00:00]

- Bloom reviews course arc: brain → Freud/Skinner → cognitive topics (development, language, vision, memory) → love (dean) → emotions, rationality, evolution → cognitive neuroscience using face recognition → human differences, behavioral genetics, nature/nurture, sex (Bloom) and food (Dr. Brownell).
- Today: morality. Upcoming: social thought and behavior; "mysteries" (sleep, laughter, religion); mental illness (two lectures on madness); happiness.
- Bloom jokes about students choosing psych over chemistry, anthropology, pre-med.

Chapter 2. Kin Selection, Cooperation and Moral Feelings [00:02:43]

Three facets of morality to be covered: moral feelings, moral judgments, moral action (Milgram).

Evolution of moral feelings: two answers.

- **Kin selection:** evolution at the level of genes; altruism preserved because it protects shared genes. Even dying to save another can be explained in cold-blooded evolutionary terms.
- **Cooperation:** animals (warning cries, grooming, food exchange) evolved to cope with prisoner's dilemmas, deception, cheating; produces emotions like guilt and anger.

Empathy

- Definition: "the feeling that your pain matters to me."
- Adam Smith (often falsely seen as a proponent of selfishness) noted empathic flinching:

> When we see a stroke aimed and just ready to fall upon the leg or arm of another person, we naturally shrink and draw back our own leg or arm and when it does fall we feel it in some measure and are hurt by it as well as the sufferer.

- Examples: cringing at someone kicked in the groin in a movie, or banging their thumb with a hammer.
- **Babies:** hearing other babies cry is aversive to them. Cynical alternative: babies are too stupid to distinguish; clever experiment played tapes of other babies vs. self crying — babies cry more to others' crying than their own, indicating sensitivity to "otherness."
- **Chimpanzees:** studies (now considered unethical) — a chimp in one room presses a lever for food; pressing shocks a non-relative chimp in adjacent room. Chimps refrain (don't starve themselves but go long periods without food). Species-specific: with a rabbit in the other room, the chimp slaps the lever repeatedly to make the rabbit scream and jump.
- **Aristotle:** empathy not logically tied to morality — seeing pain might cause you to run away or blame the sufferer. But empathy induction increases pro-social action; high-empathy individuals tend to be nicer.
- **Psychopathy:** instinctive empathy is broken. Quotes from Damon's book and others:

- Thirteen-year-old mugger of blind people: "What do I care? I'm not her."

- Gary Gilmore: "I was always capable of murder. I can become totally devoid of feelings of others, unemotional. I know I'm doing something grossly... wrong. I can still go ahead and do it."

- Ted Bundy: astonished at the fuss — "I mean, there are so many people."

In-group / out-group

- We care more about kin > friends > strangers; favor our groups in every way.
- Group memberships: men/women, Yale students, young, racial groups, etc.
- Even on-the-fly groups exert powerful moral control.

Robber's Cave study (Sherif)

- Subjects: well-adjusted, racially homogeneous, fairly rich 11–12-year-old boys at a camping program.
- Put in separate cabins, leaders assigned; boys named themselves "The Eagles" and "The Rattlers."
- Separation produced distinct cultures; competition increased within-group solidarity.
- Stereotypes: Eagles saw themselves as clean-living, no cuss words, respectful; viewed Rattlers as dirty, tough, slovenly. Rattlers viewed Eagles as goody-goody.
- Escalated to raids and violence (Eagles burnt Rattlers' banner; cuss words used). Sherif: "Excellent."
- Failed reconciliation attempts:
 - Peace talks: representatives ostracized as treasonists.
 - Individual Olympic-style competition: degenerated into Eagles vs. Rattlers.
 - Shared meals → food fights; shared movies → fights; shared firecracker fun nearly ended the experiment.
 - Religious sermons on brotherly love: kids respected them but interpreted "love thy neighbor" as loving fellow group members and not "those scummy Eagles."
- What worked: telling all kids the water line to the camp was cut and they had to defend it — a **superordinate goal** plus a common enemy.
- Bloom's aside: solution to bringing together warring countries/religions = an alien attack.

Tajfel and minimal groups

- Question: how little is needed to create a group?
- Klee vs. Kandinsky experiment: based on modern art preferences (assignments made up), people viewed in-group as more similar, smarter, and gave them more resources.
- Coin flip experiments produce the same effect (subtler): heads vs. tails groups favor own group.
- Implication: moral feelings are exquisitely attuned to group psychology, not just individuals.

Student question addressed: group experiments bear on morality because moral feelings (empathy, caring) are partially determined by group membership; obligations to in-group members are felt more strongly.

Chapter 3. Moral Reasoning and Moral Judgments [00:23:07]

- **Moral judgment ≠ mere evaluation.** "I don't like strawberry ice cream" is a judgment but not moral — no obligation, no sanctions. "I don't like baby killers" carries:

1. Obligation (we are obliged not to kill babies)

2. Disagreement = wrong

3. Sanctions (should be punished).

- Moral judgments happen constantly and often unconsciously: what to eat (moral vegetarians), charity, who to socialize with, how to treat homeless people, life goals, sex partners and conditions.
- Bloom's favorite dilemma: whether to make eye contact and greet, nod, or avoid someone you sort-of know on the street.

Universals

- Some moral intuitions appear everywhere on Earth, with weaker evidence of similar intuitions in young children and nonhuman primates (chimpanzees, capuchins, macaques): anger at cheaters, gratitude toward sharers, sense of right/wrong (consistent with prisoner's dilemma).

Cross-cultural variation (Richard Shweder)

People have found it quite natural to be spontaneously appalled, outraged, indignant, proud, disgusted, guilty and ashamed by all sorts of things.

- Long list of variable triggers: masturbation, homosexuality, sexual abstinence, polygamy, abortion, circumcision, corporal punishment, capital punishment, Islam, Christianity, Judaism, capitalism, democracy, flag burning, miniskirts, long hair, no hair, parents/children sleeping in same bed (or not), women working (or not).
- Bloom notes baby killing and reciprocity (prisoner's dilemma) appear hard-wired and not culturally variable.

Shweder's three ethics

| Ethics | Core concepts |

|---|---|

| Autonomy | Rights, equality, freedom (Western moral philosophy default) |

| Community | Duty, status, hierarchy, interdependence |

| Divinity | Purity, sanctity, pollution, sin |

- Example: rights of men and women.

- Autonomy: equal rights in all domains; restrictions need compelling argument involving others' freedom.

- Community: different rights and duties; born to certain roles.

- Divinity: religious injunctions (e.g., women not preparing food while menstruating; restrictions on who can have sex with whom rooted in pollution and sin).

- Western debates (abortion, hate speech) framed in autonomy terms.

Jonathan Haidt's work (University of Virginia)

- Even highly educated Westerners committed to autonomy show moral judgments not reducible to harm.
- The Julie and Mark scenario:

> Julie and Mark are brother and sister. They are traveling together in France on summer vacation from college. One night they are staying alone in a cabin near the beach. They decide it would be interesting and fun if they tried making love. At the very least, it would be a new experience for each of them. Julie was already taking birth control pills but Mark uses a condom too just to be safe. They both enjoy making love but they decide not to do it again. They keep that night a special secret which makes them feel even closer to each other.

- Most people say it's wrong; when asked why, they exhibit **moral dumbfounding** — they cite a birth defect risk (refuted by birth control), underage status (refuted), and ultimately "it's just wrong."
- Other Haidt-style scenarios:

- **Dead dog:** family dog hit by car; family cooks and eats it.

- **Flag/toilet:** using the American flag to clean a dirty toilet.

- **Frozen chicken:** a man buys a frozen chicken, has relations with it, then cooks and eats it.

- Among non-elite-college populations, the majority judge these as wrong; reasons offered (e.g., "disease") are post-hoc, since removing the consideration leaves the reaction.
- Conclusion: moral intuitions are governed by factors not consciously accessible.

Chapter 4. Milgram's Work in the Context of Morality [00:38:48]

- Motivation: the Holocaust — why such a thing could happen.
- Milgram biography: Yale professor, denied tenure; moved to Harvard, denied tenure; reputation as "mad doctor"; ended at CUNY as full professor at age 33; died in his early 50s. Also originated "six degrees of separation."
- Setup recap (from "Obedience" movie): subject thinks they are teaching another in a memory task; the "learner" is a confederate; subject delivers escalating shocks from "slight shock" to "XXX"; learner protests; experimenter requests obedience.
- Majority deliver fatal shocks based on instructions.

Bad explanations rejected

- Subjects were not abnormal psychopaths (replicated broadly).
- Not sadism — subjects were acutely uncomfortable, embarrassed, conflicted, stressed; nobody enjoyed it.

Follow-up findings — variables that reduce fatal shocks:

- Moving experiment off-site reduces authority of Yale → fewer fatal shocks.
- Teacher in same room as learner → fewer fatal shocks.
- Teacher must place learner's hand on shock plate → fewer fatal shocks.
- Experimenter instructs by phone → fewer.
- Instructions from an ordinary man (not white-lab-coat) → fewer.
- Witnessing rebellion by another participant → much less likely to continue.
- If participants choose their own shock level → very few go all the way.

Ethical critique

- Milgram's study is why human-subjects committees exist; the study would not be approved today.
- 2% reported regret; most reported being happy to participate, but serious damage could have been done. Participants learned about themselves that they could kill on command — Bloom argues psychologists have no right to do that.

Historical/sociological critique

- Goldhagen argued that Nazi participants were not merely obedient but enthusiastic volunteers; obedience may not be the right model for genocide (controversial).

Ingredients of the "perfect situation" to get someone to do a terrible thing

- Authority (Yale + science: "the experiment must go on").
- A self-assured experimenter who accepts responsibility (would change if nervous/unsure).
- Distance between learner and experimenter (less effect when you must touch the victim).
- A novel situation with no behavioral model.
- Knowing about Milgram's setup arms you against it in real life.

Chapter 5. Two Forces for Evil, Two Forces for Good [00:46:20]**Two forces for evil**

1. **Deindividuation of self** — diffuse responsibility:
 - In groups (smashing windows alone vs. with 20 others).

- Accepting orders ("I'm just an instrument").
- Anonymity (face paint, masks; psychologically liberating).
- Explains failure to help others. Diffusion of responsibility (Kitty Genovese case — murder in a courtyard surrounded by apartment buildings while dozens watched and did nothing).
- Practical advice: don't yell "Help" in a crowd; point to a specific person — "You in the green sweater, call the police" — and they will.

1. **Denigration of others** — make the victim matter less:

- Psychological or physical distance.
- Dehumanizing language: "cargo" instead of people; "extermination" instead of murder.
- Humor (makes target less of a person; ties in to later lecture on laughter).
- Removing names. UN Declaration of Human Rights: "Every person has a right to a name." Names mark people as persons; once you know someone's name, you're less likely to hurt them.
- Seeing them as disgusting.
 - **Disgust (Paul Rozin):** "the body and soul emotion"; human universal; basic emotion with a characteristic facial expression (cf. Paul Ekman's basic emotions); universally elicited by feces, urine, blood, vomit, rotten flesh, most meat.
 - **Charles Darwin** anecdote on disgust extending to people: "a native touched with his finger some cold preserved meat and plainly showed disgust at its softness whilst I felt utter disgust at my food being touched by a naked savage though his hands did not appear dirty."
 - **Martha Nussbaum:** "Thus, throughout history certain disgust properties have repeatedly and monotonously been associated with Jews, women, homosexuals, untouchables, lower class people. All of those are imagined as tainted by the dirt of the body."
 - Genocidal movements with written records use disgust to dehumanize.

Two forces for good

1. **Contact and interdependence**

- Extended selfish-gene logic — interconnection breeds care.
- **Robert Wright:** "One of the many reasons I don't want to bomb the Japanese is that they built my minivan."

- **Thomas Friedman's Golden Arches Theory of Human Conflict:** no two countries with McDonald's would go to war. Falsified by the NATO bombing of (Bloom recalls) Sarajevo, but the idea — interdependence promotes peace — is the spirit.
- **Contact Hypothesis:** simple contact (especially equal status, common goal, social support) increases liking. Many studies on US blacks and whites show reduced animosity when brought together properly.
- **Robber's Cave** — interconnection via shared goal then contact produced moral feeling.
- **US military** as a successful case: integrated work toward a shared goal under institutional support, on equal footing — reduced racism.
- Application to universities like Yale: debate over whether the conditions for positive contact are met (e.g., between blacks and whites, US South vs. North, US vs. international students); campus segregation debate.

1. Perspective taking

- JFK on equal rights — invited white listeners to imagine themselves in the place of those facing discrimination:
 - > If an American, because his skin is dark, cannot eat lunch in a restaurant open to the public — [and so on and so on and so on], then who among us would be content to have the color of his skin changed and stand in his place? Who among us would be content with the counsels of patience and delay?
- Nussbaum on Greek dramas: invited audiences to take perspectives of those they would never meet — "empathetic expansion."
- Bloom: "one of the greatest circles for moral good is storytelling."
- **Direct ways:** ask "How would you feel if..." (used with children).
- **Indirect ways (metaphor):**
 - Fetus described as a "pre-born child" to recruit moral concern.
 - Animal described as human-like to elicit caring.
 - Calling unrelated strangers "brothers and sisters" — family metaphor used by political movements.
 - **Steven Spielberg** chose a cute child actor (not a faceless mechanical robot) to portray a sentient AI/robot to recruit moral concern.

Assignment for next week (reading response): Suppose the Milgram experiment had never been done and was being done for the first time here. What would you

do? What do you think everyone else would do? (Short answers acceptable for passing grade.)

Notable Quotes

When we see a stroke aimed and just ready to fall upon the leg or arm of another person, we naturally shrink and draw back our own leg or arm and when it does fall we feel it in some measure and are hurt by it as well as the sufferer. — Adam Smith

What do I care? I'm not her. — Thirteen-year-old mugger of blind people (Damon)

I was always capable of murder. I can become totally devoid of feelings of others, unemotional. I know I'm doing something grossly... wrong. I can still go ahead and do it. — Gary Gilmore

I mean, there are so many people. — Ted Bundy

People have found it quite natural to be spontaneously appalled, outraged, indignant, proud, disgusted, guilty and ashamed by all sorts of things. — Richard Shweder

One of the many reasons I don't want to bomb the Japanese is that they built my minivan. — Robert Wright

Thus, throughout history certain disgust properties have repeatedly and monotonously been associated with Jews, women, homosexuals, untouchables, lower class people. All of those are imagined as tainted by the dirt of the body. — Martha Nussbaum

a native touched with his finger some cold preserved meat and plainly showed disgust at its softness whilst I felt utter disgust at my food being touched by a naked savage though his hands did not appear dirty. — Charles Darwin

Assigned Readings / References

- Course textbook chapters discussing the Robber's Cave study and the Kitty Genovese case (referenced as "discussed in detail in the textbook").
- The film "Obedience" (a movie on Milgram's conformity studies) shown on Monday.
- Reading response for next week: Hypothetical — if the Milgram experiment had never been done and were being run for the first time, what would you do, and what would others do?

Named scholars/figures referenced in lecture: Paul Bloom, Sigmund Freud, B.F. Skinner, Dr. Brownell, Adam Smith, Aristotle, Damon (book on psychopathy), Gary Gilmore, Ted Bundy, Muzafer Sherif, Henri Tajfel, Paul Klee, Wassily Kandinsky, Richard Shweder, Jonathan Haidt, Stanley Milgram, Daniel Goldhagen, Paul Rozin, Paul Ekman, Charles Darwin, Martha Nussbaum, Robert Wright, Thomas Friedman, John F. Kennedy, Steven Spielberg.

PSYC 110 - Lecture 16 - A Person in the World of People: Self and Other, Part I

Overview

This lecture opens a two-lecture sequence on social psychology, covering how we think about ourselves, other people, and groups of people. Professor Bloom argues that the most interesting aspect of our evolved minds is our capacity to understand and deal with other people, and he uses our fascination with celebrity lives (e.g., the 2005 Aniston–Pitt separation) as an entry point. The lecture works through individual differences in social connectedness (connectors and the Six Degrees concept), three core findings about the self (the spotlight effect, transparency effect, Lake Wobegon / better-than-average effect, self-serving bias, and cognitive dissonance), the fundamental attribution error and cross-cultural variation in it, and the basic determinants of liking other people (proximity / mere exposure, similarity, and physical attractiveness). Two summary morals are drawn: "enhancement of the self" and "oversimplification of the other."

Key Concepts & Definitions

- **Connectors (Gladwell):** People who are social "hubs" — they know unusually large numbers of people. Analogous to air-traffic hubs like Chicago O'Hare or Newark.
- **Six degrees of separation:** Idea, derived from Milgram's package study in Omaha, Nebraska, that people are connected to one another via short chains (max six steps in that study).
- **Bacon number:** Number of movie-collaboration steps from an actor/actress to Kevin Bacon.
- **Spotlight effect (Gilovich):** We systematically overestimate how much other people notice us.
- **Transparency effect (Gilovich):** We believe our inner states (e.g., that we are lying) are more visible to others than they actually are.
- **Lake Wobegon effect:** Systematic bias to see oneself as better than average (named after Garrison Keillor's story of a place where all children are above average).
- **Self-serving bias:** Asymmetry in attributions — successes attributed to one's own abilities, failures to external factors.
- **Cognitive dissonance (Festinger):** Unpleasant emotional state caused by inconsistency in one's beliefs/actions; we act to reduce it.
- **Confirmation bias:** Seeking out information that confirms existing beliefs (e.g., right-wing readers reading right-wing magazines).

- **Insufficient justification effect:** When external justification for behavior is small, people adjust beliefs to justify the behavior internally.
- **Attribution (Heider):** A claim about the cause of someone's behavior.
- **Person bias / Fundamental attribution error:** Tendency to over-attribute behavior to personality/disposition and under-attribute it to the situation.
- **Proximity:** Tendency to like people we are physically/spatially close to.
- **Mere exposure effect:** Simply seeing something more often makes it more likable.
- **Similarity:** Tendency to like people similar to ourselves; predicts marriage success.
- **Matthew effect:** Developmental-psychology term (from the Bible) — "to him that hath shall be given" — the advantaged get more advantages.

Detailed Notes

Chapter 1. Social Psychology and Connections between People

- Two-lecture sequence on social psychology: how we think about ourselves, other people, and groups.
- Most interesting aspect of evolved minds: capacity to understand and deal with other people.
- People are intensely interested in others' lives; the 2005 Aniston–Pitt separation cited as illustration. Psychologists must "question the natural."
- **Individual difference in social nature** introduced via a test from Malcolm Gladwell's *The Tipping Point*.
- **Stanley Milgram package study:**
 - Gave a package to 160 randomly chosen people in Omaha, Nebraska.
 - Task: get the package to a stockbroker who worked in Boston but lived in Sharon, Massachusetts.
 - Most succeeded; max six steps → origin of "six degrees of separation."
 - About **half** of the successful packages passed through only **two** people — evidence for connector "hubs."
 - Analogy: air traffic hubs (Chicago O'Hare, Newark).
 - **Gladwell's name test:**
 - 250 names randomly chosen from a Manhattan phone book, covering many ethnicities/national origins.
 - Rule: to "know" someone, they must know you back (Bloom can circle Marcia Johnson, department chair, but not Magic Johnson).

- Circle a name twice if you know more than one person with that last name.
- Bloom reports class distribution; **average among a college crowd is 21**; some score over 100.
- Higher scores correlate with age, longer time in country, being a journalist (academics not so high).
 - **Six Degrees of Kevin Bacon:**
- Average Bacon number across ~250,000 actors on IMDb = **2.8**.
- John Wayne (180 movies over 60 years) is not well-connected — mostly westerns, repeat collaborators.
- Meryl Streep also not well-connected — only "good movies," no overlap with Adam Sandler or Jean-Claude Van Damme.
- **Most connected: Rod Steiger** — *On the Waterfront*, *In the Heat of the Night*, *Carpool*, across dramas, crime serials, thrillers, westerns, horror, sci-fi, musicals.
 - Sociological implication: a reason to attend Yale is the **acquaintance network** (not just close friends) — connections matter for jobs, etc., above and beyond intellectual benefits.
 - Roadmap for next ~1.5–2 lectures: self → self vs. other → other → groups (e.g., Harvard students, gay people, black people).

Chapter 2. Aspects of the Self: The Spotlight and Transparency Effects

- Bloom's anecdote: rushed mornings; once had shaving cream in ear (noticed by a homeless man), once had a seriously misaligned shirt at a party — feels humiliating, like everyone notices.
- A *Simpsons* episode illustrates the spotlight effect (clip played in class).
- **Tom Gilovich's T-shirt experiment:**
- Subjects asked to wear T-shirts with the most embarrassing pictures.
- Worst T-shirt picture: **Hitler tied with Barry Manilow**. Best: **Martin Luther King Jr. and Jerry Seinfeld**.
- Subjects estimated how many noticed; experimenters then asked observers.
- Subjects overestimated by roughly a **factor of two** (e.g., thought 100 noticed; actually ~50).
 - **Why this matters — psychology of regret:** When elderly/dying people are asked what they regret, they regret things they **didn't try**, and the reason given is "I would look silly." Knowing the spotlight effect: people don't care as much as you think.
 - **Transparency effect demo:** Student volunteer answers three yes/no questions ("Have you been in London? Do you have a younger sibling? Do you like sushi?") and lies on a designated one. Class vote is split between two answers; student reveals she lied on #3.
 - Two takeaways:

1. People are actually quite good liars — most can succeed at this task.
2. We feel our secrets "bleed out," so we overestimate how much others detect.
 - Implication: makes teaching and storytelling hard because we overestimate how much others already know.

Chapter 3. Aspects of the Self: You're Terrific!

- If self-ratings were unbiased, average percentile self-rating should be 50%; instead, people rate themselves above average as students, teachers, lovers, and **especially drivers**.
- **Lake Wobegon effect** (Garrison Keillor) — systematic bias to view self as better than average.
- Possible explanations:
 - **Nature of feedback:** Healthy environments give more compliments than criticism → inflated self-esteem.
 - **Different criteria for goodness:** People pick the aspect they're good at (parallel parking, carefulness, risk-taking) and judge themselves on that.
 - Above and beyond these, there is a motivation to feel good about yourself.
 - **Self-serving bias:**
 - Good midterm grades attributed to being smart, hardworking, brilliant.
 - Poor grades attributed to unfair test, being busy, having better things to do.
 - Same asymmetry in professors (accepted papers = brilliance; rejected = conspiracy by jealous editors).
 - Found in athletes, CEOs, accident reports.
 - Three self-related findings so far: you think you're important (spotlight), you think your thoughts bleed out (transparency), and you think you're terrific.

Chapter 4. Aspects of the Self: Cognitive Dissonance

- **Leon Festinger's cognitive dissonance theory:** Inconsistency in the head produces an unpleasant emotional state ("dissonance"); we act to reduce it.
- **Confirmation bias:** Right-wing people read right-wing magazines; left-wing read left-wing. Bush supporters look for good news about Bush; opponents look for bad.
- **Louisa Egan's M&M study (Yale):**
 - Pretest ensures equal liking of three M&Ms.
 - Subject chooses between two; eats the chosen one (e.g., red).

- Red is removed; subject now chooses between the remaining two.
- Subjects strongly prefer the one **not** previously rejected — denigrating the rejected option to justify the choice.
- Same effect found in undergraduates, four-year-olds, and monkeys.

- **Insufficient justification effect (Festinger classic):**

- Two groups do a really boring task; one paid **\$20** (real money at the time), the other **\$1** (insultingly small).
- When asked later how fun the task was, the **\$1 group rated it more fun**.
- \$20 subjects can explain their behavior with the money; \$1 subjects rationalize: "It wasn't that bad really, it was kind of interesting, I learnt a lot."

- Dilbert cartoon illustration: dog explains cognitive dissonance — "When people are in an absurd situation their minds rationalize it by inventing a comfortable illusion." The character concludes, "I must work here because I love this work, I love this job."

- **When Prophecy Fails (Festinger):** Group convinced world would end; went on a mountain, gave away families, houses, possessions, money. When the world didn't end, instead of saying "I'm such a moron," members said going to the mountain had **delayed** the end of the world — proof they were doing the right things. People who invest heavily are extraordinarily resistant to disconfirmation.

- **Real-world applications:**

- **Hazing** (fraternities, med schools): humiliation → infer "the group must be really good." (Bloom notes hazing is illegal.)
- **Political volunteers:** Unpaid volunteers are more committed because they must justify their work to themselves ("I must think very highly of you").
- **Free therapy tends to be useless** — without paying, you don't ascribe value.
- **Children's rewards:** Two groups of kids draw pictures; one rewarded (sticker/toy), one not. The **rewarded group later values the activity less** and is less likely to do it without a reward — they explain their behavior by the reward; the unrewarded children infer intrinsic value. Caution against over-rewarding children.

- **Subtlety:** Cognitive dissonance is **not** simple inconsistency. Both \$1 and \$20 subjects had *some* justification (money). The motivation is to make ourselves look more **moral and rational** than we are — to preserve the belief that we are terrific.

- Summary of three findings about the self:

1. You believe everyone notices you (you're the hero of your story).

2. You're terrific — better than average in every way.
3. What you do makes sense — if it doesn't, you'll distort it so it does.

Chapter 5. Self and the Other

- **Attribution (Heider):** Claim about cause of someone's behavior.
- Multiple kinds of attributions possible (personality, situation, stress, ulterior motive).
- **Person bias / fundamental attribution error:** We over-attribute behavior to personality/disposition; under-weight situation/context. One of the core ideas in psychology.
- **Examples:**
 - **Professors seem smart** because they only talk about what they know; by semester's end Bloom will have told the class everything he knows.
 - **Quiz show study:** Coin flip assigns one person quiz master, the other contestant. Quiz masters ask idiosyncratic questions ("What was my dog's name?"). Observers rate the quiz master as more intelligent than the contestant — failing to discount the situation.
 - **Job talks:** If a slide projector breaks, observers blame the speaker, not the projector ("the person looked kind of upset ... they seemed like a nervous type").
 - **Actors confused with their roles (extreme cases):**
 - **Robert Young** played Marcus Welby in *Marcus Welby, M.D.* — compassionate, made house calls, saved lives. Received thousands of letters asking medical advice; later exploited the FAE in a Sanka decaf coffee ad with the famous line "I am not a doctor but I play one on TV."
 - **Sylvester Stallone** viewed as a Vietnam War hero due to Rambo; actually was in a Swiss boarding school teaching girls aged 12–15 during the war.
 - A gay-character role (movie 20 years before the lecture) was turned down by Harrison Ford, Michael Douglas, Richard Gere because they didn't want to be thought gay; eventually taken by Harry Hamlin, "kind of a B-list sort of guy." (IMDb cited.)
 - **Leonard Nimoy / Spock:** Repeatedly treated as an actual Vulcan; wrote *I Am Not Spock*, then later, after his career stalled, wrote *I Am Spock*, conceding to the FAE.
 - **Cross-cultural variation:**
 - FAE is not a universal — research compared the United States and India.
 - At about age eight, children in both cultures do **not** commit the FAE.
 - In Western cultures (ideology of personal control of destiny), the error develops.
 - Some Eastern cultures emphasize faith and situational attributions.

- **Newspaper reports of murders:** US reports emphasize personal characteristics of the accused; Indian reports emphasize the situation that drove the person.

- **Summary morals:**

- **Enhancement of the self:** when things go badly, blame the situation; when they go well, credit yourself.

- **Oversimplification of the other:** we are far less forgiving of others. "You do something stupid, that's — you're a stupid person. I do something stupid, it's an off day."

Chapter 6. How We Think about Other People

- Why we like people — partly covered in Peter Salovey's earlier ("Dean Salovey's") lecture on sexual attractiveness. Beyond honesty, kindness, intelligence, humor, three fundamental processes:

Proximity

- We like people physically/spatially close to us and people we spend time with.
- **Manhattan housing project study:** 90% said best friend lives in the same building; 50% said same floor.
- Class poll: best friend at Yale tends to be in same college; many on same floor. Joke: "If you were going to marry somebody from this class, it is the person you are sitting next to?"

Mere Exposure Effect

- Simply seeing something more often makes it more likable — becomes comfortable, safe.
- **James Cutting study:** In Intro Psych course, flashed paintings as screen-saver before lectures; said nothing about them. End of semester: subjects rated previously seen images as more likable, even without conscious memory of having seen them.
- **Mirror-image preference:** You prefer photos of yourself that match your mirror image (what you see every day); your best friend prefers the non-mirrored version (what they see every day).

Similarity

- We like people similar to us; friends and spouses are highly similar.
- Hard to separate from proximity (Yale friends similar partly because Yale students are similar to each other), but similarity has effects beyond proximity.
- Similarity predicts marriage success; couples become more similar over time (mechanism unclear).

Physical Attractiveness

- Attractive people are judged smarter, more competent, more social, nicer.
- Possibility: good-looking people actually become better because of advantages — teachers more responsive, more opportunities, more money, more access.
- **Matthew effect** (developmental psychology, from the Bible): "For unto everyone that hath shall be given and he shall have abundance ... but from him that hath not shall be taken away even that which he hath." Rich get richer; poor lose what they have.
- Studies cited:
 - Teachers rate attractive children as smarter and higher-achieving.
 - When an ugly child misbehaves, it's an "ugly soul"; an attractive child is "that little scamp, somebody must have been bothering him."
 - Bloom's neighbor's cute son in Arizona was named **Adonis**.
 - **Mock trials**: Judges give longer prison sentences to ugly people.
 - **Supermarket parking-lot study**: Parents observed to be rougher with ugly kids than good-looking kids — interpreted as the ugly child mattering less to the parent.
- Bloom critiques the parking-lot study: parents of ugly children are likely to be ugly themselves, so the finding may just be that ugly people are more violent.
- Poker quote heard on TV: "They beat me like an ugly stepchild." Bloom: the fate of the ugly stepchild is, in fact, not a very good fate.
- Lecture ends here; social psychology to continue Wednesday.

Notable Quotes

"Probably the most interesting aspect of our evolved minds is our capacity to understand and deal with other people."

"Some people are hubs. Some people are the sort of people who know a lot of people."

"I am not a doctor but I play one on TV."

"When people are in an absurd situation their minds rationalize it by inventing a comfortable illusion."

"I must work here because I love this work, I love this job."

"You do something stupid, that's — you're a stupid person. I do something stupid, it's an off day."

"For unto everyone that hath shall be given and he shall have abundance ... but from him that hath not shall be taken away even that which he hath."

"They beat me like an ugly stepchild."

Assigned Readings / References

The transcript mentions the following works/sources by name (no formal reading list was given in the transcript):

- Malcolm Gladwell, *The Tipping Point* — source of the 250-Manhattan-names test.
- Stanley Milgram — Omaha, Nebraska package experiment (origin of "six degrees of separation").
- Tom Gilovich — spotlight effect and transparency effect studies (T-shirt experiment).
- Garrison Keillor — Lake Wobegon (namesake of the better-than-average effect).
- Leon Festinger — cognitive dissonance theory; \$1 vs. \$20 boring-task study; *When Prophecy Fails*.
- Louisa Egan (Yale) — M&M choice/denigration study.
- Fritz Heider — attribution theory.
- James Cutting — Intro Psych screen-saver mere-exposure study.
- Peter Salovey ("Dean Salovey") — earlier course lecture on attractiveness referenced.
- IMDb (International Movie Database) — source for Bacon-number and casting facts.
- Leonard Nimoy — *I Am Not Spock* and *I Am Spock*.

PSYC 110 - Lecture 17 - A Person in the World of People: Self and Other, Part II; Some Mysteries: Sleep, Dreams, and Laughter

Overview

This lecture by Professor Paul Bloom continues the discussion of social psychology and then turns to two "mysteries": sleep/dreams and laughter. In the social psychology portion, Bloom examines how we form impressions of others — focusing on the power of first impressions, "thin slices," self-fulfilling prophecies (the Pygmalion effect), and the operation of stereotypes at public, private, and unconscious levels. He argues stereotypes are a normal cognitive tool for generalizing about categories but become problematic through confirmation bias, biased media data, moral concerns about profiling, and effects like stereotype threat. In the second half, Bloom treats sleep as a "minor mystery" with reasonable explanations (restoration, energy conservation), dreams as a "greater mystery" partly explained by memory consolidation, and laughter as a "true mystery" rooted in mock aggression, group cohesion, and surprise.

Key Concepts & Definitions

- **Confirmation bias:** Once you believe something, other information is encoded along lines that support what you believe.
- **Thin slices:** Brief exposures (as little as five seconds, or about one second for sexual orientation) are sufficient to form fairly accurate impressions of others.
- **Self-fulfilling prophecy / Pygmalion effect:** If I believe you have a certain characteristic, this might cause you to behave as if you have that characteristic.
- **Stereotypes:** Information we have about categories and intuitions about the typicality/frequency of certain features of categories (psychologists' broader usage).
- **Profiling:** Applying stereotypes to people in day-to-day life — viewed as morally problematic even when the stereotype is accurate.
- **Stereotype threat (Claude Steele):** If your group has a negative stereotype in a domain, being reminded of your group membership immediately before performing damages performance in that domain.
- **Public / Private / Unconscious stereotypes:** Three levels — what you publicly say, what you privately believe, and unconscious associations you may not know about.

- **Implicit Attitudes Test (IAT):** Developed by Mahzarin Banaji; measures speed differences (congruent vs. incongruent pairings) to reveal implicit attitudes.
- **Automaticity theory (Trish Devine):** Everyone holds stereotypes; these are automatically activated on contact; non-stereotyped action requires conscious effort to override.
- **EEG:** Method for studying sleep by placing electrodes on the scalp.
- **Sleep stages:** Stage I (transition), II–IV (successively deeper, with slow, irregular, high-amplitude delta waves), then back up through III and II, then REM.
- **Slow-wave (quiet) sleep:** Eyes drift slowly/separately; hard to wake.
- **REM sleep:** Rapid eye movement; brain looks awake; body paralyzed except eyes; men get erections; dreams occur.
- **Real dreams vs. sleep thought:** Real dreams are bizarre/narrative (REM); sleep thought is rumination during slow-wave sleep.
- **Lucid dreams:** Dreams in which you choose what to dream about.
- **Tickling:** Touching parts of the body in a mock attack.

Detailed Notes

Introduction (Review)

- Continuation of social psychology from prior lecture.
- Previously covered: biases in seeing ourselves; fundamental attribution error (how we see others); aspects of why we like others — proximity, similarity, attractiveness.
- Left off at the **Matthew effect:** good things compound — rich get better education, smart people are liked more, attractive people get more benefits, etc.

First and Fast: How We Form Impressions of Others

First impressions matter a lot:

- Driven in part by **confirmation bias:** once you believe something, new information gets encoded to support it.
- **Kelley's classic study:** a guest speaker was introduced to students with bios describing him as either "very warm" or "rather cold." Later impressions were heavily biased by the initial description.
- Examples of framing effects on interpretation:
 - Described as "vivacious and creative" → bouncing around confirms it.
 - Described as "drinks too much" → same behavior reads as alcoholism.

- Described as "insecure and nervous" → activity reads as nervous twitches.

- Film example: **Being There** (Peter Sellers) — character **Chauncey Gardner** is mildly retarded but reputed a genius; people interpret simple statements ("Well, I like being in the garden") as profound.
- **Washington snipers** example: initial (false) reports of a white van caused everyone to keep seeing white vans in subsequent incidents.

Thin slices:

- Literature showing impressions form very quickly.
- **Rosenthal and Ambady**: showed clips of university professors; after just **five seconds**, observers could accurately predict teaching evaluations.
- Brief exposure also lets people accurately rate others on "**The Big Five**" dimensions of personality.
- "**Gaydar**" study (same psychologists): people did not know sexual orientation was at issue; orientation was one item on a long form; interviews were filmed; viewers could detect sexual orientation in about **one second**. Everybody is better than chance; gay people are better than straight people; accuracy is only a bit better than chance.
- Combined implication: brief exposures shape much of how we think about people going forward.

Self-fulfilling prophecies / Pygmalion effect:

- Term from George Bernard Shaw's **Pygmalion**: "The difference between a lady and a flower girl is not how she behaves but how she's treated. I shall always be a flower girl to Professor Higgins because he always treats me as a flower girl and always will."
- Made into **My Fair Lady**; Bloom prefers **La Femme Nikita** as illustration.
- **Rosenthal and Jackson** classic experiment: teachers were told (randomly) which students would show an IQ "spurt." Those children did, in fact, show IQ jumps — because teachers gave them more focus and attention.

Positive Uses and Negative Effects of Stereotypes

Stereotypes are necessary cognition:

- Used in a broader sense by psychologists: information about categories and typicality.
- Essential for survival: we must make generalizations about novel things (chair → can sit on; apple → edible; dog → barks, has tail, doesn't speak English).
- Generalizations are typically (not logically) true: a chair could be explosive, an apple poisoned, a dog vegetarian — but generally these stereotypes work.

Stereotypes about people:

- We form stereotypes about men/women, children/adolescents/adults, whites/blacks/Asians, etc.
 - Not inherently bad:
- Some are positive (about creativity, intelligence, your own group).
 - Often accurate — e.g., people correctly answer who is more likely to be a lawyer (Jewish vs. Hispanic) or taller (Sweden vs. Japan).

Problems with stereotypes:**1. Confirmation bias inaccuracy:**

- If you believe gay men are effeminate, effeminate gay men confirm it; non-effeminate ones get dismissed.
- If you believe black men are criminals, you weight supporting cases more heavily.

1. Unreliable data:

- Classical music: women stereotyped as less proficient. **Blind auditions** (behind a screen) led to far more women being hired — stereotype was incorrect and had real negative effects.

1. Misleading media data:

- Media (TV, movies, plays, books) skews perceptions.
- Italian Americans portrayed as mobsters (e.g., **The Sopranos**).
- Jewish portrayals — e.g., **Shylock in Merchant of Venice**.
- Counter-portrayals: **Battlestar Galactica** — character "Starbuck" (male in the original) was transformed into a woman in the remake.

1. Moral problems / profiling:

- It's fine to stereotype chairs, apples, dogs, even dog breeds (e.g., greyhound vs. pit bull). No one calls you a "dog racist."
- Applying stereotypes to people = profiling, considered immoral even when stereotypes are accurate.
- But some stereotypes are accepted (e.g., higher auto insurance for young drivers due to accident rates with "reefer and alcohol"); charging Asians more than Europeans would be considered unacceptable.

1. Stereotype threat (Claude Steele, Stanford):

- Reminding members of a group of a negative stereotype before a task (e.g., math test) drops their performance.

- Works for black students on math; works for women on math too.

Three levels of stereotypes:

| Level | Description |

| ---|---|

| **Public** | What you say publicly; few would say a black or female candidate should be disqualified for president. |

| **Private** | What you really think but don't tell people. |

| **Unconscious** | Implicit associations the person isn't aware of but which affect thoughts about race, gender, etc. |

Data on public stereotypes:

- Proportion willing to vote for an African American president was about **half** around the time Bloom was born; now near 100%.
- White respondents endorsing negative traits of blacks is now infinitesimal.

Subliminal priming study:

- Subjects complete word fragments (e.g., "hos-") while subliminal black or white faces flash.
- Black face primes → more likely to complete as "hostile."
- White face primes → more likely to complete as "hospital."

Implicit Attitudes

- **Implicit Attitudes Test (IAT)** developed by **Mahzarin Banaji** (formerly Yale, now Harvard).
- Largest psychology experiment ever — about a million participants; available at implicit.harvard.edu.
- Class demonstration:

- **Congruent** trial: African American or bad word → "right"; white American or good word → "left."

- **Incongruent** trial: white American or bad word → "right"; African American or good word → "left."

- Across many trials, people are slower on the incongruent version, suggesting biased associations.

- Extended to: gender (women–English, men–math), age, obesity, sexual orientation.

Why care about a half-second difference?

1. **Police shooting studies:** split-second decisions about whether to shoot someone holding an unclear object are heavily affected by stereotypical attitudes.

2. **Judgment calls:** subtle effects on real decisions.

Bystander + race study:

- A scream is heard from a black or white person; subject is either alone or with others.
- Alone: nearly everyone helps regardless of race.
- With others: big difference by race of victim.
- Subjects don't say "I didn't help because the person is black"; they say "someone else would help."

Jack Dovidio hiring study (Yale colleague):

- Subjects judge hiring based on recommendations.
- Strong recommendations (1989): both African Americans and white Americans get hired at similar rates.
- So-so recommendations: white Americans hired significantly more often than African Americans.
- Same pattern in 1989, 1999, and about a year and a half ago.

Trish Devine's automaticity theory:

- Everyone holds stereotypes.
- They are automatically activated on contact with individuals.
- Overriding requires conscious effort, at both individual and group level.

Question and Answer on Stereotypes

- Q (paraphrased): What can be done?

- **Group-level interventions:**

- In psychology department job searches, deliberately surface names from disadvantaged groups (not affirmative action, but counteracting the bias by which qualified people don't come to mind).

- **Blind reviewing** in journals (because of evidence that male vs. female names affect evaluations).

- **Blind auditions** as a classic example.

- **Individual level:** harder — but be conscious of biases, recognize "I'm explicitly not racist" isn't enough, and try to set up systems where qualifications are observed absent race/sex info.

- **Q (paraphrased): Stereotypes vs. own-group bias?**

- Both play a role.

- Stereotypes operate above and beyond in-group favoritism — evidence: African Americans also show the same IAT bias against African Americans and in favor of white Americans.

The Minor Mystery of Sleep

- Sleep is a **motivation** (like food or drink); depriving someone of sleep is a form of torture.
- Big individual differences in how much sleep people need.
- "Sleep macho" used to mean very little sleep; now means a lot.

Studying sleep — EEG:

- Electrodes on scalp record brain electrical activity.
- Wakeful attention: irregular **beta waves**.
- Awake but non-attentive: large, regular **alpha waves**.
- Sound asleep: deep **delta**.

Sleep stages:

- **Stage I:** transition while falling asleep.
- **Stages II–IV:** successively deeper; slow, irregular, high-amplitude **delta waves**.
- After Stage IV, you ascend back through III, II.
- **REM sleep** emerges: brain looks wide awake, body relaxed, rapid eye movements, dreams occur.

- A good night = **four to five sleep cycles** alternating non-REM and REM.

Two types of sleep:

| Type | Features |

| --- | --- |

| **Slow-wave / quiet sleep** | Eyes drift separately and slowly; hard to wake up. |

| **REM sleep** | EEG similar to waking; body paralyzed except eyes; men get erections; dreams. |

- Sleep-researcher joke (which Bloom told the class not to write down): dreams fly through the ether and erections serve as antennas.

Why do we sleep?

- Body is worn out; sleep restores it.
- Growth-producing hormone is released; brain and organs get restored; less food needed; immune system active.
- If you stop sleeping, you don't die dramatically — you get sick and then die.
- Analogy: like turning off a car's engine to repair it.
- Related view: sleep evolved to preserve energy and keep us out of trouble at night. The two views are not incompatible.

Sleep disorders:

- Bloom briefly lists them.
- **Ambien side effects:** some users go downstairs, open the refrigerator, and eat huge amounts of food while asleep; a more recent and serious side effect is sleep-driving.

The Greater Mystery of Dreams

- Hamlet on dreams: deciding whether to kill himself; reasons for: "the slings and arrows of outrageous fortune," "a sea of troubles," "the heartache in the thousand natural shocks that flesh is heir to." Only reason against: **nightmares** — "To sleep, perchance to dream..."
- Philosophical questions: Are you dreaming right now? (Descartes; see also **The Matrix**.) Can you be immoral in a dream? (Augustine.)

- Bloom's answer: you are not dreaming now; you usually can't be immoral in a dream. Exception: **lucid dreams**, where you choose what to dream — possibly immoral if it encourages bad habits of thought.

Real dreams vs. sleep thought:

- **Real dreams** (REM): e.g., "in a submarine wrestling a chicken while your grandmother looks disapprovingly on."
- **Sleep thought** (slow-wave): rumination, e.g., "did I take the garbage out?"

Facts about dreams:

- Everyone dreams (3–4 times a night, depending on sleep duration).
- Not everyone remembers them; dreams leave fragile memories that fade quickly.
- **Dream diary** recommended — write dreams down upon waking.

Dream content (from dream reports, e.g., Hill's 50,000-report collection at dreambank.net):

- Most dreams are mildly negative (on 1–10 scale, below 5.5).
- Report misfortunes.
- **Tribal societies** > industrialized societies in physical aggression.
- **Men** > women in aggressive dreams.
- **Americans** > Europeans in aggressive dreams.

What people want to dream about:

- Women: romance and adventure.
- Men: sex with strangers.
- After adolescence, about **10% of dreams have explicit sexual content**.

Most common dream:

- Not falling, not flying, not public speaking, not naked in public.
- **Being chased** — a "primal" puzzle for evolutionary psychologists and cognitive neuroscientists.

Theories of dreams:

- Freudian (disguised wish-fulfillment): not much support.

- Popular theory: dreams are a **side effect of memory consolidation**. Memories get played over to consolidate them across brain regions — like backing up a computer. Random events flash to consciousness and get woven into a story.
- On this view, dreams serve **no adaptive function**; they are **epiphenomena** — byproducts of another system.

The True Mystery of Laughter

- Bloom got interested through watching his own son **Zachary** laugh.
- A baby-laughing clip from the web shown as another example.
- From an evolutionary standpoint, the puzzle: why make this weird noise in bizarre circumstances?

A bad neuroscience-textbook explanation:

"We laugh when there is incongruity between what we expect and what actually happens unless the outcome is frightening."

- Bloom: about as wrong as anything can be.
- Doesn't explain why incongruity should cause a loud, distinctive noise of staccato segments (each ~1/15 sec separated by ~1/5 sec).
- A bowl of fruit appearing would be incongruous; nobody shrieks with laughter at it.
- Lots of laughter isn't caused by incongruity.

Robert Provine's descriptive work (book: **Laughter**):

- Sent himself and students to places like shopping malls to record what made people laugh.
- Recorded over **1,000 laughter-initiated situations**.
- Most things that triggered laughter were not jokes or humor.
- Sample laughter-initiating lines:
 - "I'll see you guys later."
 - "Look. It's Andre."
 - "Are you sure?"
 - "I know."

- "How are you?"
- "I try to lead a normal life."
- "It wasn't you."
- "We can handle this."
 - Only ~10% of comments could later be re-coded as actually humorous, e.g.:
- "Poor boy looks just like his father."
- "You smell like you've had a good workout."
- "Did you find that in your nose?"
- Dorm food references.
- "He has a job pulling back skin in the operating room."

Properties of laughter:

- **Universal:** every society has it.
- **Early-emerging:** infants do it.
- **Social/communicative:** it's loud — loud noises evolve to communicate.
- **Involuntary noisemaking and contagious:**
- **Laugh track** invention made TV/movies funnier.
- **Rim shot** is a pre-technological attempt to simulate laughter.
- Children especially affected by contagious laughter.
 - **Other primates:** monkeys laugh when they attack (gathering to kill/eat); chimpanzees laugh when they tickle each other.

Tickling:

- Touching parts of the body in a **mock attack**.
- Has to be recognized as mock; being mugged is not funny.

Theories — laughter as mock/collective aggression:

- Laughter often signals teasing, kidding, insults, mock attack — aggression that is attenuated and not real. Laughter signals it isn't real.
- Collective aggression: mob assaults, executions, lynchings, rapes are accompanied by laughter — signaling group solidarity.

- **Plato:** laughter as bonding against a common enemy; sound of group cohesion against a common enemy.

Comedians on aggression in laughter:

- **Dave Barry:** "The most important humor truth of all is that to really see the humor in a situation you have to have perspective. Perspective is derived from two ancient Greek words, 'pers' meaning something bad that happens to someone else and 'ective' meaning ideally somebody like Donald Trump."
- **Mel Brooks:** "Tragedy is when I cut my finger. Comedy is when you fall in an open sewer and die."

Ingredients of humor:

- **A target who experiences some harm** — enemy, friend, or self.
- **Harm not so severe** that it elicits fear, grief, or pity. A stranger slipping on a banana peel is funny; splitting his head open with blood is not (unless you dislike him).
- Damage often involves **embarrassment, sex, scatology, banana peel, pie in the face, pants falling** — where empathy doesn't kick in.
- Aggression is unleashed at someone.
- **Surprise** must be present — this is what made the baby laugh (unpredictable sound), and what made Zachary laugh at **Winnie the Pooh and the Blustery Day** (Rabbit climbs a shelf, it crashes down, he's stunned but not dead).

Why can't we tickle ourselves?

- No surprise, no mock aggression, and likely a general deadening of self-stimulation.
- Some people claim they can — Bloom has heard this multiple times.

Final assignment:

- Reading response: think of an interesting testable idea about either dreams or laughter; can be one sentence or longer. TFs will send Bloom their one or two most interesting remarks; winner gets a small literary or food prize.

Notable Quotes

"The difference between a lady and a flower girl is not how she behaves but how she's treated. I shall always be a flower girl to Professor Higgins because he always treats me as a flower girl and always will."

"We laugh when there is incongruity between what we expect and what we actually — what actually happens unless the outcome is frightening."

"The most important humor truth of all is that to really see the humor in a situation you have to have perspective. Perspective is derived from two ancient Greek words, 'pers' meaning something bad that happens to someone else and 'ective' meaning ideally somebody like Donald Trump."

"Tragedy is when I cut my finger. Comedy is when you fall in an open sewer and die."

"To sleep, perchance to dream, for in that sleep of death what dreams may come when we have shuffled off this mortal coil might give us pause."

"Right now many of you are showing irregular beta waves suggesting intense comprehension and great intellectual focus."

"From this perspective, dreams serve no adaptive function at all but rather they're epiphenomena. They are the byproducts of another system."

Assigned Readings / References

- George Bernard Shaw, *Pygmalion* (quoted).

- Film: **My Fair Lady**.
- Film: **La Femme Nikita**.
- Film: **Being There** (Peter Sellers; character Chauncey Gardner).
- TV: **Battlestar Galactica** (original and remake; character "Starbuck").
- TV: **The Sopranos** (referenced).
- Shakespeare, **Merchant of Venice** (Shylock).
- Shakespeare, **Hamlet** ("to be or not to be").
- Film: **The Matrix** (in connection with Descartes).
- Augustine (theological question about sin in dreams).
- Film: **Winnie the Pooh and the Blustery Day**.
- Robert Provine, **Laughter** (book).
- dreambank.net — Hill's collection of ~50,000 dream reports.
- implicit.harvard.edu — Mahzarin Banaji's Implicit Attitudes Test.
- Studies/researchers cited: Kelley (warm/cold speaker), Rosenthal and Ambady (thin slices), Rosenthal and Jackson (Pygmalion in classroom), Tajfel (minimal group), Robber's Cave study, Claude Steele (stereotype threat), Mahzarin Banaji (IAT), Jack Dovidio (hiring study), Trish Devine (automaticity theory), Robert Provine (laughter).
- Final assignment: reading response with a testable idea about dreams or laughter.

PSYC 110 - Lecture 18 - What Happens When Things Go Wrong: Mental Illness, Part I (Guest Lecture by Professor Susan Nolen-Hoeksema)

Overview

This guest lecture by Professor Susan Nolen-Hoeksema provides an overview of how modern clinical psychology conceptualizes, diagnoses, and treats mental disorders, using mood disorders (depression and bipolar disorder) as a case example. She emphasizes that there is no biological test for any known mental disorder; diagnosis relies on behavioral criteria matched against the DSM, with judgments influenced by social norms, characteristics of the target person (e.g., gender), and context. The lecture covers the "three Ds" heuristic — distress, dysfunction, and deviance — and walks through DSM criteria for major depression and manic episodes, illustrated by video clips of patients. Statistics on age and gender differences in depression are reviewed, along with biological, cognitive, and interpersonal theories and treatments. The lecture closes with comparative effectiveness data for SSRIs and cognitive behavioral therapy (CBT), noting that CBT substantially reduces relapse rates.

Key Concepts & Definitions

- **Behavioral criteria:** A set of symptoms a person reports about how they feel and think, plus observations about behavior and how typical/atypical it is; matched against DSM criteria via subjective judgment.
- **Three Ds:** Distress, dysfunction, deviance — heuristics for labeling behavior as abnormal.
- **Distress:** Behaviors that cause the individual or others significant distress.
- **Dysfunction:** Behaviors that prevent the person from functioning in daily life.
- **Deviance:** Behaviors or feelings that are highly unusual (most controversial; heavily influenced by social norms).
- **Antisocial personality disorder:** Individual has no regard for the rights of other people, no hesitation to steal or hurt others, no empathy/sympathy, and no distress over this; causes other people distress.
- **DSM (Diagnostic and Statistical Manual):** Manual for diagnoses in U.S. clinical psychology/psychiatry, fourth revision; since 1980 emphasized objective, observable criteria.
- **Unipolar depression:** Depression only.

- **Bipolar disorder:** Cycling between depression and mania (formerly called manic-depression).
- **Anhedonia:** Diminished interest or pleasure in usual activities.
- **Insomnia / Hypersomnia:** Trouble sleeping / sleeping all the time.
- **Psychomotor retardation/agitation:** Slowed movement (retardation; more common) or hyperactivity (agitation; rare).
- **Mania:** Abnormally and persistently elevated, expansive, or irritable mood, lasting at least one week, with associated symptoms.
- **Flight of ideas:** Racing thoughts that the manic person cannot speak fast enough to express.
- **Hypomania:** A more mild version of mania with the same symptoms.
- **Monoamines:** Class of neurotransmitters implicated in mood disorders — serotonin, norepinephrine, dopamine.
- **Short allele on the serotonin transporter gene:** A polymorphism that, combined with stress (e.g., maltreatment), predicts depression.
- **Negative cognitive triad** (Aaron Beck): A negative view of the self, the future, and the world.
- **All-or-nothing thinking:** Thinking things are either all good or all bad.
- **Emotional reasoning:** e.g., "If I feel like a loser I must be a loser."
- **Personalization:** Self-blame seen in depressed people.
- **Attributional style in depression:** Internal (self-blame), stable (lasting forever), global (affecting many areas).
- **Anticipatory coping:** Anticipating situations that trigger negative feelings and preparing coping responses.
- **Cognitive Behavioral Therapy (CBT):** Structured, focused, relatively short-term therapy based on Beck's theory.
- **Interpersonal Therapy (IPT):** Less structured, more focused on past relationships and how upbringing fostered negative self-views.
- **Walking wounded:** Depressed people who haul themselves through the day acting normal while suffering.

Detailed Notes

Introduction (Chapter 1)

- Professor Paul Bloom introduces Susan Nolen-Hoeksema as the fourth and final guest lecturer of the semester.
- She is Professor in the Department of Psychology and Director of Graduate Studies.

- Known for work in clinical psychology, especially research on depression, its nature/causes, and sex differences in depression.
- A noted scientist, award-winning teacher, textbook author, and popular writer.
- Will teach a course in clinical psychology next year.

Behavioral Criteria for Assessing Mental Disorders (Chapter 2)

- Goal: brief overview of how modern clinical psychology views mental disorders, characteristics that cut across them, then mood disorders (depression and bipolar disorder) as examples.
- The fundamental question: "What is abnormality?" — where to draw the line between normal/healthy/typical and abnormal/atypical/deviant/unhealthy/maladaptive.
- **There is no biological test for any of the known mental disorders right now.**
- Instead: behavioral criteria — symptoms reported by the person plus observations of behavior, matched against existing criteria; comes down to a subjective judgment call.

Factors influencing diagnostic judgments:

1. **Social norms** — e.g., a woman wearing a veil is typical/prescribed in a Muslim community but until recently was often considered atypical in non-Muslim cultures.
2. **Characteristics of the target person (e.g., gender)** — a man crying is seen as unusual in our culture; a woman crying less so. Conversely, a woman beating up someone is unusual; less so for a man. Gender stereotypes/roles influence judgments.
3. **Context** — "paranoia" and hyper-vigilance for threat is adaptive in downtown Baghdad but abnormal in a quiet farm in Central Connecticut.

The Three Ds:

- **Distress** — e.g., depression is a miserable state; antisocial personality disorder causes others distress.
- **Dysfunction** — depression causes complete decline in functioning (can't go to class/work, withdraw socially, lose job, flunk out).
- **Deviance** — highly unusual behaviors/feelings; most controversial because heavily influenced by social norms.

The DSM:

- Diagnostic and Statistical Manual; fourth revision.
- Around since the '50s; early editions (1950s–60s) were highly subjective and based on Freudian theory.
- Since 1980, real effort to make criteria objective, observable, and reliable across clinicians.

- Provides lists of symptoms, required symptoms, and number of symptoms; deviation, dysfunction, and distress are built into the criteria.

Prevalence:

- As many as **1 in 4 women** will have an episode of serious depression in her life.
- **About 13% of men** will have an episode of serious depression.
- College years are one of the peak times of **first onset** of depression.
- **Late adolescence / early 20s** is the peak onset time for bipolar disorder.

Unipolar Disorders (Chapter 3)

Mood disorders divide into unipolar (depression only) and bipolar (cycles between depression and mania).

DSM Criteria for Major Depression:

- **Criterion 1 (required):** Sadness OR diminished interest/pleasure in usual activities (**anhedonia**).
- Plus at least **four of the following** symptoms:

#	Symptom	Notes
1	Significant weight or appetite change	Lose interest in eating; or binge eating (her friend gained 50 lbs in a year, binge eating especially at night)
2	Sleep disturbances	Insomnia (esp. waking at 3–4 a.m. unable to return to sleep) or hypersomnia (e.g., sleeping 20–22 hours a day)
3	Psychomotor retardation or agitation	Retardation more common: slowed movement, reaction times, speech; prone to accidents. Agitation much rarer.
4	Fatigue / no energy	Can't get up or get moving
5	Feelings of worthlessness or excessive guilt	Can become psychotic (e.g., believing they are Satan; that random events like a flood are their fault). More commonly unrealistic negative self-esteem.
6	Diminished ability to concentrate or indecisiveness	Reading passages repeatedly without processing; can't decide on a paper topic
7	Suicidal ideation or behavior	Occurs in all psychopathology but especially common in depression

- Symptoms must persist for **at least two weeks** to receive the diagnosis.

- Average length of an untreated episode is **at least six months**.

Video clip of "Tara":

- Differentiates everyday sad moods from her debilitating, overwhelming, vegetative depression.
- There is a continuum from normal bad moods to disorder, with no clear demarcation line.
- More moderate forms still warrant a diagnosis and can morph into more serious forms if untreated.
- **"Walking wounded"** — depressed people who haul themselves through the day pretending to be normal, often out of shame or belief they shouldn't need treatment; can persist for years before falling apart.

Bipolar Disorders (Chapter 4)

Bipolar disorder involves periods of depression and distinct periods of the opposite — **mania**.

Criteria for a Manic Episode:

- **Criterion 1:** Abnormally and persistently elevated, expansive, or irritable mood for **at least one week** — not just a good day.
- Plus **three or more** of the following:

#	Symptom	Notes
1	Inflated self-esteem / grandiosity	"If you can't keep up with me it's your fault."
2	Decreased need for sleep	Only a couple of hours, up "raring to go"
3	More talkative; pressured speech	Talk rapidly, hard to interrupt
4	Flight of ideas	Racing thoughts; can't talk fast enough to express them
5	Highly distractible	
6	Increase in goal-directed activity	Grand schemes — cash out family bank accounts, sell house/car, finance irrational ventures
7	Pleasurable but dangerous activities	Sexual promiscuity, drug abuse, gambling, feeling on a "hot streak"

- Mood often irritable and trigger-fire; can become violent due to agitation/irritability.

Video clips:

- An older woman in a manic episode at the hairdresser — pressured speech, increasing agitation, then flipping into a depressed state. **Lithium** mentioned but not stabilizing her.

- "Bernie" — a man with bipolar disorder, not currently in episode. **Hypomania** introduced as the more mild version with the same symptoms.

Continuum and famous cases:

- Mania runs along a continuum from mild to severe/psychotic.
- Psychotic mania: grandiose delusions — believing they are the Messiah, Albert Einstein reincarnated, or have supernatural powers; grandiose hallucinations.
- Some chronically mildly manic people, especially with talent/intelligence, "make it work for them."
- **Kay Jamison** (Johns Hopkins) wrote a book chronicling historical figures (Robert Schumann, Winston Churchill, well-known authors/poets/musicians/politicians) who likely had mild bipolar disorder and channeled mania through extraordinary talent.
- Some very successful CEOs may be chronically slightly manic — function on a few hours of sleep, grandiose, self-confident, channeled productively.

Consequences and motivations for treatment:

- Mania can lead to STDs, drug activity, arrest, bankruptcy.
- Negative consequences and the eventual plunge into depression motivate help-seeking (since mania itself can feel pleasurable).

Prevalence comparison:

- Depression: ~22% of women, ~13% of men lifetime (note: earlier in lecture she said "1 in 4 women"; in Chapter 4 stated as "22% of women").
- **Bipolar disorder: ~1% of population; equally prevalent in women and men.**

Statistics for Depression (Chapter 5)

Age differences (nationwide study, ages 15–55, several thousand people, community sample; percentage with major depression episode in the past month):

- Highest in **15–24** age range.
- Decline somewhat thereafter, although 35–44 remains fairly high.
- Rates of major depression in the **elderly** are quite low by most national statistics, true up to about age 80 or 85.

Three explanations for low elderly rates:

1. As you get older you get wiser → lower depression.
2. Cohort/historical effect — current younger generations are more prone to depression for life compared to grandparents, due to changes in social support, family networks, etc.

3. Depression negatively impacts physical health (higher rates of cardiovascular disease, stroke, immune system diseases); people with lifelong depression die earlier, leaving a healthier elderly sample.
4. All three may be partly true.

Gender differences:

- Data: compilation of hundreds of studies of children and adolescents, using self-report measures (e.g., **Beck Depression Inventory** and its kiddie version) on several thousand children.
- **Prior to age ~13:** boys and girls have similar levels of depression.
- **Around puberty:** girls' rates rise dramatically; boys' rates stay the same or decline.
- **By 18–20:** nearly a **2:1 ratio** of depressed girls to depressed boys, persisting through the adult lifespan.

Hypotheses for gender difference:

- Biological — hormones.
- Sociological — stress, particularly abuse, in girls' lives.
- Likely a combination.

Biological, Cognitive, and Interpersonal Theories and Treatments (Chapter 6)

Genetics:

- Strong evidence for bipolar disorder.
- **Monozygotic (identical) twin concordance:** >60% if co-twin has bipolar disorder.
- **Dizygotic (fraternal) twin concordance:** ~12%.
- **Second-degree relatives:** ~2% (barely above general population ~1%).
- For major depression: stronger genetic component in **early onset depression** (first episodes in childhood/very early adolescence); weaker for depressions clearly triggered by major life events (trauma, loss).

Neurotransmitters — the monoamines:

- Serotonin, norepinephrine, dopamine.
- Both bipolar disorder and depression linked.
- Old view: too little neurotransmitter (especially serotonin).
- Current view: **receptor functioning** — receptors don't function efficiently, so neurons can't make use of available neurotransmitter. Antidepressants improve neurotransmitter functioning.

Gene × Stress interaction (serotonin transporter gene):

- Researchers at Yale: **Julia Kim-Cohen** (recently joined); **Joan Kaufman** in psychiatry.
- **Classic study by Avshalom Caspi and colleagues:** people with one or two **short alleles** on the serotonin transporter gene who were confronted with stress (e.g., maltreatment) were more likely to develop depression.
- Genotype alone did not predict depression in the absence of stress.
- Replicated in at least **four different studies** with other samples and other forms of major trauma.
- **Gene × environment intersection** is a major risk factor.

Brain areas implicated:

Area	Function	Finding in mood disorders
Prefrontal cortex	Higher-order thinking, problem solving, goal-directed behavior	Lowered activity in depression; may explain concentration, planning, emotion regulation difficulties
Amygdala	Processing of emotional information	Overactive responses to emotional information (depression and bipolar)
Hippocampus	Memory and concentration	Shrinkage in chronic depression; may relate to concentration/attention problems
Anterior cingulate	Responses to stress and choice of behaviors	Dysregulation may underlie poor coping and difficulty adjusting behaviors

Drug treatments:

- **Monoamine oxidase inhibitors (MAOIs)** — older class.
- **Tricyclic antidepressants** — still used; ~60% respond; many side effects; fatal in overdose.
- **SSRIs (selective serotonin re-uptake inhibitors)** — Paxil, Prozac, etc.; **Prozac introduced 1987** in U.S.; took over the market; not much more effective than older antidepressants, but fewer side effects.
- **SNRIs (selective serotonin/norepinephrine re-uptake inhibitors)** — prevent re-uptake of serotonin (and norepinephrine) back into the sending neuron, increasing synaptic neurotransmitter; reasonably effective, but a substantial number do not respond; may have to cycle through drugs.
- **Lithium** — drug of choice for bipolar disorder; stabilizes mood swings by stabilizing multiple neurotransmitter systems; major side effects (gastrointestinal), dangerous

during pregnancy for fetal development; often combined with antidepressants because lithium primarily affects manic episodes.

- **Antipsychotic medications** — used in mood disorders when person has lost touch with reality.

Cognitive Behavioral Therapy (CBT) — Aaron Beck's Negative Cognitive Theory:

- Aaron Beck — psychiatrist at the University of Pennsylvania; founded CBT.
- Applies to depression alone (not bipolar disorder).
- **Negative cognitive triad:** negative view of self, future, and world.
- Fed by cognitive distortions:

- **All-or-nothing thinking.**

- **Emotional reasoning** ("if I feel like a loser I must be a loser").

- **Personalization** (self-blame).

- Depressed people make attributions for negative events that are **internal, stable, and global.**

Prospective evidence (Temple University + University of Wisconsin study):

- Identified first-year college students with a negative cognitive/attributional style who had never been depressed.
- Tracked over two and a half years.
- Substantially higher rate of major depression in negative-cognitive-style group vs. comparison group.

CBT therapy components:

1. Identify themes in negative thoughts and triggers.
2. Help the person challenge those thoughts — what's the evidence, are there other interpretations, how could they cope if a bad thing did happen.
3. Recognize negative beliefs/assumptions and challenge their truth value.
4. Change aspects of the environment related to depressive symptoms via active problem-solving.
5. Teach mood-management skills to prevent decline.

CBT vs. Paxil study (240 patients with major depressive disorder):

- 4 months of acute treatment: CBT or Paxil (SSRI); placebo (sugar pill) control.
- **8 weeks:** Paxil had a slight edge over CBT.
- **16 weeks:** Paxil and CBT essentially even — about **60%** of people no longer depressed in each.

Relapse over 12 additional months:

| Group | Relapse rate |

|-----|-----|

| Placebo (off Paxil; no CBT) | ~**80%** in first year |

| Stayed on Paxil | ~**50%** |

| Received CBT | ~**35%** |

- CBT teaches coping skills for new stressors; reduces relapse dramatically. Finding replicated repeatedly.

CBT video clip:

- Therapist demonstrated: **Aaron Beck's daughter**, his heir in CBT practice and development.
- Role play: depressed man who recently lost his job.
- Therapist has him generate his own challenges to negative thoughts and write them down.
- Major premise: most of the work happens **between** sessions, not in session.
- Helps him develop phrases to use when discouraged.
- Teaches **anticipatory coping** — anticipating triggering situations and pre-planning responses.

Interpersonal Therapy (IPT):

- Other major psychotherapy for depression.
- Theory: negative self-views and relationship expectations are rooted in upbringings that fostered them.
- Helps clients understand negative self-views and how they're rooted in past relationships.
- Less structured than CBT; more focused on the past.
- CBT is focused on the present.
- Less research on IPT than on CBT; positive alternative especially for those whose depression is tied up in recurrent themes in relationships.

Closing note:

- The good news: multiple drug therapies and at least two psychotherapies (CBT and IPT) shown by controlled research to be helpful; people have choices and there is no need to stay in depression.

Notable Quotes

"First of all, there is no biological test for any of the known mental disorders right now."

"If you're paranoid and hyper-vigilant, looking for threat in downtown Baghdad, that's considered very adaptive behavior these days because it could prevent you from getting hurt or killed. Whereas, if you're in a quiet little farm in Central Connecticut, being extremely paranoid and believing there's someone who's going to shoot you around the corner is not considered as normal or as acceptable or adaptive behavior."

"As many as one in four women will have an episode of serious depression at some time in her life, and about 13% of men will have an episode of serious depression in their lives."

"It's like the emotion has been sucked out of their life altogether."

"There is this characteristic of a lot of depressed people that I call the 'walking wounded.' They just haul themselves through the day trying to act normal."

"So there is no problem with self-esteem, thank you very much. 'If you can't keep up with me it's your fault.'"

"It's not at all uncommon for them to cash out all the family bank accounts, to sell the house, to sell the car, to sell the kids so that they can finance this great scheme for making a zillion dollars on the Internet tomorrow."

"Genes do not determine the disorder, but the intersection of genes and stress seems to be a major risk factor for the disorder."

"If you just take drugs to get out of a current episode and you go off of the drugs without having any kind of psychotherapy, your rate of relapse tends to be very high."

Assigned Readings / References

- **Kay Jamison** (Johns Hopkins) — book chronicling historical biographies of figures with mild bipolar disorder (Robert Schumann, Winston Churchill, well-known authors/poets/musicians/politicians). Professor offered to send the citation by email if requested.
- **Avshalom Caspi and colleagues** — classic study on the short allele of the serotonin transporter gene $\times$ stress predicting depression.
- **Aaron Beck** — Negative Cognitive Theory of Depression (University of Pennsylvania).
- **Joan Kaufman** (psychiatry) and **Julia Kim-Cohen** (Yale) — current work on gene $\times$ neurotransmitter $\times$ stress intersections.
- **Beck Depression Inventory** — self-report depression questionnaire (and its kiddie version) referenced for gender-difference data.
- **DSM (Diagnostic and Statistical Manual), 4th revision** — diagnostic manual referenced throughout.

PSYC 110 - Lecture 19 - What Happens When Things Go Wrong: Mental Illness, Part II

Overview

This lecture continues the unit on clinical psychology (also called abnormal psychology or psychopathology), framing it as a domain of tremendous scientific and personal importance. Professor Bloom argues that, although mental illness labels have historically been misused to ostracize social deviants, some conditions are real illnesses in the medical sense — they damage functioning, and treated people typically improve and do not wish to return to their illness. He walks through four major categories of disorders: schizophrenia, anxiety disorders, dissociative disorders, and personality disorders, reviewing symptoms, subtypes, prevalence, causes, and controversies. He raises the boundary problem (how to distinguish illness from eccentricity, evil, or unusual altruism) and closes by previewing therapy, noting the gruesome history of past treatments and the unresolved question of whether current therapies will look equally barbaric.

Key Concepts & Definitions

- **Clinical psychology / abnormal psychology / psychopathology:** the study of mental illness.
- **Hallucination:** a sensory experience that isn't real (most typically auditory in schizophrenia).
- **Delusion:** a belief that isn't right / that you shouldn't be having.
- **Ideas of reference:** a delusional theme in which one believes events, conversations, or media coded messages revolve around oneself.
- **Disorganized speech:** babbling/garbled speech in schizophrenia.
- **Catatonic:** extreme disorganized motor behavior in which the person doesn't move, often freezing in a position.
- **Positive symptoms (schizophrenia):** things present that are unusual (hallucinations, delusions, disorganized speech, disorganized behavior).
- **Negative symptoms (schizophrenia):** absence of normal thought or affect (emotion).
- **Paranoid schizophrenia:** believes others are spying/plotting; often has delusions of grandeur.
- **Catatonic schizophrenia:** unresponsive to surroundings; often repeats what people say without generating own speech.
- **Disorganized schizophrenia:** delusions, hallucinations, babbling, unable to help self.

- **Amphetamine psychosis:** schizophrenia-like symptoms (hallucinations, delusions) induced by drugs that shoot up dopamine.
- **Anxiety disorder:** primary disturbance is anxiety — persistent, uncontrollable, unreasonable anxiety, or maladaptive behaviors to reduce it.
- **Generalized anxiety disorder:** constant worrying; ~1 in 20 people get it; can be paralyzing with physical symptoms (headaches, stomachaches, muscle tension, irritability).
- **Phobia:** intense, irrational fears focused on objects, events, or social settings.
- **Preparedness theory:** we evolved to be sensitive to certain phobic objects that were dangerous in our evolutionary history.
- **Obsession:** irrational disturbing thoughts that intrude into consciousness.
- **Compulsion:** repetitive actions performed to alleviate the obsessions.
- **Obsessive-compulsive disorder (OCD):** affects ~2–3% of population; linked to heightened neural activity in the caudate nucleus.
- **Comorbid:** disorders that often go together.
- **Dissociative disorder:** involves dissociation of memory — unaware/separated from part of identity or history.
- **Dissociative amnesia (psychogenic amnesia):** memory loss following a terrible event; can be selective or global.
- **Dissociative fugue (psychogenic fugue):** global amnesia plus identity replacement; person leaves home and develops a new identity.
- **Dissociative identity disorder (formerly multiple personality disorder):** two or more distinct people in one head.
- **Personality disorder:** a personality so bad it veers into mental illness (e.g., narcissistic, avoidant, dependent, histrionic, borderline, paranoid).
- **Antisocial personality disorder / psychopathy / sociopathy** (formerly "moral insanity"): selfish, callous, impulsive, sexually promiscuous; lacks love, loyalty, affiliation, compassion; easily bored, seeks stimulation.
- **"Going postal" / "running amok":** terms (American; Papua, New Guinea Stone Age) for a humiliated person who tries to regain status through one act of terrible violence.

Detailed Notes

Chapter 1. Identifying Mental Illness

- Clinical psychology is the final topic; described as "what psychology is really about" for many.

- Personal relevance: many students currently in therapy or on medications (Prozac, Zoloft, Ambien, Wellbutrin).
- Estimate: about **one half of all college graduates** in the US will have some mental illness serious enough to require treatment.
- Even those not directly affected will see loved ones develop conditions (Alzheimer's, schizophrenia, depression, anxiety).
- Scope of "mental disorders" is broad — includes:
 - prototypical schizophrenic on the streets;
 - alcohol/cocaine and other addictions;
 - Down syndrome, autism;
 - old person losing memory;
 - teenager in deep depression;
 - severe social phobia.
- Hard cases at the boundary:
 - John Gotti / a killer without a conscience — mentally ill?
 - The Kramer character / actor on "Seinfeld" — zaniness vs. illness.
 - Extreme altruism — Batman (sleeps one hour a night, fights crime); a New Yorker case of a wealthy man with loving family who had a kidney removed to help a stranger ("I have two kidneys... I could save someone's life") whose wife called him mentally ill.
- Past/wrong views to dispense with:
 - **Demonic possession** (Gospels: Jesus exorcised demons) — not true.
 - **Social deviance view (Thomas Szasz)**: labeling someone mentally ill is a social decision to ostracize deviants and rid them of moral agency.
- Evidence that the social-deviance view has merit:
 - Dissidents in many countries labeled mentally ill and put in asylums.
 - Slaves in the US who tried to escape were described as mentally ill.
 - Until **1973**, being gay counted as mental illness in official classifications.
 - US presidents (including Bush and especially Clinton) called "psychopath," "schizophrenic," "delusional" by opponents.
- But going too far: some people are mentally ill in the same real sense as cancer — it damages functioning; ill people are not more creative/productive/vivacious; treatment makes them more competent and happier, and they don't choose to return to illness.

- Modern view: disorders are like medical disorders; schizophrenia is as much a disease as cancer.
 - Prior coverage: amnesia (in memory unit), autism (in social reasoning unit).
 - Sex differences in DSM categories:
- Women: more anxiety disorders and mood disorders.
 - Men: more substance disorders, particularly alcoholism; predominantly male antisocial personality disorder.
 - Schizophrenia: roughly evenly matched.
- Today's focus: schizophrenia, anxiety disorders, dissociative disorders, personality disorders (skipping mood disorders — covered in Dr. Nolen-Hoeksema's lecture).

Chapter 2. Schizophrenia

- **Prevalence:** about 1% of the world's population; most common reason for being in a mental hospital due to severity.
 - Called "the lepers of the twentieth century" — shunned, rejected, no known good treatment.
 - Etymology: "split" + "mind" — a **split from reality**, NOT split/multiple personality. (Multiple personality is a dissociative disorder.)
 - Genders roughly equally affected; strikes men earlier; can first occur in college/university age.
 - **Five main symptoms** (four positive, one negative):
1. **Hallucinations** — sensory experiences that aren't real; typically auditory (hearing voices, particularly telling them to do things). Some try to block them by humming, counting, or holding mouth open.
 2. **Delusions** — beliefs that aren't right. Often weird and clearly wrong; often believe they are famous people (many believe they're Jesus Christ). In 1959 a Michigan hospital had **three Jesus Christs** who would meet and talk. Richard Dawkins' *The God Delusion* is mentioned as a contested use of the term.
 3. **Ideas of reference** — newspapers contain coded messages for them; whispers are about them; aliens/FBI/CIA/government conspiring against them.
 4. **Disorganized speech / behavior** — babbling; odd motor movements; extreme = **catatonic** (frozen).
 5. **Negative symptom** — absence of normal thought or affect; quietness, stillness, no caring.
- **Three major subtypes** (of five):
 - **Paranoid schizophrenia** — others spying/plotting; delusions of grandeur; supernatural powers; "God" or "messiah."
 - **Catatonic schizophrenia** — unresponsive; echoes others rather than generating speech.

- **Disorganized schizophrenia** — most stereotypically "insane"; delusions, hallucinations, babbling, dangerous, unable to function.

- **Core dysfunction:** inability to put thoughts and perceptions together, to sequence/coordinate them, to impose logical structure and realistic temporal sequence.

- **Vicious circle:** cognitive problem → loss of social contact → less reality checking → cognitive problem worsens.

- **Genetics:**

- Strong genetic component.

- Identical twin schizophrenic → ~**50%** odds of becoming schizophrenic.

- Not 100%, so environment must contribute.

- Genes create vulnerability; whether expressed depends on environmental triggers.

- **Environmental triggers:**

- Birth trauma.

- Viral infections — more schizophrenics born in winter (more illness in winter); epidemics/plagues correspond to jumps in schizophrenic births.

- **Toxoplasmosis** (carried by cat feces): asked parents of schizophrenics "Did you own a cat when your child was born?" — yes responses correlated with higher odds.

- **Stressful family environments** — but cause vs. effect uncertain (per "worst study in the world" cautions); schizophrenic children may also create stressful family environments.

- **Excess dopamine theory** (older):

- Drugs reducing dopamine reduce symptoms.

- Drugs that boost dopamine → **amphetamine psychosis** (hallucinations, delusions).

- But incomplete: doesn't explain negative symptoms; brain structure differences also found (enlarged cerebral ventricles, reduced frontal lobe activity).

- **Cross-cultural mystery** (from Gray textbook): symptoms and prevalence similar everywhere, but **less industrialized countries have a better recovery rate** than industrialized countries.

Three possible explanations:

1. More latitude / less criticism in families.

2. Less use of antipsychotic medication (which may impair recovery while helping symptoms).

3. Cultural framing of schizophrenia as a transient disorder may make transient outcome more likely.

Chapter 3. Anxiety Disorders

- Primary disturbance: anxiety — persistent, or maladaptive behaviors to reduce it.
- Normal to have anxiety; becomes a disorder when too much, uncontrollable, unreasonable, life-disrupting.

Generalized Anxiety Disorder (GAD)

- ~1 in 20 lifetime prevalence.
- Constant worry; can be paralyzing.
- Physical symptoms: headaches, stomachaches, muscle tension, irritability.
- Genetic component; related to major depression.
- Possible roots in childhood trauma → hyper-vigilance → more prone to GAD after difficult life events.

Phobias

- Intense irrational fears about objects, events, social settings.
 - Diagram referenced (proportion afraid of each object).
 - **Snakes: ~40% afraid.**
 - Mice: many afraid ("mice are the worst things in the world").
 - Cats: very few afraid.
 - **Classical conditioning model is inadequate:**
- Many snake-phobics never had bad snake experiences.
- Many people with bad experiences (car crashes, electrocution, witnessing shootings) do not develop phobias.
- **Preparedness theory:** evolved sensitivity to evolutionarily dangerous objects.

Obsessive-Compulsive Disorder (OCD)

- Obsessions = intrusive irrational thoughts; compulsions = repetitive actions to alleviate them.
- ~2–3% of population.
- Examples: obsessed with being dirty → compulsive washing; obsessed with God's anger → compulsive prayer.
- Common themes: cleanliness, religion.
- Sufferers often know rationally the behaviors are unreasonable.
- Professor's personal example: worrying he left his door unlocked, running back to check.
- **Most common compulsions:** checking and washing.
- Neuropsychological correlate: heightened activity in the **caudate nucleus**.

- Treatable with drugs affecting **serotonin** (a neurotransmitter), despite seeming "Freudian."

Chapter 4. Q&A on Schizophrenia and Anxiety Disorders

- **OCD vs. Tourette's:** Tourette's is a specific neurophysiological syndrome with involuntary tics, tremors, sometimes shouted obscenities/taboo words — no obsessive thoughts. OCD is broader; its behaviors are in service of thoughts.
- **Can individuals have multiple disorders?** Yes — disorders can be **comorbid**. Severe depression often co-occurs with anxiety disorders.
- **Superstition:** depends on severity. Mild ("Step on a crack, break your mother's back"; broken mirror) is harmless. When rituals escalate (must redo walk if a crack is stepped on; "must do X or terrible things happen"), it can verge into OCD; OCD often has religious/magical manifestations.
- **Are schizophrenics dangerous?** Statistically, not as a rule — more often victims than perpetrators; vulnerable; can't defend themselves. Paranoid schizophrenics could develop delusions to harm someone, but rare.
- **How permanent are medication effects?** Generally **temporary**. With mild depression, drugs (Prozac, Wellbutrin) can give time to rebuild coping skills; without coping skills relapse is likely (per Nolen-Hoeksema). Schizophrenia medication effects are also temporary. Drugs help people get out of a problem.

Chapter 5. Dissociative Identity Disorders

- Lecture opens with a movie clip; informal poll: how many think the depicted person is faking?
- **Dissociation** = dissociation of memory; separated from parts of identity/history, except sometimes under special circumstances.
- **Normal dissociative experiences** (from Nolen-Hoeksema's abnormal psychology textbook checklist):
 - "Not sure whether one has done something or only thought about it."
 - "So involved in the fantasy that it seems real."
 - "Feeling as though one's body is not one's own."
 - Pharmaceutical experiments can produce these.
 - "Driving a car and realizing that one doesn't remember part of the trip."
 - "Talking out loud to oneself when alone."
 - "Not recognizing one's reflection in a mirror" (uncommon but still in normal range).

Three types of dissociative disorders:

| Disorder | Description |

|---|---|

| Dissociative amnesia (psychogenic amnesia) | Memory loss after seeing something terrible; selective or global; over half of people charged with homicide claim some degree (confounded by alcohol/drugs and possible lying). |

| Dissociative fugue (psychogenic fugue) | Global amnesia + identity replacement. Example: man's wife leaves him for another; six months later he's tending bar in Miami Beach calling himself **Martin**. When fugue wears off, old identity returns and new is forgotten. Bloom: "This is my favorite mental disorder." |

| Dissociative identity disorder (DID; formerly **multiple personality disorder**) | Two or more distinct people in one head. Example: woman shifts to a 6-year-old personality named **Donna**. |

- **DID is rare and controversial.**
- Featured in *Primal Fear* (Ed Norton's first big movie).
- Used as criminal defense — the **Hillside Strangler** claimed to be two people but was convicted (both).
- Typically starts early; mostly women; often involves recollection of torture or sexual abuse.
- Co-occurs with **PTSD**.
- Cause hypothesis: severe abuse + genetic/biological predisposition to dissociate. DID patients are unusually easy to hypnotize → may begin as **self-hypnosis** to cope.

• **Doubts about DID:**

- In a recent poll, **less than one quarter** of psychiatrists believe DID exists.
- **Between 1930 and 1960** — only **2 cases** in the US.
- **In the 1980s** — **20,000 cases**.
- Largely an American phenomenon.
- Highly therapist-dependent: some hospitals never see a case; some therapists see it in nearly every patient.

- **Suggestion hypothesis:** DID may be real to the patient but induced by therapists who, gripped by theories of repression and multiple personalities, tacitly or overtly encourage patients to develop separate personalities.
- **Continuum view:** philosopher **Dan Dennett** and psychologist **Judith Harris** note that we shift selves across contexts (friends, family, strangers). DID may be an extended/play-acted version of normal multi-self psychology, partly to please therapists/doctors.

Chapter 6. Q&A on Dissociative Identity Disorders

- **Dissociative amnesia vs. retrograde/anterograde amnesia:** the latter result from brain damage and are long-lasting/severe. Dissociative amnesia is caused by specific life events and is often short-lived. All are brain events, but dissociative amnesia is more "psychological."
- **What happened around 1960:** the famous **case of Sybil**, made into a movie, hugely influenced beliefs that DID was real.

Chapter 7. Personality Disorders

- Not actually covered in the Gray textbook.
- Personality = your way of dealing with the world, particularly with other people. Disorders = personalities so bad they veer into mental illness.
- Examples listed:
 - **Narcissistic** — extreme self-focus and self-praise.
 - **Avoidant**.
 - **Dependent**.
 - **Histrionic**.
 - **Borderline** — "really bad... just awful to be with."
 - **Paranoid personality disorder** — not paranoid schizophrenia (no hallucinations); just paranoid beyond normal levels.

Violence and crime

- Most murderers are **not mentally ill** in the medical/clinical sense — driven by normal desires (rage, jealousy, hate) taken to extremes.
- Mass murderers also not substantively different psychologically.
- Cross-cultural pattern: a humiliated male, repeatedly losing status, tries to recover it through one act of terrible violence — gains posthumous status. Called "**going postal**" (American); "**running amok**" (Stone Age Papua, New Guinea); present in every society.

Serial killers: many have mental illnesses, but varied:

- **Jerome Brudos** — severe foot fetish; killed young women, severed their feet, kept them.
- **Son of Sam** — paranoid schizophrenic; murders ordered by a barking dog.
- **Jeffrey Dahmer** — cannibal killer; killed to eat victims. (Joke from a colleague: "He has a severe eating disorder.")
- Other named: **Ted Bundy**, **John Wayne Gacy**, imaginary **Hannibal Lecter**.

- Many killers claim **dissociative disorders** ("it wasn't me, it was my alter ego Fred"); credibility unclear.

Antisocial personality disorder (psychopathy / sociopathy)

- Formerly "moral insanity."
 - Bloom blends psychopath and sociopath into one category and uses "psychopath."
 - Typically male.
 - Traits: selfish, callous, impulsive, sexually promiscuous; lack love, loyalty, normal feelings of affiliation and compassion; easily bored, seek stimulation.
 - Not necessarily unattractive: the **most famous psychopath is James Bond** ("a perfect psychopath in every regard"), as played by the original actor and Sean Connery — but Roger Moore's and Timothy Dalton's versions were not psychopaths.
 - **Is it an illness?**
- Psychopaths don't seek treatment — they don't experience a problem.
 - Many are reasonably successful.
 - The psychopaths studied by psychologists are by definition the unsuccessful ones.
 - Argument: the **successful psychopaths** "run the world" — excel in every field; cold-blooded, ambitious; conceal traits well enough not to be imprisoned.

Chapter 8. Brief History on Therapy

- Central question: **Does therapy work?**
 - History has been "gruesome and unsuccessful":
- Madness viewed as being in league with the devil → tortured to death, burnt, sent to sea.
 - **18th century**: viewed as degenerates, sent away from society.
 - **19th century**: brief compassion — **Pinel** tried to establish mental hospitals.
 - Subsequent treatments largely unsuccessful (summarized in a video clip).
 - Bloom speculates whether current therapies will, in 100 years, look as barbaric.
 - Next lecture: what therapies currently work; then discussion of **happiness**.
 - Optional assignment: email Bloom the most interesting thing covered in the class (subject heading "Intro Psych").

Notable Quotes

"By one estimate, one half of all college graduates in the United States... one half of you will have some sort of mental illness in your life serious enough to require some sort of treatment."

"Schizophrenia is as much a disease as is cancer and should be thought of in the same way."

"Schizophrenics have been described as the lepers of the twentieth century."

"Cleanliness and religion are common themes of obsessive-compulsive disorder."

"This is my favorite mental disorder. If I had to get a serious mental disorder, I would get this because I'd get to travel." — on dissociative fugue.

"Between 1930 and 1960, there were two cases in the United States. In the 1980s, there were 20,000 cases."

"The most famous psychopath, of course, is James Bond who is a perfect psychopath in every regard."

"I often wonder a hundred years from now how they're going to look at our current therapies and then whether they'll see them as equally barbaric and stupid as we look at the therapies in the past."

Assigned Readings / References

- **Gray textbook** — referenced for: list of major DSM disorder categories; the mystery of cross-cultural recovery rates for schizophrenia. Noted as NOT covering personality disorders.
- **Dr. Nolen-Hoeksema's abnormal psychology textbook** — source of the dissociative experiences checklist; also her prior guest lecture is referenced (mood disorders; coping skills and relapse).
- **Richard Dawkins, *The God Delusion*** — mentioned in passing on the contested boundary of "delusion."
- **Thomas Szasz** — psychiatrist cited for the social-deviance critique of mental illness labels.

- **Dan Dennett** (philosopher) and **Judith Harris** (psychologist) — cited for the continuum view of multiple selves.
- *Diagnostic and Standard Manual* — referenced as source of major disorder categories (not required for students).
- Film referenced: *Primal Fear* (Ed Norton).
- Optional student email assignment: subject heading "Intro Psych," to Professor Bloom.

PSYC 110 - Lecture 20 - The Good Life: Happiness

Overview

This final lecture wraps up the prior discussion of clinical psychology by examining whether therapy works, then turns to the emerging field of positive psychology and the science of happiness. Professor Bloom argues that therapy in general does work, partly because of "nonspecific factors" like support and hope, and that different disorders respond best to different therapies. He then surveys happiness research, noting that most people rate themselves as happy, but that simple self-report measures can be skewed by trivial circumstances. Three surprising findings frame the happiness discussion: happiness has a strong heritable set point and people adapt to most events ("hedonic treadmill"); happiness is relative to others' status and salary; and memories of past pleasure or pain are skewed toward peaks and endings (Kahneman). Bloom closes with humility about open questions in psychology and optimism about its scientific methods.

Key Concepts & Definitions

- **Regression to the mean:** A statistical phenomenon where extreme states tend to trend back toward an average over time; cited as a reason people may falsely attribute improvement to therapy.
- **Control group:** A comparison group (e.g., waiting-list patients) used to test whether treatment has a real effect.
- **Nonspecific factors:** Properties shared by virtually all therapies — e.g., support and hope — that contribute to therapeutic benefit independent of the specific method.
- **Placebo effect:** Benefit that comes from belief that something is being done to help, rather than from a specific therapeutic mechanism; can be a self-fulfilling prophecy.
- **Positive psychology:** A movement focused on human flourishing, psychological strengths, and virtues — taking people who are okay and making them better — rather than only treating the broken.
- **DSM-IV (The Diagnostic and Statistical Manual of Mental Disorders):** Reference catalog of mental disorders.
- **The Handbook of Positive Psychology:** Reference work listing psychological strengths and virtues.
- **Lake Wobegon effect (with happiness):** Most people rate themselves as happier than average — statistically improbable.

- **Affective forecasting:** Predicting how happy or sad we will be in the future based on what's happening to us; people are bad at it (Dan Gilbert's work).
- **Set point (happiness):** A genetically influenced natural happiness level or range to which people return.
- **Hedonic treadmill:** Adaptation to good and bad events such that they lose their effect on happiness — "you keep on running but no matter how fast you run you stay where you are."
- **Peak-end rule (Kahneman):** Memory of pleasure/pain is highly skewed toward peaks and endings rather than total amount.

Detailed Notes

Procedural / Course Logistics

- Reading responses and essays completed; grades available next week.
- Experimental participation requirement due by May 7th.
- Final exam on Monday; old final posted (after a teaching fellow called Bloom "damn lazy" for not posting one).
- Special accommodations handled by Judy York, same as midterm.
- Review sessions tomorrow and Friday; Thursday session moved to this room, 6 to 8 p.m., run by teaching fellows.

Chapter 1. How and Why Therapy Works

- The lecture finishes the clinical psychology discussion begun previously (which had ended with the gruesome, unsuccessful, arbitrary history of treating mental illness). Dr. Nolen-Hoeksema had reviewed therapies for depression; the textbook covers schizophrenia, anxiety disorders, etc.
 - **Does therapy work?** Hard to tell because self-report of improvement may be due to **regression to the mean**.
- Mood plots up and down around an average; extremes trend back to average regardless of intervention.
- People go to therapy when feeling unusually bad, so subsequent improvement is statistically likely even without effect.
- Bloom's illustration: "On the worst day of your life you could do naked jumping jacks on the roof of your college for ten minutes. I guarantee you your next day would probably be better."
- **Proper test:** compare to a control group of comparable people who don't receive treatment.

- Example study: depressed patients statistically equal at pre-therapy; some randomly assigned to cognitive training, others to waiting list. Cognitive training group had lower depression scores.

- **General conclusions:**

- Therapy by and large works; treated people do better than untreated, not merely because they sought help.

- Different therapies suit different problems:

- **Everyday unipolar depression** → cognitive behavioral therapy + possibly SSRIs.

- **Bipolar depression** → CBT useless; medication is best bet.

- **Anxiety disorder** → CBT can help.

- **Schizophrenia** → CBT not of much help.

- **Therapist quality varies** — large individual differences in therapist efficacy for reasons not fully understood.

- **Nonspecific factors** (why therapy generally works):

- **Support:** Acceptance, empathy, encouragement, guidance, human touch — someone who really cares about you.

- **Hope:** Enthusiasm/belief that treatment may work; can become a self-fulfilling prophecy.

- **Placebo effect** is often dismissed but powerful — even if the therapist "runs around and dances," belief alone can produce benefit.

Chapter 2. Question and Answer on Therapy

- **Q: Isn't regression to the mean arbitrary?**

- A: It assumes most people have a stable average and fluctuate around it; this holds for things like mood. Statistically, after a bad day you trend up; after the happiest day, you trend down. Human behavior isn't a random walk — we seem to have set points.

- **Q: Could the control group (waiting list) experience anxiety from being told they're not chosen?**

- A: Yes, a plausible confound. Other studies use designs where the control group doesn't know they are controls (e.g., low-scoring depression-inventory students intervened upon while others are unaware). Multiple methods are needed.

Chapter 3. Happiness and Positive Psychology

- Most psychology focuses on human misery — clinical interventions aimed at bringing broken/sad people up to normal.
 - **DSM-IV** catalogs everything that can go wrong, "from Aspergers syndrome to fetishes to paranoid schizophrenia." Browsing it can make one feel mentally unstable.
 - **Positive psychology**: Movement asking how psychology can give insight into flourishing, strengths, and virtues. Has its own **Handbook of Positive Psychology**.
- Some of the work is "real crap" — new age banalities tied to grant funds and Time magazine; attracts self-help hucksters.
- Other work is "quite neat, quite interesting and quite promising."
- **Recommended books on happiness:**
- Marty Seligman, *Authentic Happiness* — Seligman is the pioneer of positive psychology.
- Jonathan Haidt (brilliant young scholar; also disgust/morality, the "sex with dead chicken study") — happiness work.
- Nettle, *Happiness* — smart, beautifully written, extremely short.
- Dan Gilbert, *Stumbling on Happiness* — funny, smart, on NYT bestseller list.
- **How happy are you (1 to 10, 5 = average)?**
- Most common answers are high — 7 or 8.
- **Lake Wobegon effect**: most people think they're happier than most people.
- **Cross-national data** (study of 42 countries):
- No country reported being unhappy on average.
- **Happiest**: the Swiss. (One suggestion offered: chocolate.)
- **Saddest**: the Bulgarians.
- **Americans**: 7.71 — pretty happy.
- **Reasons for caution about these numbers:**
- **Dime-on-photocopier study**: People asked "How happy are you with your entire life?" right after using a copy machine reported greater life satisfaction when a dime had been planted on top.
- **Sunny day study**: Phone interviews found people reported greater life happiness on sunny days — but the effect disappeared if "How's the weather?" was asked first, prompting them to discount weather as the source.

Chapter 4. Getting Used to Happiness

- **What is happiness for?** An evolutionary answer: happiness is a goal state we evolved to pursue, a signal that needs have been satisfied. Like food/satiation.
 - **Steven Pinker's evocative summary:** "We are happier when we are healthy, well-fed, comfortable, safe, prosperous, knowledgeable, respected, non-celibate, in love."
 - **The puzzle:** Modern Americans are healthier and richer than any prior generation but no happier.
- People in the 1950s — poorer, sicker, shorter-lived — are/were as happy as people today.
 - Parents and grandparents were as happy as current students.
 - Poor countries: little effect of national wealth on happiness.
 - Large individual differences in happiness even among those whose basic needs are met.

Three surprising findings about happiness:

1. Happiness doesn't change as much as you think.

- **Strong heritable basis** — a genetically determined set point or range.
 - Identical twins are very similar in happiness, but not identical.
 - **Life events have surprisingly small lasting effects:**
- **Paralysis from the neck down:** Initially devastating but happiness returns roughly to baseline after about a year.
 - **Lottery winners (many millions of dollars):** Initially elated but not as happy a year later. Can even backfire — wrenches winners from family, work, friends, leading to depression.
 - **Tenure (Dan Gilbert's research):** Assistant professors overestimate the happiness boost from getting it and the misery from being denied.
 - **Pre-2004 election (Bush vs. Kerry):** People radically overestimated how much the outcome would affect their happiness.
 - **Consumer goods (Xbox 360, flat screen TV):** Joy fades almost immediately.
 - **Affective forecasting (Dan Gilbert):** we are bad at predicting future feelings.
 - **Day-to-day irrelevance:** Most of daily life isn't spent thinking about who's president, prizes won, etc. Bloom: even after winning Nobel + Guggenheim + MacArthur in one day, a month later "the plumber's not coming and my kids don't respect me."
 - **Adaptation / set point logic:** We get used to good and bad things alike.
 - **Exceptions to adaptation:**

- **Noise:** We don't habituate to ongoing noise (e.g., construction); happiness drops and stays down.
- **Cosmetic surgery (breast enhancement/reduction):** Makes people happier and they stay happier — explanation: appearance matters constantly and you don't get used to looking different.
 - **Hedonic treadmill:** keep running, stay in place; habituate like to a hot or cold bath.
 - **Biblical illustration (Ecclesiastes):** A king with gardens, parks, vineyards, castles, slaves, and male and female concubines yet says: "I hated life. All this vanity and a chasing after wind and there is nothing to be gained under the sun."
 - **Advice from happiness books on dealing with the treadmill:**
- Possessions are not the key — we adapt quickly.
- **Endless novelty** strategy: keep doing new things (sex with someone new, climb Everest, become accountant, scuba dive...).
- **"Old guy" alternative:** step off the treadmill — seek substantial goods like friends, family, long-term projects, rather than quick fixes of delight.

2. Happiness is relative.

- National wealth matters little once above starvation, but **personal relative salary** matters.
- **H.L. Mencken:** "A wealthy man is one who earns a hundred dollars more than his wife's sister's husband."
- Experimental preference: People prefer **\$70,000 when others make \$65,000** over **\$75,000 when others make \$80,000** — relative position wins.
- **British civil servants research:** happiness, health, relationship quality depend on relative position, not absolute salary.
- Humans are status-conscious primates; hierarchy level affects happiness.
- **Maria Callas and Stanley Fish anecdote:** Fish, on being hired, reportedly said he didn't want to talk salary — he just wanted \$100 more than the top person in the department.
- **Q on solitary confinement/prison and habituation to social losses:**
- Hard initially due to shifts in status and social relationships.
- But if you become popular within the new setting (e.g., head of the prison book club), a cheerful person tends to remain cheerful.

Advice to the king (Ecclesiastes):

- Mild unipolar depression → SSRIs + CBT.
- Move castle to a quiet part of the kingdom (noise stress).
- Give up concubines; find a queen.

- Develop social relationships, join a club, get involved in charity, take up a hobby.

3. Memory of pleasure and pain is skewed (Daniel Kahneman).

- Kahneman: one of the few living psychologists with a Nobel prize; previously discussed under rationality research.
- **Dental torture thought experiment** (after the movie *Marathon Man*):
 - **A**: an hour of severe pain.
 - **B**: an hour of severe pain + a few additional minutes of mild pain.
 - Logically A is less painful in total, but **B leaves a less bad memory**.
 - **Peak-end principle**: Remembered experience is skewed to peaks and endings, not total amount.
 - **Empirical work**: Kahneman tested this in the lab (cold-water immersion) and clinically with patients undergoing extremely painful colonoscopy procedures.
 - **Implication**: Topping a painful event off with mild pain gives a less bad memory.
 - **Party example**:
 - **A**: 90% great, last 10% someone slaps you and pours dip on you.
 - **B**: 90% slapping/dip, last 10% great.
 - Simple summing favors A; memory favors B because endings matter.

Chapter 5. Closing Remarks

- Bloom thanks the teaching fellows: **Sunny Bang, Erik Cheries, Jane Erickson, Izzat Jarudi, Greg Laun, Koleen McCrink**.
- The course's opening promissory note — students should now be able to address topics like dreams, testimony, disgust, memory, depression, language, humor, and even good and evil.
- Psychology at Yale is broad; many follow-on courses exist. Bloom is on leave in the fall, not taking sophomore advisees, but open to advice meetings.
- Joke: "Some of you will choose cognitive science instead."

Two closing themes:

- **Humility**: Many basic questions about the mind remain unanswered.
 - We know the brain produces mental life but not how a "lump of meat" produces conscious experience.
 - About half of personality variance is genetic; the other half is experiential — but we lack good theories of which experiences make one adventurous, timid, bitter, or satisfied.

- We understand social influences toward evil but not why some are immune and act heroically.
- **Optimism:** Scientific methods — evolutionary, neurological, developmental, computational; experimental and observational testing — will continue to make progress on questions about beliefs, emotions, decisions, even moral sense.
- Pushback on the worry that scientific study diminishes humanity's specialness: serious scientific study deepens appreciation of the mind's complexity, uniqueness, and beauty.

Notable Quotes

"On the worst day of your life you could do naked jumping jacks on the roof of your college for ten minutes. I guarantee you your next day would probably be better."

"We are happier when we are healthy, well-fed, comfortable, safe, prosperous, knowledgeable, respected, non-celibate, in love." — Steven Pinker

"A wealthy man is one who earns a hundred dollars more than his wife's sister's husband." — H.L. Mencken

"I hated life. All this vanity and a chasing after wind and there is nothing to be gained under the sun." — Ecclesiastes (the king)

"You keep on running but no matter how fast you run you stay where you are, you get used to it."

"The more you look at the mind and how it works from a serious scientific point of view the more you come to appreciate its complexities, uniqueness, and its beauty."

"Placebo effect is often used sort of in a dismissive way, 'Oh, it's just a placebo,' but placebos can be powerful."

Assigned Readings / References

Books recommended by Bloom during this lecture (not formal assignments):

- Marty Seligman, *Authentic Happiness*.
- Jonathan Haidt — work on happiness (also known for disgust and morality research, including the "sex with dead chicken study").
- Nettle, *Happiness*.
- Dan Gilbert, *Stumbling on Happiness*.
- *DSM-IV* (The Diagnostic and Statistical Manual of Mental Disorders).
- *The Handbook of Positive Psychology*.
- Course textbook (referenced for therapies for schizophrenia, anxiety disorders, etc.).